

ZIG

NASA-Eyes-Level Local 3D Rendering Research

Deep technical R&D for an Android astronomy viewer (Kotlin + Jetpack Compose)
Fully local, offline after install. Scientific accuracy before visual polish.

REVISION 3 - SCOPE: a single-object inspector. Thirteen object buttons, one render at a time, orbit plus pinch zoom, inspect anywhere. No orbits, no simulation, no multi-object rendering.

35 sections: NASA Eyes investigation | engine recommendation | master resource table | per-object research with #1/#2/#3 rankings and an INSPECTION SPEC per object (Sun, Mercury, Venus, Earth, Moon, Mars, Jupiter, Saturn, Uranus, Neptune, Milky Way, ISS, black hole) | atmospheres | rings | light geometry | intrinsic data sources | open-source code | asset pipeline | packaging | performance and memory | licensing and legal safety | gap analysis | recommended stack | per-object winners | roadmap | risks | complete source list | Appendix B (per-button budget).

Scope: research and verification only - no implementation code.
Every URL was fetched or queried during the research; licences were checked at source.
Items flagged NOASSERTION or LICENSE UNCERTAIN must be re-verified before shipping.

This PDF is a typeset rendering of ZIG_NASA_EYES_LEVEL_RESEARCH.txt, which is the canonical plain-text version. Prose is re-wrapped and wide tables are re-typeset with wrapped cells; no content differs between the two files.

=====

ZIG – NASA-EYES-LEVEL LOCAL 3D RENDERING RESEARCH Deep technical R&D report – Android / Kotlin / Jetpack Compose Revision 2 – compiled 2026-09-14 (all URLs and asset facts verified on that date) Scope: research only. No application code is provided or implied by this document.

=====

HOW TO READ THIS REPORT

- Every factual claim about a third-party asset, licence, dataset or codebase is tied to a URL that I fetched and checked during this research.
- Where I inspected a live deployed application (NASA's Eyes web app), I say so explicitly and give the exact file I read. Those findings are labelled [OBSERVED BUNDLE] and are reproducible by anyone with curl.
- Where I could not verify a claim from a primary source, it is marked UNVERIFIED or LICENSE UNCERTAIN – DO NOT USE UNTIL VERIFIED.
- Nothing here recommends copying proprietary code. NASA Eyes source code is NOT published (Section 3.8 proves this).

REVISION 3 (2026-09-14) – SCOPE: THE SINGLE-OBJECT INSPECTOR

The product is now an OBJECT INSPECTOR, not a solar-system simulator:

- The UI is a list of buttons, one per object.
- Exactly one object is on screen at any time.
- The user can orbit it, zoom into it, and inspect any part of it.
- There are no orbits, no ephemerides, no time simulation, no multi-object scenes, no inter-body eclipses, and no network in the renderer.
- Each object must look as close to reality as its best free, legally usable scientific data allows, at the highest quality the device can sustain.

This revision re-derives the whole plan under that scope. Sections 19, 21-24, 26-28, 30-34 were rewritten or amended; every object section gained an INSPECTION SPEC with its real data ceiling; Appendix B is the per-button asset budget.

THE CENTRAL TRADE, IN ONE SENTENCE Removing orbits and multiple objects frees the entire GPU budget for a single body, which roughly quadruples the usable texture resolution (16K instead of 8K), allows 4-8x the triangles, and makes full-resolution atmosphere, ring and shadowing effects affordable - but it also makes the SCIENTIFIC DATA the binding constraint rather than the GPU, so the app must know exactly where each object's real data ends and say so (Sections 2.4 and 2.5).

WHAT CHANGED SINCE REVISION 2

- R3-1. Scope rewritten: sections 1.4, 1.5, 2 (all), 4.5, 5.2 added or replaced.
- R3-2. Per-object INSPECTION SPEC added to all thirteen objects (6.7, 7.5, 8.5, 9.5, 10.5, 11.5, 12.5, 13.4, 14.4, 15.4, 16.5, 17.5, 18.8).
- R3-3. Section 23 rewritten as SELF-SHADOWING AND LIGHT GEOMETRY, with the horizon-map technique for terrain shadows and the sun-direction control that replaces orbit-driven lighting.
- R3-4. Section 24 rewritten as INTRINSIC OBJECT DATA; the ephemeris backbone (SPICE runtime, Horizons validation, baked trajectories) is removed.
- R3-5. Section 26 gained Phase K, the local tile store, and explicit deltas that supersede the floating-origin phase and the simulation clock.
- R3-6. Section 27 gained the button-to-pack map: thirteen buttons, thirteen packs, with a complete offline base module.
- R3-7. Section 28 rewritten with corrected ASTC arithmetic (an error of about 33% in revision 1 and 2 is corrected in 28.2) and new single-object budgets.
- R3-8. Sections 30-34 rewritten for the inspector; Appendix B added.
- R3-9. Two asset findings carried from the second-pass review remain in force: the Mercury 166 m/px MESSENGER BDR/LOI mosaics (Section 7) and the LROC licence correction (Section 10.1b). A third is added here: LANDSAT as the Earth hero layer (Section 9.5), verified as USGS public domain with no restrictions on use or redistribution.

R3-10. Rule R0 from revision 2 remains the first legal rule: no licence file means all rights reserved, not public domain (Section 29.1).

FINAL RECOMMENDATION SUMMARY

PRODUCT	A single-object inspector: 13 buttons, one object per screen, orbit + pinch zoom + tap-to-focus, a real rotation spin control and a physical sun-direction control. Fully offline after install.
ENGINE	Google Filament (Apache-2.0) used directly, with SceneView (Apache-2.0) optional for Compose plumbing and loading. Rejected: raw OpenGL ES / Vulkan (a year of work for no quality gain - the hard parts here are materials, textures and HDR, all of which Filament already solves), Unity/Unreal (cost, weight, Compose friction), WebView/three.js (violates the offline requirement).
GEOMETRY	Locally generated UV spheres or ellipsoids with real radii and real oblateness for all planets; DEM-driven adaptive displacement for Moon, Mars, Mercury and Venus; NASA glTF only for the ISS.
DATA CALLS	Sun SDO/AIA + HMI (CC0) + procedural photosphere + corona Mercury MESSENGER MDIS BDR/LOI 166 m/px + USGS DEM 665 m Venus Magellan C3-MDIR + GTDR + Fresnel + slope (radar, labelled) Earth Blue Marble NG (12 months) + Black Marble + Landsat hero tiles at 15-30 m/px (USGS, public domain, no restrictions) Moon LROC WAC 100 m/px global + LOLA ldem_64 + NAC 0.5 m/px hero patches (all via PDS or USGS, not the LROC website) Mars Murray Lab CTX 5 m/px regions + MOLA + HiRISE 0.5 m/px Giants OPAL-derived maps (or JPL PIA), zonal-wind shaders Milky Way SVS 4851 deep map (non-Gaia layers) + USNO/Yale stars ISS NASA VTAD ISS_stationary.glb (44.5 MB, 247547 tris) Black hole own implementation of the Bruneton BSD-3 method
RENDERING	One object at unit radius; 16K albedo tiles, up to 4M-triangle displacement, full-resolution LUT or ray-marched atmosphere, tau-driven rings with analytic shadows, offline-baked horizon maps for grazing-sun terrain shadows, RGBA16F HDR with bloom and AgX or PBR Neutral tone mapping, TAA.
THE DATA WALL	Published: Mercury 166 m/px, Moon 0.5 m/px at hero sites, Mars 0.25-0.5 m/px at sites, Earth 15-30 m/px, Jupiter about 125 km/px. The app reports the current data resolution on screen and labels anything it invents beyond it. This is a feature, not a caveat.
PACKAGING	Play Asset Delivery: base under about 200 MB renders all thirteen objects offline; install-time and on-demand packs (up to about 1.5 GB each) upgrade objects to maximum fidelity.
LEGAL	Rule R0 (no licence file means all rights reserved) leads; then no Gaia/ESA-archive pixels in a commercial build; prefer mission data over presentation images for Hubble/OPAL; no GPL/MPL code; no CC BY-NC or CC BY-SA assets; credit everything on a Data & Credits screen; and never fetch imagery from a non-archive redistribution page whose terms differ from the archive (the LROC case).
OPEN ITEMS	(a) choose the Hubble/OPAL route; (b) verify the KTX-Software licence file; (c) obtain permission for, or replace, the licence-free black-hole references; (d) confirm Landsat tile selection and its per-tile credit strings.

1. EXECUTIVE SUMMARY

1.1 THE HEADLINE ANSWERS

- 1) NASA's Eyes on the Solar System ("EOSS") is a closed-source application. There is no public repository, no source distribution, and its runtime asset tree is partly hotlink-protected. You cannot and must not reuse it. What you CAN reuse is: (a) the subset of models NASA republishes through NASA 3D Resources, (b) the public scientific data pipelines (NAIF/SPICE, JPL Horizons, NASA GIBS, USGS Astrogeology, PDS), (c) the techniques, which are ordinary, well-documented WebGL/PBR techniques. ZIG will match or exceed EOSS visually because EOSS is deliberately conservative (billboard rings, ~2-4 px/deg planet textures streamed, LDR images, point-sprite starfield).
- 2) The correct rendering foundation for ZIG on Android in 2026 is Google Filament (Apache-2.0), used directly through its Kotlin/Java API, with SceneView (Apache-2.0) as an optional Compose-ergonomics + asset-loading layer. Raw OpenGL ES and raw Vulkan are rejected not because they are "hard", but because they add 6-18 months of work to reach Filament's baseline (clustered-forward PBR, IBL, PCSS/DPCF/VSM shadows, HDR bloom, ACES/AgX/PBR-Neutral tone mapping, TAA/FXAA/MSAA, SSAO, depth of field, colour grading, screen-space lens flares) with no quality advantage for this project.
- 3) Filament's material system is a first-class custom-shader platform, not a sealed box. It exposes: `sampler2d` / `sampler2dArray` / `samplerCubemap`, ``shadingModel : unlit | lit | subsurface | cloth``, a fully replaceable skybox material (``vertexDomain : device``, ``eyeDirection`` variable), and ``customSurfaceShading : true`` with a user-supplied ``surfaceShading(MaterialInputs, ShadingData, LightData)`` function per light. [VERIFIED in Filament docs/Materials.md.html and filament/src/materials/skybox.mat] That means: a physically-based planetary atmosphere, a gravitational-lensing star dome, and a relativistic accretion-disk shader can all live inside Filament as materials, and still get Filament's post-processing and tone mapping on top.
- 4) Asset quality is where ZIG wins. The best free, legally usable, scientifically defensible assets are overwhelmingly NOT "3D models". They are scientific raster products that you must convert:
 - Moon: USGS LROC WAC global mosaic 100 m + NASA SVS "CGI Moon Kit" LOLA displacement (public domain).
 - Mars: Caltech Murray Lab Global CTX Mosaic, 5 m/px, 5.7 terapixels, "grants free use of the V01 version of the mosaic for all purposes" – an explicit all-purposes grant (NASA/JPL/MSSS/The Murray Lab).
 - Mercury: USGS MESSENGER MDIS 250 m mosaic + MD3/MD3-color + USGS DEM 665 m (public domain).
 - Venus: USGS Magellan SAR FMAP / C3-MDIR colorized + GTDR topography + Fresnel reflectivity (public domain) – radar, not visible light.
 - Earth: NASA Visible Earth Blue Marble Next Generation (public domain) + NASA Black Marble night lights + NASA GIBS VIIRS true colour.
 - Sun: SDO/AIA + SDO/HMI via JSOC, released under a CC0 waiver – the cleanest licence in the entire project.
 - Gas giants: no perfect single source exists (see Section 12-15); ranking is Hubble OPAL annual global maps (with a copyright caveat), JPL Photojournal cylindrical maps (clean, lower resolution), then Solar System Scope CC-BY-4.0.
 - Rings: do not use a ring PNG. Build the ring texture yourself from Cassini UVIS/RSS optical-depth profiles published by the PDS

- Ring-Moon Systems Node (COUVIS_8001, 438 MB).
- Milky Way: NASA SVS "Deep Star Maps 2020" (16K EXR) for the diffuse glow + a real, public-domain star catalogue rendered as GPU point sprites for the stars. Do NOT use Gaia data (non-commercial).
- ISS: NASA VTAD `ISS\_stationary.glb` (44.5 MB, 247,547 triangles, 28 materials, 26 PNG textures) – I downloaded and measured it.
- Black hole: build it yourself from BSD-3-Clause reference code plus published physics. Bruneton's precomputed-table beam tracer is the only published real-time method designed for constant cost per pixel and it is exactly what a mobile GPU needs.

- 5) Four licensing landmines dominate this project. Read Section 29 before shipping anything:
- (a) Gaia / ESA Space Science Archives data = CC BY-NC 3.0 IGO → commercial use requires written authorisation from ESA (data.licences@esa.int).
 - (b) AURA/STScI Hubble images = copyrighted, commercial use requires STScI permission. ESA/Hubble images are CC BY 4.0 instead. The same OPAL map may exist under both regimes, and which one you use is a legal decision.
 - (c) HYG star database = CC BY-SA 4.0 → share-alike propagates into your data asset. Shadertoy default = CC BY-NC-SA 3.0 → most black-hole shaders on Shadertoy are unusable commercially.
 - (d) GPL-3.0 (rantonels/starless), GPL-2.0 (Stellarium, Celestia) code may not be linked into a proprietary app. Many "great black hole renderers" on GitHub fall into this bucket. Verify each one (Section 25).
 - (e) NO LICENCE FILE AT ALL = ALL RIGHTS RESERVED, not public domain. Copyright is automatic on creation; a repository without a licence file grants you nothing beyond viewing and forking on the host itself. This applies to several widely-copied black-hole renderers (Section 18.5, Section 25). The only legitimate routes are: ask the author for a licence (often granted on request for small projects), reimplement the published physics yourself, or use an MIT/BSD alternative. See Section 29.1, rule R0.

1.2 WHAT QUALITY LEVEL IS ACTUALLY ACHIEVABLE

With the recommended stack, on a 2024-2026 flagship Android phone:

- Planetary surfaces: 8K-16K equirect source data downsampled to 4096-8192 texels across (i.e. 0.1-0.05 px/deg), ASTC-compressed. This is 10-30x the on-disk quality of what EOSS streams for most bodies and vastly better than any "free solar system model pack".
- Rocky bodies can carry real displacement/normal relief derived from laser altimetry or stereo photogrammetry (Moon LOLA, Mars MOLA/CTX, Mercury USGS DEM, Venus GTDR) – EOSS does not do this per-body at this fidelity.
- Atmospheres: a physically-parameterised single/Multiple-scattering model (Bruneton-style precomputed LUTs, BSD-3-Clause reference implementation) with per-body Rayleigh/Mie parameters, ozone-like absorption for Earth, and limb sunset reddening driven by the actual sun direction. This is a genuine step above EOSS's Rayleigh-with-shadow-entity approximation.
- Rings: optical-depth-driven procedural ring textures with per-ringlet structure from Cassini occultation data, forward-scattering and shadow casting. Far beyond EOSS's two billboard PNGs.
- Starfield: real catalogue positions + photometric magnitudes/colours as HDR point sprites (millions of stars at 1 vertex each), plus a NASA public-domain deep star map for the unresolved Milky Way glow.
- Black hole: relativistic lensing with exact Schwarzschild light bending via precomputed tables (constant cost/pixel), blackbody accretion disk with Doppler beaming + gravitational redshift + photon-ring higher-order images.

This is the Interstellar-class image, achieved with a legally clean BSD-3 technique instead of the film's proprietary DNGR.

- Cinematic pipeline: HDR half-float render target → bloom → ACES/AgX/PBR-Neutral tone map → colour grade → TAA. All first-party Filament.

1.3 WHAT IS NOT ACHIEVABLE / WHERE YOU MUST BE HONEST

- No true-colour full-sphere map of the Sun's photosphere exists (you only ever see one hemisphere). Full-sphere solar maps are EUV (false colour).
- No visible-light global map of Venus's surface exists. All Venus surface maps are radar. Venus's visible appearance is a nearly featureless sulphuric-acid cloud deck; anything else is artistic.
- Uranus/Neptune have no high-resolution imagery other than Voyager 2 flybys and Hubble OPAL maps; their "true colour" was only settled in 2024.
- Jupiter/Saturn surface maps are snapshots of a dynamic fluid; "correct" is time-stamped, not timeless.
- Interstellar's Gargantua is NOT physically faithful in photometry (moderate spin $a/M=0.6$ as visualised, Doppler asymmetry deliberately suppressed, 6500 K white balance). Reproducing "the movie look" and "physically correct" are different targets; Section 18 gives both.

1.4 WHAT THE SINGLE-OBJECT SCOPE CHANGES (Revision 3)

The product is no longer a solar-system simulator. It is an OBJECT INSPECTOR: a menu of buttons, one object per button, exactly one object on screen at a time, and the user free to orbit, zoom and inspect any part of it at the highest fidelity the source data supports.

WHAT IS NOW OUT OF SCOPE (and therefore deleted from the plan):

- Orbital mechanics, ephemerides, SPICE kernels at runtime, Horizons validation, trajectory and trail rendering, spacecraft mission timelines.
- Multi-object scenes, inter-object shadowing, transits and eclipses between bodies, floating-origin camera maths for solar-system scale.
- Time scrubbing across a simulation clock. (A per-object spin control and a sun-direction control replace it - see 2.4 and 23.4.)
- The "dynamic" half of NASA Eyes: the time manager, trajectory manager, distance manager and comparison manager observed in its bundle are simply not features of ZIG.

WHAT THAT BUYS (this is the important part - the budget is now spent on one object instead of a system):

1. FULL VRAM TO ONE BODY. Earlier budgets reserved capacity for many bodies. With a single object resident, a flagship can hold a 16K equirectangular albedo (about 60-180 MB depending on ASTC block size, see 28.2) plus a 16K normal map plus an 8K height field and still have headroom. That is roughly a 4-8x increase in texture budget over a multi-body scene.
2. FULL FILL RATE TO ONE BODY. Atmosphere scattering, ring transmission, terrain self-shadowing and the relativistic pass no longer compete with other objects for fragment throughput. Full-resolution ray-marched atmospheres become affordable instead of half-resolution LUT lookups.
3. NO LOD JUGGLING ACROSS OBJECTS. There is one object, so "LOD" now means only "how much of THIS object's detail do I stream in", which is exactly the virtual-texturing problem - and it is a better problem to have.
4. BESPOKE SHADERS. With 13 objects instead of a generic pipeline, each object can have a purpose-built material: a solar granulation shader, a Venusian cloud-deck shader, a ringlet shader, a lensing shader. No generalisation tax.
5. INSTANT, PREDICTABLE LOADING. One object load is a bounded, cacheable,

- measurable operation - which makes a per-object asset pack the natural unit of download (Appendix B).
6. THE INSPECTION AXIS BECOMES THE PRODUCT. Because there is no simulation, all engineering effort goes into the one thing the user will judge: how close can I get, and does it still look real.

THE NEW CENTRAL CONSTRAINT: THE DATA WALL. When the user can zoom anywhere, the limiting factor is no longer the GPU - it is the resolution of the real scientific data. ZIG must therefore (a) maximise real data depth per object, (b) know exactly where each object's data ends, and (c) tell the user when they hit that boundary rather than inventing detail. Section 2.4 is the per-object ceiling table; Appendix B is the per-object disk budget. A "maximum real detail reached" indicator in the HUD is a headline feature of the inspector concept, not a caveat.

1.5 OBJECT LIST (the buttons)

Sun | Mercury | Venus | Earth | Moon | Mars | Jupiter | Saturn | Uranus | Neptune | Milky Way | International Space Station | Black hole. Thirteen objects, one render each, one pack each. The Milky Way button is the "look around" object: it is an inside-out sphere rather than an outside-out one (Section 19). The ISS is a standalone inspectable model with no orbit attached (Section 17).

2. DEFINITION OF THE TARGET QUALITY

"NASA Eyes level or better" is the wrong benchmark for this product. Eyes is a solar-system simulator with modest per-object assets; ZIG is a per-object inspector, so the correct benchmark is the best available scientific dataset for each body, rendered as well as a phone can render it. The target is therefore stated as: THE HIGHEST-FIDELITY TRUE REPRESENTATION OF EACH OBJECT THAT ITS BEST FREE, LEGALLY USABLE DATA SUPPORTS, INSPECTABLE AT ANY ZOOM UP TO THE DATA'S OWN LIMIT.

2.1 MEASURABLE TARGETS (per object, 2026 high-end Android)

Criterion	Target
Albedo texture	8192-16384 texels across for solid bodies; 4096 minimum; ASTC 4x4-6x6
Colour fidelity	sRGB-correct, per-source transfer documented; no false colour presented as true
Relief	normal + height derived from the best real altimetry or stereo DTM available for that body
Displacement	adaptive tessellated patch over the sphere, 0.5-4M triangles, driven by the DEM
Atmosphere	full-resolution scattering (LUT-based single+multiple, or ray-marched on flagships)
Lighting	physically-scaled sun, user-rotatable light direction, correct terminator and limb
Self-shadowing	terrain self-shadow at grazing sun; ring shadows on planet and planet shadows on rings
Materials	metallic-roughness PBR with measured albedo/roughness anchors (Section 20.3)
HDR	RGBA16F, bloom, AgX or PBR Neutral tone mapping, TAA, dithering
Interaction	orbit, pinch zoom, tap-to-focus, turntable spin, real rotation period
Honesty	on-screen indication of the data's own resolution limit, from 2.4

2.2 THE QUALITY HIERARCHY (unchanged priority order)

1. Scientific accuracy: provenance documented per texture, transfer and wavelength stated, false colour labelled, and no invented detail beyond the measured data limit.
2. Visual quality: PBR-correct shading, HDR, correct limb and terminator, sub-texel relief from real height data, no visible tiling or seams.
3. Premium presentation: cinematic camera moves, real rotation, 60 fps, no hitching while tiles stream.
4. Data/geometry fidelity: real radii, real oblateness, real rotation periods, measured albedo and roughness, real ring optical depths.
5. Local Android compatibility: no network in the viewer, no WebView, no cloud.
6. Performance: 60 fps flagship / 30 fps floor, quality tiers that degrade effects and never facts.
7. Package size: deprioritised. Per-object packs of 0.1-2.5 GB each.

2.3 WHAT "LOOKS EXACTLY LIKE REALITY" MEANS, PRECISELY

Every clause below is a testable requirement:

- COLOUR: the RGB channels must be traceable to stated wavelengths and a stated transfer. A filter composite (OPAL 631/502/395 nm) is labelled "approximately true colour"; a radar mosaic is labelled "radar"; an EUV image is labelled with its wavelength; a synthetic colourisation is labelled "simulated".
- SHAPE: oblateness is applied (Jupiter 1/15.4, Saturn 1/10.2, Uranus and Neptune about 1/58), axial tilt is correct, and the rotation axis is drawn from IAU rotation elements - not eyeballed.
- LIGHT: the sun is a physical illuminant, so brightness scales correctly with distance (Earth 1361 W/m², Mars about 590, Jupiter about 50, Neptune about 1.5) and exposure is the user's control, not a fudge. Two objects lit the same way that are not the same distance apart is a bug, not a style.
- LIMB: airless bodies have a hard limb; atmospheric bodies have a soft limb with scattering past the geometric terminator; the Sun has limb darkening.
- NIGHT: Earth has real night lights; the Moon has honest darkness; no other body gets invented city lights.
- SURFACE: relief is real (LOLA, MOLA, USGS DEMs), so shadows fall where shadows fall. Terrain detail finer than the DEM is interpolation, not invention - and if the app adds procedural micro-detail it says so.

2.4 THE INSPECTION CEILING TABLE (the most important table in this report)

Per object: the finest real data available, what that means for the user, and where the app must stop zooming or start labelling.

Object	Best real ground resolution	What the user can see	Where the data ends
Sun	340 km/px (SDO AIA/HMI 4096 px full disk)	granulation, sunspot groups, limb darkening, active regions	below about 100 km/px the surface is procedural; label it
Mercury	166 m/px (MESSENGER MDIS BDR/LOI, 256 px/deg, 92160x46080)	craters, scarps, hollows, ejecta rays	166 m/px; relief beyond 665 m is interpolated from the DEM
Venus	75 m/px radar on about 17% of the surface; 4.6 km/px global radar	radar texture of the surface; a featureless cloud deck in visible light	visible-light detail does not exist; the cloud deck is procedural
Earth	15 m/px pan / 30 m/px multispectral (Landsat, PD) over land since 2013	coastlines, cities, farmland, glaciers, individual fields	ocean floor: only about 450 m bathymetry (ETOPO2022), no optical imagery
Moon	0.5 m/px (LROC NAC) over large areas; 100 m/px global WAC	boulders, rover tracks, lander hardware at hero sites	global colour 100 m/px; relief 474 m/px (LOLA ldem_64); NAC DTMs 1-2 m at sites
Mars	0.25-0.5 m/px (HiRISE) at sites; 5 m/px (CTX) nearly global	dunes, rover tracks, layered deposits, fault scarps	relief 463 m/px (MOLA) global; HiRISE DTM about 1 m at hero sites
Jupiter	about 125 km/px (OPAL 3600 px; Cassini map about 120 km features)	belts, zones, vortices, the Great Red Spot	hit the wall fastest of all: at full-disk-plus-4x the map is soft
Saturn	about 105 km/px body; rings to 1 km radial (Cassini UVIS)	banded atmosphere; richly structured rings	rings are far sharper than the planet; say so, it is interesting
Uranus	about 45 km/px (OPAL)	a nearly featureless pale cyan disk with a bright polar cap	almost nothing to see; the tilt is the show
Neptune	about 43 km/px (OPAL); Voyager 2 mosaics for spot epochs	dark spots and bright companion clouds at the observed epochs	sparse epochs; date-labelled spots only
Milky Way	16384 x 8192 map (about 1.3 arcmin/texel) plus catalogue stars	the galactic band, dust lanes, named stars, constellations	angular, not spatial: stars are point sources at every zoom
ISS	its model's own texture resolution (26 textures, 247547 tris)	modules, truss, radiators, solar array blankets	below the texture resolution the geometry looks like a model
Black hole	resolution independent (analytic geodesics)	photon ring, higher-order disk images, beaming asymmetry	never - this is the only object with unlimited meaningful zoom

2.5 HOW THE CEILING IS PRESENTED TO THE USER

- A discreet "DATA: 166 m/px (MESSENGER MDIS)" readout that updates as the user zooms, so the achievement of real detail is visible rather than hidden.

- When the user passes the data limit, a soft "beyond published data - procedural detail" indicator appears, and the procedural layer can be toggled off to see exactly what the science supports. This one feature is worth more in reviews and credibility than any single texture upgrade.

3. NASA EYES TECHNICAL INVESTIGATION

3.1 METHOD

I did not rely on second-hand descriptions. I fetched the deployed web application and read its code and asset tree directly on 2026-09-14 using curl.

Findings below marked [OBSERVED BUNDLE] come from these files:

```
https://eyes.nasa.gov/apps/solar-system/      (index.html, 3,071 bytes)
https://eyes.nasa.gov/apps/solar-system/config.js (128 bytes)
https://eyes.nasa.gov/apps/solar-system/vendors.js (2,529,206 bytes)
https://eyes.nasa.gov/apps/solar-system/app.js (1,740,432 bytes)
https://eyes.nasa.gov/apps/solar-system/preload.js (133,068 bytes)
https://eyes.nasa.gov/assets/...              (asset probes)
```

3.2 ARCHITECTURE [OBSERVED BUNDLE]

- The DOM root for the 3D view is literally:

```
<div id="pioneer"></div>
```

 i.e. "Pioneer" is the internal name of JPL's scene/render engine. The UI is a separate DOM layer (#ui / #static-ui / #dynamic-ui), i.e. DOM/HTML for UI and WebGL canvas for the 3D scene – the standard pattern of a browser app, not a game engine UI.
- Runtime asset roots are injected at load:

```
config = { staticAssetsUrl: 'https://eyes.nasa.gov/assets/static',
            dynamicAssetsUrl: 'https://eyes.nasa.gov/assets/dynamic' }
```

 so the app can be re-hosted (this is how the education "embed" builds work).
- The app loads four bundles: commons.js, preload.js, vendors.js, app.js.

3.3 THE UNDERLYING RENDERER IS THREE.JS [OBSERVED BUNDLE]

vendors.js contains the three.js library plus:

```
GLTFLoader, KTX2Loader, DRACOLoader, MeshoptDecoder, Basis transcoder,
KHR_texture_basisu, KHR_draco_mesh_compression, KHR_meshopt_compression,
KHR_mesh_quantization, KHR_materials_clearcoat / sheen / transmission /
volume / ior / iridescence / anisotropy / dispersion,
KHR_texture_transform, KHR_lights_punctual,
EXT_color_buffer_float / half_float, OES_texture_float_linear,
WEBGL_multisampled_render_to_texture, EXT_texture_filter_anisotropic,
KHR_parallel_shader_compile, KHR_SUPERCOMPRESSION_ZSTD.
```

app.js also references ShaderChunk (three.js shader chunk system), ACES tone mapping and Bloom. Practical reading: the NASA Eyes web app is a three.js-based PBR viewer with a JPL-authored scene/data layer (Pioneer) on top. Everything in that stack is either MIT (three.js) or Apache-2.0 (Draco, BasisU)

– the open-source technologies are reusable; JPL's layer is not.

3.4 THE SCENE-GRAPH / COMPONENT MODEL [OBSERVED BUNDLE]

app.js contains a component/manager architecture. Component classes observed (count = occurrences in the bundle):

```
TrailComponent (116), ModelComponent (80), DivComponent (69),
SpheroidComponent (35), AtmosphereComponent (38), SpheroidLODComponent (32),
CameraComponent (30), LabelComponent (27), SpriteComponent (26),
ShadowConeComponent (21), OrbitLineComponent (20), WMTSCComponent (19),
CMTSCComponent (19), LightSourceComponent (18), ParticleSprayComponent (16),
ConnectedSpriteComponent (15), SpoutComponent (14), RingsComponent (14),
StarfieldComponent (13), CometTailComponent (12), IndicatorComponent (11),
```

DynamicEnvironmentMapComponent (9), ConstellationsComponent (9),
SkyboxComponent (8), FrustumComponent (8), DistanceLineComponent (8).

Application managers observed:

time, camera, scene, trajectory, wmts, label, selection, layer, content,
search, link, distance, comparison, autoplay, title, spout, router.

This is the single most useful public insight into EOSS: it is a *data-driven component system* where each astronomical entity is assembled from declarative components (spheroid LOD + texture sets + atmosphere + rings + trails + sprites), and the time/camera/trajectory managers drive everything. ZIG should copy the *architecture idea* (entity + components + a single simulation clock + double-precision body frames), not the code.

3.5 TEXTURES AND LOD: CMTS AND WMTS [OBSERVED BUNDLE]

Two distinct texture subsystems exist:

- CMTSComponent: a pyramidal LOD texture stack for bodies. Asset URLs are templated as

```
$DYNAMIC_ASSETS_URL/cmts/{body}/color
$DYNAMIC_ASSETS_URL/cmts/{body}/height
$DYNAMIC_ASSETS_URL/cmts/{body}/normal
$DYNAMIC_ASSETS_URL/cmts/{body}/specular
$DYNAMIC_ASSETS_URL/cmts/{body}/night
$DYNAMIC_ASSETS_URL/cmts/{body}/cloud
```

with ``setMaxLevel(...)`` (a LOD pyramid) and ``setShadowEntities([...])`` (the entity list that casts shadows onto that body).

→ EOSS planets are spheroids with a colour/height/normal/specular/night/cloud texture pyramid. Note the separate "night" channel: night-side lighting is a first-class feature in EOSS for at least some bodies.

- WMTSComponent: OGC Web Map Tile Service streaming, used for Earth. The manifest I fetched –

```
https://eyes.nasa.gov/assets/dynamic/earth/data/visibleEarth/dataset_manifest.json
(533,061 bytes; "generated": "2026-09-14"; "type": "wmts"; frequency
"daily"; dates 2016-01-01 → 2026-09-13; layer
"VIIRS_SNPP_CorrectedReflectance_TrueColor"; per-date "blankAreaPct") –
```

proves that EOSS's Earth is NASA GIBS VIIRS SNPP Corrected Reflectance True Colour tiles, i.e. the *real* NASA Earth imagery service, with a per-day availability and cloud-blank-ratio check.

The component also parses ``maxLevel``, ``spheroidLOD``, ``floatingClouds``, ``cmts.textures[...]`, and ``night`` flags.

- Access reality: ``cmts/*`` and several ``/models/...`` and ``/env_maps/...`` paths return HTTP 403 "AccessDenied" (XML) to unauthenticated non-app requests, i.e. hotlink protection. Some (e.g. ``models/sc_iss/ISS_stationary.glTF``) are downloadable. Do not build a pipeline on scraping these paths: it is both fragile and legally undefined.

3.6 STARFIELD, RINGS, SUN IN EOSS [OBSERVED BUNDLE]

- Starfield: six discrete magnitude/LOD bins

```
$STATIC_ASSETS_URL/stars/stars.{0..5}.bin    (116,545 / ... / 173,171 bytes)
plus
```

```
$STATIC_ASSETS_URL/stars/galaxies.0.bin      (27,857 bytes)
$STATIC_ASSETS_URL/stars/constellations.bin  (10,952 bytes; header text
contains constellation names, e.g. "Aquila")
```

and an optional skybox image

```
$STATIC_ASSETS_URL/env_maps/starmap_{resolution|2048}.jpg (403 to me)
```

with a ``setParticleSize(...)`` control.

→ Conclusion: EOSS renders stars as a small number of GPU point-sprite particle systems from binary catalogues (hundreds of KB total), with an optional low-res star map for the diffuse background. It is NOT a huge photoreal Milky Way. This is a low bar for ZIG to beat.

- Rings: ``sprites/saturn_rings_top.png``, ``saturn_rings_bottom.png``,

- `uranus\_rings.png`, `neptune\_rings.png` – sizes I actually downloaded:
saturn_rings_top.png = 3,588 bytes. These are BILLBOARD sprites (top and bottom planes), not volumetric or optically simulated rings.
→ Huge headroom for ZIG (Section 22).
- Sun: a SpriteComponent with `sprites/sun\_glow.png` (size 100x100 px units, colourMultiplier (1,1,0.5), renderOrder -2), a light source, and separate `Corona`/coronal content in mission modules. The sphere itself has LightSourceComponent + glowing sprite.
- ISS: `models/sc\_iss/ISS\_stationary.gltf` (+ 22.2 MB .bin). I downloaded and parsed it: generator "Khronos glTF Blender I/O v4.2.60", 149 meshes, 150 nodes, 47 materials, 25 images (JPEG/PNG), ~264,809 triangles, and it uses KHR_materials_clearcoat, KHR_materials_specular, KHR_materials_ior. That is a genuinely good spacecraft asset – and a near-identical model is republished by NASA at a stable, documents-based URL (Section 17).

3.7 DATA PIPELINE [OBSERVED BUNDLE + UNVERIFIED DETAILS]

- Trajectories/ephemerides: the bundle builds URLs of the form
\$DYNAMIC_ASSETS_URL/dynamo/{mission-url}
so the mission-data service is internally called "dynamo". Component names (TrailComponent, OrbitLineComponent, ShadowConeComponent, CometTailComponent) show what is built from that data.
- Coordinate handling: `LatLonAlt` type observed, proj4 library present (projections/tile maths), moment.js present (time handling), hammer.js present (touch gestures), plus a `time` manager and a `trajectory` manager.
- UNVERIFIED: JPL has not published a public specification of "Dynamo" that I could locate as a primary technical document during this research. The observable behaviour (URL naming, per-mission data files, DSN/Horizons-class positions) is consistent with precomputed SPICE/Horizons-derived ephemeris products, but do not state that as fact in any ZIG documentation.
- Public equivalents that ARE documented and usable:
NAIF/SPICE (kernels + toolkit) – <https://naif.jpl.nasa.gov/naif/rules.html>
JPL Horizons – <https://ssd.jpl.nasa.gov/horizons/>
NASA GIBS (Earth imagery) – <https://nasa-gibs.github.io/gibs-api-docs/>
PDS (planetary data) – <https://pds.nasa.gov/>
NASA 3D Resources (models) – <https://science.nasa.gov/3d-resources/>

3.8 IS THE NASA EYES SOURCE CODE PUBLIC? – NO. EXPLICIT STATEMENT.

- There is no official NASA/JPL repository for Eyes on the Solar System. I queried the GitHub API directly:
GET /search/repositories?q="eyes on the solar system" → total_count = 2,
both third-party hobby projects.
GET /search/repositories?q=eyes+org:nasa → total_count = 0.
One of those two hits is a third-party re-upload of EOSS spacecraft models with NO licence file.
 - The application ships only minified bundles under eyes.nasa.gov. No source archive, no build instructions, no licence granting reuse of the application.
 - Wikipedia lists the application's licence as "public domain" (US Government work). Treat that as a statement about the *work being a government work*, NOT as a grant of source code, and not as a statement that every asset inside it (ESA data, third-party imagery, and hotlink-protected runtime assets) is free of third-party rights.
- CONCLUSIONS (mandatory):
- DO NOT copy, decompile, or reuse NASA Eyes code or its minified bundles.
 - DO NOT scrape eyes.nasa.gov runtime assets as a content pipeline; several are 403-protected, and the ones that are not carry no redistribution licence from the app itself. Use NASA 3D Resources and the scientific

archives instead.

C) DO NOT claim ZIG is "based on NASA Eyes".

3.9 WHAT NASA OFFICIALLY PUBLISHES (THE REUSABLE FOUR-FIFTHS)

- A) Assets: NASA 3D Resources (science.nasa.gov/3d-resources/, and the legacy nasa3d.arc.nasa.gov/models) – spacecraft, ISS, rovers, asteroids; many in .glb. Each model page states a Source credit line (often "NASA/JPL/Eyes on the Solar System"). NASA's own usage guidelines explicitly cover "files used in the rendition of 3-dimensional models, such as texture maps and polygon data in any format" (see 3.10).
- B) Data: NAIF/SPICE kernels + toolkit, JPL Horizons, PDS (all nodes), USGS Astrogeology Astropedia, NASA GIBS, SD0/JSOC, NASA SVS, NASA Visible Earth, MAST/ESA archives, Caltech Murray Lab.
- C) Techniques: documented in the open literature and in open-source code – PBR/PBRT-style shading ([three.js/Filament docs](https://three.js/filament/docs)), KTX2/Basis (Khronos), Draco/meshopt (Khronos/zeux), atmospheric scattering (Bruneton & Neyret; Hillaire), relativistic rendering (Luminet 1979; James et al. 2015; Bruneton 2020).
- D) Open-source technology that reproduces equivalent results: Filament (Apache-2.0), three.js (MIT), KTX-Software (Apache-2.0), Draco (Apache-2.0), Meshoptimizer (MIT), Bruneton atmosphere + black hole shaders (BSD-3-Clause).

3.10 THE NASA USAGE-STATEMENT NUANCE (READ THIS TWICE)

From NASA's official "Images and Media" guidelines

(<https://www.nasa.gov/nasa-brand-center/images-and-media/>):

- "NASA content – images, audio, video, and media files used in the rendition of 3-dimensional models, such as texture maps and polygon data in any format – generally are not subject to copyright in the United States."
- "NASA content used in a factual manner that does not imply endorsement may be used without needing explicit permission. NASA should be acknowledged as the source."
- "For use of NASA images clearances may be necessary for images that include any NASA logos or NASA employees... Otherwise, NASA imagery can be generally used editorially within published works that are not promotional in nature."
- "The NASA Insignia, Logotype, identifiers, and imagery are not in the public domain." (The NASA "meatball" logo and identifiers are protected by law; 14 CFR 1221 restricts their use.)
- "NASA occasionally uses copyright-protected material of third parties with permission... Those images will be marked identified as copyright protected with the name of the copyright holder."
- If an identifiable person appears, commercial use may require permission.

Practical engineering rules for ZIG:

1. Credit every NASA dataset/asset on an in-app "Data & Credits" screen.
2. Never place the NASA logo/insignia in ZIG, and never structure marketing as "powered by NASA" or "NASA-approved".
3. Never use an asset marked as third-party copyrighted by NASA without a separate licence.
4. Treat ambiguity as a blocker: if a texture's provenance is unclear, mark it LICENSE UNCERTAIN – DO NOT USE UNTIL VERIFIED.

3.11 WHAT ZIG DELIBERATELY DOES NOT COPY FROM EYES

Revision 3 removes any temptation to imitate the simulation parts of EOSS. The following components observed in the Eyes bundle are explicitly NOT features of ZIG, and nothing in this report should be read as requiring them: the time manager (simulation clock and time scrubbing), the trajectory and trail managers, the orbit-line and distance-line components, the comparison manager, the autoplay manager for mission timelines, the dynamic environment map (used for real-time lighting as the scene evolves), and the WMTS streaming component (network dependence). What ZIG does carry over is architectural discipline: one entity, declarative components, a single scene, and a strict separation between data and presentation. Section 30 is rewritten accordingly.

4. RECOMMENDED ANDROID 3D ENGINE

4.1 CANDIDATES AND SCORES (higher = better; weights chosen for THIS project, where scientific accuracy > visual quality > everything else)

Scoring key: SC=scientific/engineering capability, VQ=visual quality ceiling, PBR, GLTF, HDR/IBL, SHAD=shadows, POST=post-processing/tone mapping, CUST=custom shader capability, PERF=performance on Android, MEM=memory control, COMP=Android/GPU coverage, CMPZ=Jetpack Compose integration, MAT=maturity/ maintenance, LIC=licence cleanliness. Each /10.

Engine	SC	VQ	PBR	GLTF	HDR	SHAD	POST	CUST	PERF	MEM	COMP	CMPZ	MAT	LIC	TOTAL/140
Google Filament (direct, Kotlin/Java API)	10	10	10	10	10	9	10	9	9	8	10	7	10	10	132
Filament + SceneView (Compose wrapper)	10	10	10	10	10	9	10	9	9	8	10	10	8	10	133
Godot 4 (native lib, Compose host)	8	9	8	8	9	9	9	9	8	7	9	5	9	9 (MIT)	116
Raw OpenGL ES 3.2 (custom engine)	9	10	10	10	10	10	10	6	6	10	6	5	n/a (own code)	118	
Raw Vulkan 1.3 + VMA (custom engine)	9	10	10	10	10	10	10	7	7	8	5	5	n/a	118	
three.js in WebView / Filament.js WASM	6	7	7	8	7	6	7	7	5	5	9	6	8	9	91
Unity / Unreal	10	10	10	9	10	10	10	9	8	6	10	5	10	3-5 (proprietary)	121
BGFX / OGRE-Next / OSG (C++ renderer libs)	7	8	8	8	8	8	8	9	6	6	8	4	6	8	108

* Asterisks = "only if you personally write it, over many months, and get it right."

4.2 WHY THE SCORES LOOK LIKE THAT (the parts that matter)

- Filament is Google's production real-time PBR engine (20,489 stars, pushed 2026-09-13, Apache-2.0). Documented features include: physical light units, directional/point/spot lights, screen-space refraction, global fog, HDR bloom, depth-of-field bokeh, multiple tone mappers (PBR Neutral, AgX, ACES, filmic, generic custom), colour/tone management (luminance scaling, gamut mapping), colour grading (exposure, night adaptation, white balance, channel mixer, ASC CDL, contrast, saturation), TAA/FXAA/MSAA, screen-space lens flares. Backends: OpenGL ES 3.0+ (Android default), Vulkan 1.0 (Android), Metal, OpenGL 4.1+, WebGPU, WebGL 2. Current Maven artifact verified: com.google.android.filament:filament-android 1.71.5.
- Filament is not a closed black box, which is the usual objection. Verified in docs/Materials.md.html and filament/src/materials/skybox.mat:
 - * parameter types include sampler2d, sampler2dArray, samplerCubemap;
 - * shadingModel ∈ {lit, subsurface, cloth, unlit, specularGlossiness};
 - * `customSurfaceShading : true` lets you supply


```
vec3 surfaceShading(const MaterialInputs, const ShadingData, const LightData)
```

 invoked per light – i.e. you can implement your own BRDF/atmosphere response while keeping Filament's light management;
 - * the skybox is an ordinary material (`vertexDomain : device`, `depthWrite : false`, variable `eyeDirection`), so a full-screen ray-marched lensing or atmospheric pass is legal inside Filament.
 - * Material compilation happens offline with `matc` → `.filamat` (no runtime shader compilation cost, no driver-dependence surprises).
- SceneView (<https://github.com/SceneView/scenview>, Apache-2.0, 1,316 stars, pushed 2026-09-12; Maven io.github.scenview:scenview 2.3.x) adds a

- Compose-native API (nodes as composables), async model/material/environment loaders, gesture delegates, geometry generators, HDR environment loading, and post-processing options (bloom, DOF, SSAO, vignette, colour grading, tone mapping) documented in its README. Risk to manage: it is a convenience layer; ZIG must be able to bypass it for custom materials and render targets. Recommendation: depend on Filament directly for the physics-heavy materials, and use SceneView only for Compose plumbing/loading where it does not obstruct custom `.filamat` materials and custom RenderTargets.
- Why NOT raw OpenGL ES or Vulkan: the argument is not "it's hard", it is "it buys zero quality and costs a year". Everything the project needs (PBR, IBL, HDR, bloom, ACES/AgX, TAA, shadows, material variants, glTF ingestion, GPU memory tracking) is already implemented and battle-tested by Filament on Android GPUs, with an open licence and a Kotlin API. A hand-written Vulkan engine would still need all of it, and would then be worse on the tricky Android GPU-compatibility details (Mali/Adreno/PowerVR difference in MSAA resolve, VSM, texture swizzle, timer queries – Filament's release notes are essentially a chronicle of these fixes).
 - Why NOT WebView/three.js/WASM: explicitly excluded by the project (no web page as final architecture, no streaming), and it also caps quality (JIT shader compile, no Vulkan on all devices, sandbox limits on big ArrayBuffers, no true compute).
 - Why NOT Unity/Unreal: baseline licence cost and royalties, huge APK, poor Compose integration, and no scientific advantage.

4.3 THE ONE LEGITIMATE REASON TO TOUCH A LOWER-LEVEL API

If you later want *screen-space* relativistic effects at full resolution with fp32 or compute-based ray marching, Filament still lets you:

- render to a half/quarter-res RenderTarget, do the heavy pass in a custom material (or a native compute dispatch if you add Vulkan compute yourself through Filament's `Engine`-adjacent Vulkan extensions – advanced, not required), then composite.

Plan ZIG so that the relativistic pass is a pluggable module behind an interface; then exchanging the implementation later costs nothing.

4.4 VERDICT

Winner: ****Google Filament, used directly, with SceneView as an optional Compose/ergonomics layer.**** Reasons, in the project's priority order:

1. Scientific accuracy is served by custom materials (custom surface shading + custom skybox) rather than by engine features you'd have to build anyway.
2. Visual quality ceiling is competitively high and, crucially, *known-correct* PBR with physical light units – mandatory when you claim physical accuracy.
3. Licence is Apache-2.0 across the engine, so proprietary/commercial shipping is unproblematic; no share-alike, no royalties.
4. Android coverage is the best available (GLS3 default, Vulkan capable).
5. KTX2/Basis + Draco + meshopt are first-class, which is exactly the packaging strategy needed for gigabyte asset sets.
6. Maintenance: pushed within the last 24 hours of writing this report.

4.5 RE-EVALUATION UNDER THE SINGLE-OBJECT SCOPE

The scope change was re-tested against the engine decision, because one object per screen removes the usual arguments for a general-purpose engine.

Does the case for raw OpenGL ES or Vulkan improve? Partially - the scene graph, culling, multi-object sorting and LOD management that a general engine provides are now nearly irrelevant, so a custom renderer would be a much smaller project than before. It is still the wrong choice, for three reasons.

1. The hard parts of this project are NOT scene management; they are the material effects (atmospheric scattering, ring transmission, relativistic lensing), the texture pipeline (KTX2/ASTC, mip correctness, tiling) and the HDR pipeline (bloom, tone mapping, TAA). A custom renderer saves you none of

that work, and Filament hands you all of it solved and tested on Android GPUs.

2. Filament's material system is genuinely open at the level that matters here: customSurfaceShading, a replaceable skybox material with eyeDirection, and sampler2d/sampler2dArray/samplerCubemap parameter types. Every bespoke shader this report calls for can be expressed as a Filament material compiled offline by matc.

3. Driver variance on Mali, Adreno and Xclipse is where custom engines die.

Filament's release history is a long record of exactly those fixes.

What DOES change is how the engine is used: with one object resident, ZIG should push Filament harder than a normal app would - full-resolution post-processing, larger RenderTargets, higher sample counts, bespoke materials per object, and offline-baked LUTs - because there is no other object competing for the frame. The engine verdict is unchanged: Google Filament, with SceneView as an optional Compose and asset-loading layer.

5. MASTER RESOURCE TABLE

Notes on reading the table:

- "PD-US" = public domain in the US as a US Government work / USGS product (USGS Astropedia pages state "Access Constraints: public domain" or "CC0").
- "NASA-CONTENT" = NASA-produced content subject to NASA's media guidelines ("generally not subject to copyright in the US"; credit required; no logo; no endorsement implication; third-party elements excluded).
- Exact per-object detail, ranking and rationale are in Sections 6-19; exact licence links are in Section 29; full URL list is in Section 35.

OBJECT	RECOMMENDED SOURCE #1	TYPE	RES/FORMAT	SCI PROVENANCE	LICENCE	COMMERCIAL	APK REDIST	MODIFY
Sun	SDO (AIA+HMI) via JSOC + procedural photosphere shader	data+shader	4096x4096 frames; JPG/FITS	NASA SDO science data	CC0 waiver (JSOC) + NASA-CONTENT	YES (CC0)	YES	YES
Mercury	USGS MESSENGER MDIS Basemap BDR/LOI 166 m + MD3 color + USGS DEM 665 m	raster	4 GB / 772 MB / 506 MB TIF	MESSENGER MDIS (APL/ASU), cartography USGS	PD-US	YES	YES	YES
Venus	USGS Magellan SAR FMAP/C3-MDIR colorized + GTDR topo + Fresnel reflectivity	raster	96 MB-82 GB TIF/PNG	Magellan SAR/altimetry, USGS	PD-US	YES	YES	YES
Earth	NASA Visible Earth Blue Marble NG + Black Marble 2016 + GIBS VIIRS	raster	21600x10800; 86400x43200	MODIS/VIIRS NASA	PD-US/NASA-CONTENT	YES	YES	YES
Moon	USGS LROC WAC 100 m mosaic + NASA SVS CGI Moon Kit (LOLA)	raster	5.5 GB TIF; ldem 23040x115 20	LROC (ASU), LOLA (GSFC), SVS	PD-US (LROC/LOLA data), NASA-CONTENT (SVS kit)	YES	YES	YES
Mars	Caltech Murray Lab CTX Global Mosaic V01 (5 m/px) + USGS MOLA DEM 463 m + HiRISE DTMs	raster	4°x4° tiles TIF; 2 GB; HiRISE 1 m	CTX/MRO (MSSS/JPL), MOLA, HiRISE (UA)	"free use for all purposes" (Murray Lab) + PD-US + PD (HiRISE)	YES	YES	YES
Jupiter	Hubble OPAL annual global map (NASA-hosted page) OR JPL PIA07782 Cassini cylindrical map	raster	3600x1598 TIF; PIA TIFF 18.56 MB	HST WFC3 OPAL; Cassini NAC	MIXED: NASA-CONTENT vs STScI/AURA; PIA = NASA/JPL photo policy	CONDITIONAL	CONDITIONAL	YES
Saturn	Hubble OPAL annual global map OR JPL Cassini products; rings from PDS COUVIS_8001	raster + data	maps ~0.1°/px; 438 MB profiles	HST OPAL; Cassini UVIS/RSS	same as Jupiter + PD-US (PDS)	CONDITIONAL	CONDITIONAL	YES
Uranus	Hubble OPAL 2022-2024 maps + Voyager 2 frames + CIE-corrected colour (Irwin et al. 2024)	raster	~0.1°/px TIF	HST OPAL; Voyager 2 ISS	same as Jupiter	CONDITIONAL	CONDITIONAL	YES

OBJECT	RECOMMENDED SOURCE #1	TYPE	RES/FORMAT	SCI PROVENANCE	LICENCE	COMMERCIAL	APK REDIST	MODIFY
Neptune	Hubble OPAL maps + Voyager 2 frames + corrected colour (Irwin et al. 2024)	raster	~0.1°/px TIF	HST OPAL; Voyager 2 ISS	same as Jupiter	CONDITIONAL	CONDITIONAL	YES
Milky Way	NASA SVS "Deep Star Maps 2020" (16K EXR) + real catalogue point stars from PD catalogues	raster+catalog	16384x8192 EXR; binary catalogs	Tycho-2/Hipparcos/UCAC3/Gaia DR2 (per SVS credits)	NASA-CONTENT (+ ESA credit line for Gaia)	CONDITIONAL-SAFE	CONDITIONAL	YES
ISS	NASA VTAD `ISS_stationary.glb` (44,495,916 bytes; 247,547 tris; 28 materials; 26 PNGs)	model	GLB, glTF 2.0	NASA VTAD / JSC-derived	NASA-CONTENT	YES (editorial, credit)	YES w/ credit	YES
Black hole	Own implementation: Bruneton 2020 LUT beam tracing (BSD-3) + blackbody disk + Doppler/redshift; optional MIT/CC-BY reference code	code+math	512x512 RG32F + 64x32 RG32F LUTs	Luminet 1979; James et al. 2015; Bruneton 2020; EHT 2019/2022	BSD-3 + own code	YES	YES	YES

Per-object detail tables (with #1/#2/#3 and scores) follow in Sections 6-19.

5.2 INSPECTION ASSET TABLE (per object: deepest real data and how to hold it)

"Tile pyramid" means a locally stored, mip-like pyramid of ASTC tiles streamed from device storage (never the network); only visible tiles are resident, which is what makes 0.5 m/px Moon or 5 m/px Mars inspectable on a phone.

Object	Deepest real source	Pyramid target	On-disk (ASTC 6x6 + mips)	Resident at once	Notes
Sun	SDO AIA/HMI 4096x4096	4096 disk frames + 8K procedural base	0.03-0.25 GB	30 MB	procedure extends past the wall
Mercury	MDIS BDR/LOI 166 m/px	16384 base + optional 92160 full	0.02 GB / 2.5 GB full	60-400 MB	full-res global is feasible as an optional pack
Venus	Magellan C3-MDIR 4.6 km/px; 75 m stereo 17%	8192 base + local stereo patches	0.05 GB + 0.1-0.5 GB	40 MB	radar labelling is mandatory
Earth	Landsat 15 m/px (PD)	16384 base + curated Landsat tiles	0.6-1.4 GB curated	150-500 MB	the deepest urban detail in the app
Moon	LROC NAC 0.5 m/px; WAC 100 m/px global	32768 base pyramid + NAC hero patches	3.5 GB global + 0.5-1 GB patches	200-800 MB	hero sites: Apollo, Tycho, Aristarchus, Reiner Gamma
Mars	HiRISE 0.25-0.5 m/px sites; CTX 5 m/px	CTX pyramid + HiRISE patches	1-3 GB curated	200-800 MB	Jezero, Valles Marineris, Olympus flank, polar layers
Jupiter	OPAL 0.1 deg/px	4096 map + wind LUT	0.01 GB	20 MB	smallest data set, most zoom-limited object
Saturn	OPAL body + UVIS ring profiles	4096 body + 16384 radial ring	0.02 GB	30 MB	rings can carry 32768 radial detail via an array
Uranus	OPAL 0.1 deg/px	4096 map	0.01 GB	20 MB	two appearance epochs
Neptune	OPAL epochs + Voyager 2	4096 map + per-epoch spots	0.02 GB	25 MB	spots are date-labelled
Milky Way	SVS 4851 16K + USNO/Yale catalogue	8192 skybox + sprite VBO	0.05-0.15 GB	60 MB	zoom is angular
ISS	NASA VTAD glb + KTX2 re-encode	model + 3 LODs	0.15 GB	150 MB	the only true 3D model object
Black hole	none needed (analytic)	LUTs + disk textures	0.01 GB	15 MB	unlimited zoom

6. SUN

6.1 WHAT IS ACTUALLY AVAILABLE (and the honesty constraint)

There is no photograph of "the far side of the Sun at visible wavelengths" – the Sun only ever shows one hemisphere. Any "Sun texture map" you find online is either (a) a composite of EUV images from multiple viewpoints, (b) an artistic painting, or (c) a single image stretched onto a sphere. State this in-app.

- The only genuine full-sphere solar map I could find from a primary source: NASA SVS "Full Map of the Sun's Surface", ID 30362, released 2013-10-22, file `euvi\_aia304\_2012\_carrington.tif`, 4104x2304, 12.4 MB, credit line in the SVS item. This is a CARRINGTON map built from STEREO/EUVI + SDO/AIA 304 Å data – i.e. extreme-ultraviolet, FALSE COLOUR (304 Å $\approx$ 30.4 nm, not visible light). URL: <https://svs.gsfc.nasa.gov/30362>
- SDO's own data: the Joint Science Operations Center (JSOC) publishes a CC0 waiver for "data stored and distributed from JSOC data series names sdo.X, hmi.X and aia.X ... as well as any images or movies found on the JSOC web site unless explicitly declared otherwise", with the requested attribution "Courtesy of NASA/SDO and the AIA and HMI science teams". URL: http://jsoc.stanford.edu/JSOC_Data_Policy.html (verified) SDO rules page: <https://sdo.gsfc.nasa.gov/data/rules.php> This is the single cleanest licence in the whole ZIG asset set: CC0 + courtesy attribution. AIA images are 4096x4096; HMI continuum and magnetograms are 4096x4096.
- NASA SVS also curates "SDO 4k Content" (ID 12613 / gallery sdo4k-content) with 4096x4096 frame sequences and movies for rotation studies – useful as a reference for realistic rotation/granulation appearance and as a source for the "live Sun" mode.

6.2 WHY A TEXTURED SPHERE IS THE WRONG ANSWER FOR THE SUN

Physically, what you see at the limb and the disk is:

- photosphere: granulation (typical granule $\sim 1,000$ km $\approx 1.4''$, lifetime 8-20 min), supergranulation network ($\sim 30,000$ km cells), sunspots with umbra/penumbra structure, faculae and network bright points;
- limb darkening: the disk is bright at centre and falls off towards the limb;
- chromosphere/corona: visible at the limb as a thin red H α rim plus the optically thin corona (K-corona, electron scattering, $\sim 1e-6$ of photospheric brightness, extends millions of km, co-rotates, structured by the magnetic field);
- an HDR dynamic range problem: a real photospheric view is $\sim 1e5$ - $1e6$ cd/m 2 scale against a black sky.

A static texture cannot express any of this at close range, and at distance the correct answer is a physically scaled emissive sphere with limb darkening.

6.3 RECOMMENDED CONSTRUCTION

#1 BEST – "Procedural photosphere + real data + LUT atmosphere/corona"

- Spheroid: locally generated UV sphere, radius 695,700 km (IAU nominal solar radius), oriented with the Carrington frame (IAU WGCCRE solar rotation elements; 25.38 d sidereal Carrington period, differential rotation law).
- Photosphere shader (Filament `shadingModel : unlit` + custom emissive or a `customSurfaceShading` variant):
 - * 3-octave domain-warped Voronoi/noise granulation in a sun-fixed frame (prevent "boiling" by animating a 3rd time dimension, not by scrolling UVs);
 - * supergranular network mask (large-scale cellular noise) to modulate granule contrast, plus magnetic network bright points;
 - * sunspot model: dark umbra ($T \approx 4,000$ K) + filamentary penumbra ($T \approx 5,000$ - $5,500$ K), inserted from a sunspot table with real latitudes/longitudes/sizes/areas read from SDO/HMI active-region data (processed offline into a small JSON: for each Carrington rotation, spot centres, umbra/penumbra radii, tilt);
 - * limb darkening: use the standard 1D coefficients. Use a published limb-darkening law (e.g. the quadratic/LIMB-DARK form of Neckel & Labs or Claret's tables) rather than eyeballing;

- * emission in physical units: set Filament directional light to the solar constant at 1 AU (1,361 W/m² luminance equivalent) or simply calibrate so the disk renders at the HDR target's top end and let ACES/AgX tone map.
 - Corona: 2-3 composited layers: (a) an optically thin shell/shader with an analytic K-corona radial falloff (intensity $\propto r^{-2}..r^{-2.5}$ outside $\sim 1.2 R_{\odot}$, with streamer/helmet structure driven by a noise field modulated by a real HMI synoptic magnetogram exported as a small 1024x512 texture), (b) a screen-space lens flare (Filament has screen-space lens flares) for the cinematic look, (c) optional HDR bloom from Filament.
 - If you want a literal "look, that's the real Sun" mode: overlay the live or archival SDO/AIA disk image as an albedo/emissive layer in a separate mode, with the wavelength honestly labelled ("SDO/AIA 304 Å – extreme ultraviolet, false colour"). AIA 171/193/304 and HMI continuum all available.
 - Corona/atmosphere scientific anchor: NASA's own descriptions of the corona's temperature and structure; use published SDO/Parker Solar Probe science pages as the textual source for in-app explanations.
- #2 BACKUP – NASA SVS "Full Map of the Sun's Surface" (Carrington EUV map, 4104x2304)
- Use ONLY in an explicitly EUV/304 Å-labelled mode. It gives full 360° coverage, which no visible-light dataset can.

- #3 BACKUP – Solar System Scope 8K sun texture (CC BY 4.0)
- <https://www.solarsystemscoope.com/textures/> – the standard CC BY 4.0 set (2k/8k). It is an artistic composite, not a scientific product; use only as a distant-view fallback and credit it. Verified licence via Wikimedia Commons category and the Qt 3D attribution page (both state CC BY 4.0 for solarsystemscoope textures).

6.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
SDO/JSOC data + procedural shader (own)	10	9	9	n/a	10	9	8	10 (CC0)	9.4
SVS 30362 Carrington EUV map	8	7	9	7	9	9	9	9	8.4
Solar System Scope 8k	4	8	9	8	3	9	9	9 (CC BY 4.0)	7.3

#1 wins because it is the only option that is simultaneously licence-clean (CC0), physically parameterised (limb darkening, granulation scale, spot temperatures), close-up capable, and free of false-colour claims.

6.5 ANDROID COST

Procedural solar surface + corona: 1 sphere (low poly: 64x32 is enough because the detail is in the fragment shader; use adaptive tessellation only if you do displacement), 3-5 noise textures (256-512², R8) + 1 magnetogram (1024x512, R8) + LUTs $\approx$ < 8 MB VRAM. Full-screen corona shell costs $\sim$ 1 extra pass. The expensive part is fill rate at close range; use a screen-space radial analytic corona rather than a large volumetric shell.

6.7 INSPECTION SPEC (single-object viewer)

- DEPTH: SDO AIA and HMI frames at 4096x4096, i.e. about 340 km/px across the 1.39 million km disk. That is the deepest real solar surface data that exists.
- MAX USEFUL ZOOM: about 2x the SDO disk scale (roughly 170 km/px) before the surface becomes procedural. At 340 km/px a typical granule (about 1000 km) is three pixels; the shader will make it look better than three pixels, and the HUD must say "below published resolution - procedural granulation".
- THE FAR-SIDE PROBLEM, STATED HONESTLY: no instrument has ever imaged the far side at visible or EUV wavelengths, so a full-sphere Sun texture is necessarily either (a) a Carrington composite (SVS 30362, EUV, false colour, labelled), or (b) a single-epoch disk applied to the whole sphere (an assumption, labelled), or (c) filled with JSOC helioseismic far-side maps at their real low resolution. Use (a) in "EUV mode", (b) by default with a note,

- and offer (c) as a "far-side context" overlay at honest low weight.
- TOGGLES: photosphere (procedural + HMI continuum), AIA 171 / 193 / 304 with wavelength labels and false-colour warning, HMI magnetogram (white/black polarity), a sunspot table from real active-region data, and a corona intensity control.
- SUN ANGLE: irrelevant for the Sun itself (it emits); keep the corona and the limb glow locked to the camera, not to a light.

7. MERCURY

7.1 AVAILABLE DATA (all USGS Astropedia, "Access Constraints: public domain")

Product	Resolution	Size	Direct file
Mercury MESSENGER MDIS Basemap BDR 166 m (RECOMMENDED albedo, 750 nm, Kasseleinin-Shkuratov photometric model)	166 m/px, 256 px/deg, 8-bit, 92160x46080	4 GB	https://planetarymaps.usgs.gov/mosaic/Mercury_MESSENGER_MDIS_Basemap_BDR_Mosaic_Global_166m.tif
Mercury MESSENGER MDIS Basemap LOI 166 m (low-incidence-angle variant)	166 m/px, 92160x46080	4 GB	https://planetarymaps.usgs.gov/mosaic/Mercury_MESSENGER_MDIS_Basemap_LOI_Mosaic_Global_166m.tif
Mercury MESSENGER MDIS Global Mosaic 250 m (mono fallback)	250 m/px	2 GB	https://planetarymaps.usgs.gov/mosaic/Mercury_MESSENGER_mosaic_global_250m_2013.tif
Mercury MESSENGER MDIS Global Color Mosaic 665 m	665 m/px	772 MB	https://planetarymaps.usgs.gov/mosaic/Mercury_MESSENGER_ClrMosaic_global_665m_v3.tif
Mercury MESSENGER MDIS Basemap MD3 Color Global Mosaic 665 m (end-of-mission 3-colour)	665 m/px	760 MB	https://planetarymaps.usgs.gov/mosaic//Mercury_MESSENGER_MDIS_Basemap_MD3Color_Mosaic_Global_665m.tif
Mercury MESSENGER Basemap Enhanced Color Global Mosaic 665 m (FALSE COLOUR)	665 m/px	759 MB	https://planetarymaps.usgs.gov/mosaic/Mercury_MESSENGER_MDIS_Basemap_EnhancedColor_Mosaic_Global_665m.tif
Mercury MESSENGER Global DEM 665 m (v2)	665 m/px, 16-bit	506 MB	https://planetarymaps.usgs.gov/mosaic/Mercury_Messenger_USGS_DEM_Global_665m_v2.tif

Page examples (verified): https://astrogeology.usgs.gov/search/map/mercury_messenger_mdiss_global_mosaic_250m
https://astrogeology.usgs.gov/search/map/mercury_messenger_mdiss_global_color_mosaic_665m
https://astrogeology.usgs.gov/search/map/mercury_messenger_global_dem_665m

Note: the MD3/Enhanced/Color mosaics place the 1000 nm, 750 nm and 430 nm filters in R,G,B. That is NOT what a human eye sees; the mosaic pages say so explicitly ("This is not what Mercury would look like to the human eye"). Do not present the enhanced-colour mosaic as literal colour. Product pages to cite: BDR 166 m

https://astrogeology.usgs.gov/search/map/mercury_messenger_mdiss_global_basemap_bdr_166m
 and LOI 166 m

https://astrogeology.usgs.gov/search/map/mercury_messenger_mdiss_basemap_loi_global_mosaic_166m Both are 256 px/deg (166.31 m/px) simple-cylindrical, planetocentric, positive-east, projected onto the MESSENGER global DEM, with "Access Constraints: None / Use Constraints: Please cite authors" (originators NASA, JHUAPL, ASU, Carnegie Science; publisher USGS Astrogeology). Data files verified live on 2026-09-14 (HTTP 200 after the standard S3 redirect).

7.2 RECOMMENDED CONSTRUCTION – TYPE C + D (sphere + displacement + material layers)

- Geometry: locally generated UV sphere, radius 2,439.7 km (IAU nominal).
- Base colour: 166 m BDR (or LOI) mosaic converted to sRGB with a calibrated photometric conversion; add subtle colour by mixing the MD3 colour mosaic at low saturation (research-backed: Mercury's visible colour variation is subtle and dominated by maturity/space weathering; the 430/1000 ratio map is the right way to inject that subtly).
- Displacement/normal: USGS 665 m DEM → derive both a displacement map for close-range geometry and a 2K-4K normal map. Mercury's relief is ~7 km range (Caloris basin etc.), so displacement is meaningful at limb views.
- Material: albedo ~0.1 (Mercury is dark), roughness high (0.8-0.95), metallic 0, no specular highlight; add a subtle regolith "backscatter" via a custom surface-shading tweak if you want the classic moon-like opposition surge.
- Atmosphere: essentially none (exosphere, ~1e-15 bar). Do NOT add a Rayleigh atmosphere. A very faint sodium tail/exosphere haze is optional and only justifiable with a citation; default = none.

- Lighting: harsh, high-contrast sun at 0.39 AU (1.9x Earth insolation) – set the directional light intensity accordingly and let tone mapping handle it.

7.3 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
USGS MESSENGER BDR 166 m + DEM 665 m (+MD3 tint)	10	9	9	10	10	9	9	10	9.5
USGS Color Mosaic 665 m only	9	7	7	7	10	9	9	10	8.2
Solar System Scope 8k Mercury (CC BY 4.0)	4	8	9	8	2	9	9	9	7.0
#1 wins: it is the actual MESSENGER cartographic product with an elevation model, it is public domain, and at 166 m/px it is 1.5x finer than the 250 m mosaic and 4x finer than the 665 m colour mosaics.									

7.4 PRACTICAL PROCESSING

250 m mosaic = 23054 px wide (x 11527). Downsample with a proper Lanczos filter to 8192x4096 (0.05 px/deg) and encode ASTC 6x6 (~0.5-0.75 bytes/px → ~24-37 MB VRAM with mips). The original is 8-bit; keep the DEM as 16-bit source, convert to R16F or a normalised R8 + separate normal map for runtime.

7.5 INSPECTION SPEC (single-object viewer)

- DEPTH: the BDR and LOI mosaics are 92,160 x 46,080 px (4.25 Gpx, 256 px/deg, 8-bit). That is the deepest global coverage of any rocky body in this project: ASTC 6x6 with mips is about 2.5 GB on disk and streams as a tile pyramid.
- TIERED DELIVERY: ship a 16,384-wide base (about 937 m/px) as the default Mercury pack (tens of MB), and offer the full 166 m/px pyramid as an optional 2.5 GB download for users who want to zoom to the wall.
- WHAT THE USER SEES: at 166 m/px a 1440-px screen at 1:1 spans about 240 km. The smallest feature that can be trusted is roughly 3-4 texels, about 500 m - so yes, individual fresh craters, scarps, hollows and ejecta rays.
- RELIEF CEILING (state it): albedo is 166 m/px but the global DEM is 665 m/px, so relief detail is capped at 665 m. Where MDIS stereo coverage supports a finer local DEM, use it and credit it; do not invent relief from albedo.
- TOGGLES: BDR (74 deg illumination, morphology) versus LOI (45 deg, low-incidence); MD3 true-ish colour versus enhanced colour (explicitly labelled false colour); DEM/normal overlay; sun-angle slider.
- CORRECTION TO REVISION 2: the "23040x11520" figure previously printed in the Moon row was the LOLA ldem_64 height field, not any Mercury product. Mercury BDR is 92,160 x 46,080 as stated above.

8. VENUS

8.1 AVAILABLE DATA (USGS, public domain)

Product	Meaning	Size	File
Venus Magellan SAR FMAP Stereo Look Global Mosaic 75 m	radar, 17% coverage only	82 GB	https://asc-pds-services.s3.us-west-2.amazonaws.com/mosaic/Venus_Magellan_StereoLook_mosaic_global_75m.tif
Venus Magellan Global C3-MDIR Synthetic Color Mosaic 4641 m	radar, colourised to "simulate the surface colour"	96 MB	https://planetarymaps.usgs.gov/mosaic/Venus_Magellan_C3-MDIR_Colorized_Global_Mosaic_4641m.tif
Venus Magellan Global Topography 4641 m (GTDR v2)	altimetry, -2,951..11,687 m	65 MB	https://planetarymaps.usgs.gov/mosaic/Venus_Magellan_Topography_Global_4641m_v02.tif
Venus Magellan Global Fresnel Reflectivity 4641 m	Hagfors-model radar reflection coefficient	33 MB	https://planetarymaps.usgs.gov/mosaic/Venus_Magellan_FresnelReflectivity_Global_4641m.tif
Venus Magellan Global Meter-Scale Slope 4641 m	radar-derived roughness	33 MB	https://planetarymaps.usgs.gov/mosaic/Venus_Magellan_MeterScaleSlope_Global_4641m.tif

Pages verified: https://astrogeology.usgs.gov/search/map/venus_magellan_global_topography_4641m

https://astrogeology.usgs.gov/search/map/venus_magellan_global_c3_mdir_synthetic_color_mosaic_4641m

https://astrogeology.usgs.gov/search/map/venus_magellan_global_fresnel_reflectivity_4641m

I verified the GTDR file is a real 8192x4096 16-bit TIFF served from AWS S3 (HTTP 206 + TIFF header read).

8.2 THE HONEST STATEMENT ABOUT VENUS

- At visible wavelengths Venus is a nearly featureless, opaque sulphuric-acid cloud deck (the surface is invisible; the cloud tops are what you see), with the classic UV "Y"-shaped markings only in the ultraviolet.
- Therefore the correct ZIG presentation has TWO modes:
 - (a) VISIBLE MODE (scientifically correct look): a smooth, high-albedo (0.7-0.8) cloud sphere with subtle banding and the meridional cloud rotation (super-rotation, ~4 days atmosphere vs 243-day surface), wrapped in a thick CO₂/SO₂ atmosphere rendered with a scattering model (Mie-dominated, high single-scattering albedo, Rayleigh + absorption). There is NO free, high-resolution, scientifically sourced map of Venus's cloud tops; a smooth cloud layer plus controlled procedural banding is more honest than pasting a "Venus texture".
 - (b) RADAR/GLOBE MODE (explicitly labelled "Magellan SAR radar imagery"): surface geometry + albedo from the C3-MDIR/FMAP mosaics; topography from GTDR; roughness from the slope map (drive a sheen/gloss variation); reflectivity from the Fresnel map (Venus's surface has radar-bright highland areas that are genuinely a property of the surface at radio wavelengths).
- The "scientifically accurate colour" of Venus's clouds is a pale yellow-cream; published false-colour imagery (purple/blue composites) must be labelled as false colour if used at all.

8.3 RECOMMENDED CONSTRUCTION – TYPE D + E (sphere + multi-layer material + shader)

- Sphere radius 6,051.8 km.
- Layer 1 (surface, optional mode): radar mosaic base + DEM displacement + slope-driven roughness.
- Layer 2 (cloud deck): opaque cloud top at ~65-70 km altitude, albedo ~0.75, roughness ~0.95, with: super-rotating zonal banding (period ≈ 4-5 days), low-contrast polar "dipole" haze structure (documented in Venus Express/Akatsuki imagery), plus a slow morphing noise to avoid a static look.
- Layer 3 (atmosphere): the Bruneton scattering model (Section 21) with Venus parameters: CO₂-dominated Rayleigh coefficient (larger than Earth's per mole because of the higher density), Mie from sulphuric-acid droplets with asymmetry $g \approx 0.7-0.75$, plus SO₂/UV absorber for the upper haze. Result: the correct bright, hazy limb and the correct lack of visible surface.
- Night side: no artificial city-light style content; Venus's night side is dominated by infrared thermal emission (not visible), so keep it dark.

8.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
Magellan C3-MDIR + GTDR + Fresnel + slope, procedural cloud/atmosphere	10	9	8	8	10	9	8	10	9.3
Solar System Scope 8K "Venus surface" + "Venus atmosphere" (CC BY 4.0)	5	8	9	8	3	9	9	9	7.2
Any "Venus true colour" JPEG found online	2	7	8	7	1	9	9	UNCERTAIN	UNUSABLE

#1 wins because it is the only combination that is simultaneously real data, correctly labelled (radar), and capable of a physically-plausible cloud deck.

8.5 INSPECTION SPEC (single-object viewer)

- DEPTH: radar 4,641 m/px globally (C3-MDIR, colourised from SAR) and 75 m/px stereo products covering about 17% of the surface. Topography 4,641 m/px,

- Fresnel reflectivity 4,641 m/px, meter-scale slope 4,641 m/px.
- MAX USEFUL ZOOM: about 4.6 km/px globally and about 75 m/px inside the 17% stereo footprint - then stop. There is no finer Venus surface data, and there is no visible-light surface data at all.
- THE CLOUD MODE IS THE DEFAULT, and it is deliberately bland because Venus is bland: an opaque sulphuric-acid deck, albedo about 0.75, super-rotating, UV-marked structures only in the ultraviolet. Procedural banding must be subtle and labelled "procedural appearance".
- TOGGLES: cloud deck (default, labelled) | radar surface (Magellan SAR, labelled radar) | topography (hypsometric tint from GTDR) | Fresnel reflectivity | slope/roughness. The "peel" transition between cloud and radar is the single best didactic feature in the whole app: it shows why Venus hides its surface.
- RELIEF: 16-bit GTDR drives both displacement and a normal map; the Fresnel map legitimately drives a specular/roughness variation in radar mode (it is a real reflectance measurement, not decoration).

9. EARTH

9.1 THE BEST FREE DATA (all NASA, public domain / NASA content)

Layer	Dataset	Source page	Notes
Day surface	Blue Marble: Next Generation (BMNG), 12 monthly cloud-free composites, up to 21600x10800 (and community re-tilings to 86400x43200)	https://visibleearth.nasa.gov/collection/1484/blue-marble and https://earthobservatory.nasa.gov/features/BlueMarble	PD-US; "NASA image courtesy Reto Stöckli and Robert Simmon"
Night surface	Black Marble / Earth at Night 2016, up to 86400x43200	https://science.nasa.gov/earth/earth-observatory/earth-at-night/ and SVS 30919 (86400x43200 file, 868.6 MB)	PD-US; VIIRS Suomi-NPP; credit NASA Earth Observatory/Joshua Stevens
Live/daily	NASA GIBS WMTS: VIIRS_SNPP_CorrectedReflectance_TrueColor, daily since 2016	https://nasa-gibs.github.io/gibs-api-docs/	Open policy + acknowledgement text; this is exactly what NASA Eyes uses
Topography/bathymetry	ETOPO 2022 (NOAA, 60"/30"/15") and NASADEM/SRTM (30 m, land, 56°S-60°N)	https://www.ncei.noaa.gov/products/etopo-global-relief-model ; https://www.earthdata.nasa.gov/	US Government PD; SRTM has latitude coverage gaps, ETOPO is global incl. oceans
Clouds	No official NASA "CGI cloud map". Options: (a) derive a cloud alpha by differencing GIBS VIIRS daily true colour against a BMNG surface composite during preprocessing (fully NASA-data-derived), (b) Solar System Scope 8K clouds (CC BY 4.0), (c) procedural	—	(a) is the scientifically defensible route and is a preprocessing task, not a runtime task
Atmospheric constitution	Earth atmosphere parameters for a scattering model (Rayleigh + Mie + ozone)	Bruneton reference implementation + published coefficients	BSD-3 reference code (Section 21)

9.2 RECOMMENDED CONSTRUCTION – TYPE D + E (the most complex of the planets)

- Sphere radius 6,371.0 km (mean) / WGS84 ellipsoid if you want geodesy-grade accuracy; at planetary scale the 1/298.257 flattening is visible at the limb if you claim accuracy, so use the ellipsoid (Filament handles the mesh fine).
- Land/ocean material separation: derive a water mask from the BMNG blue/land classification (BMNG already has "with/without topography and bathymetry" variants); use it to drive: roughness (ocean 0.02-0.1, land 0.8-1.0), metallic 0, and a Fresnel-driven specular sun glint. This is the single biggest realism win on Earth and EOSS also has a specular channel for it.
- Night lights: Black Marble as an emissive layer, modulated by a physically plausible "night" mask ($\text{dot}(\mathbf{N}, \mathbf{sun}) < 0$ with a soft terminator), and dimmed where clouds are present (an approximation you should document).
- Clouds: separate translucent shell at ~2-3 km geometric offset (or a shell at

- cloud-top altitude ~8 km with alpha), double-sided, lit by the sun with a fraction of transmittance. Optional second high thin cirrus layer.
- Atmosphere: Bruneton-style single+multiple scattering LUT with correct ozone absorption (this is what produces the blue limb and the orange sunset band); Earth's parameters (Rayleigh scale height ~8 km, Mie ~1.2 km, $g \approx 0.76$, ozone layer ~10-30 km) are published and are the reference case for the reference implementation.
- Special effects from real geometry: specular sun glint, the correct 23.44 deg axial tilt, and a real rotation period. The sun direction is the user's control (Section 23.4) rather than an ephemeris, because there is no orbit simulation in ZIG.
- Optional premium feature that no free app has: SEASONAL MONTHS. Because BMNG ships 12 monthly composites, the user can cross-fade January to December and watch the snowline and vegetation change. Presented as a labelled month selector ("Blue Marble, February 2004"), not as "today", since these are fixed monthly climatological composites rather than live imagery.

9.3 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
BMNG + Black Marble + derived GIBS clouds + scattering LUT	10	10	10	10	10	8	7	10	9.5
Solar System Scope 8K day/night/clouds/specular/normal set (CC BY 4.0)	5	9	9	8	3	9	9	9	7.6
EOSS-style GIBS WMTS streaming	9	9	9	8	10	6 (needs network)	7	10	rejected: violates offline requirement

#1 wins: maximum resolution, full NASA provenance, PD licence, seasonal capability, and the only set with a credible night-lights layer.

9.4 MEMORY REALITY CHECK FOR EARTH

8K (8192x4096) ASTC 6x6 day map $\approx 81924096(128/36)/8$ bits ≈ 22.3 MB (plus ~33% mips). Night map 4K ASTC 6x6 ≈ 5.6 MB. Clouds 4K ASTC ≈ 5.6 MB. Normal/roughness masks can be 2K (≈ 1.4 MB each). Ocean mask 2K R8 ≈ 1 MB. Total Earth ≈ 45 -60 MB VRAM – comfortable. Going to 16K day map ≈ 89 MB, still feasible on a 12 GB phone but pointless: at full-sphere viewing the screen shows ~2K pixels of the disk, so 8K is already beyond Nyquist for a phone screen; only surface-level camera modes justify more, and then you should stream tiles from disk via virtual texturing rather than hold everything resident.

9.5 INSPECTION SPEC (single-object viewer)

- DEPTH, THREE TIERS: (1) Blue Marble Next Generation 21,600 x 10,800 (about 1.85 km/px) as the global base, in 12 labelled monthly versions; (2) Black Marble 2016 night lights up to 86,400 x 43,200 (about 460 m/px) for the night appearance; (3) LANDSAT Collection 2 as the hero layer - 30 m/px multispectral and 15 m/px panchromatic, land only, since 2013, USGS public domain with no restrictions ("There are no restrictions on the use or redistribution of Landsat data... Permission is not required for use").
- THE LANDSAT UPGRADE IS THE HEADLINE FEATURE OF THIS OBJECT: it is roughly 60-120x finer than the Blue Marble base and it is unambiguously free for commercial use. Curate 64-128 tiles over the places users actually want (cities, deltas, volcanoes, glaciers, reefs, deserts, megaprojects) at 15-30 m/px; on a 1440-px screen at 1:1 that is about 20-45 km across, which means individual fields, streets and city blocks.
- OCEAN CEILING (state it): there is no optical imagery of the sea floor. Below the waterline the honest limit is bathymetry - ETOP0 2022 at 15 arc-seconds (about 450 m) - plus sea-surface colour. A "bathymetry" toggle with a labelled hypsometric ramp is the correct treatment; a fake textured sea floor is not.
- TOGGLES: month selector (12 BMNG composites), day/night crossfade with real

night lights, cloud layer on/off with opacity, ocean specular sun glint, bathymetry overlay, Landsat hero tiles, atmosphere on/off.

- SUN ANGLE: the sun-direction control (23.4) drives the terminator, the sunset band and the size of the sun glint - three effects that make this object the best showcase of the lighting engine.

10. MOON

10.1 THE BEST FREE LUNAR ASSETS (this is the strongest object in the project)

Asset	Content	Resolution/Size	Licence	URL
USGS "Moon LRO LROC WAC Global Morphology Mosaic 100m"	LROC WAC grayscale morphology, 15,000+ images, orthorectified to GLD100/LOLA	5.5 GB TIF; 23040x11520 (303.2 px/deg)	PD-US	https://planetarymaps.usgs.gov/mosaic/Lunar_LRO_LROC-WAC_Mosaic_global_100m_June2013.tif ; https://astrogeology.usgs.gov/search/map/moon_lro_lroc_wac_global_morphology_mosaic_100m https://svs.gsfc.nasa.gov/4720
NASA SVS "CGI Moon Kit" (ID 4720)	LROC-based colour map + LOLA displacement, explicitly made for 3D rendering	colour up to 16384x8192 TIFF (909 MB) / EXR 942.8 MB; displacement 23040x11520 (506 MB uint, 1012 MB float)	NASA-CONTENT (NASA SVS/GSFC; credit Ernie Wright, LRO/LROC/LOLA)	
USGS "Moon LROC WAC DTM GLD100 118m"	stereo DTM 79°S-79°N, 98.2% coverage	10 GB TIF	PD-US	https://planetarymaps.usgs.gov/mosaic/Lunar_LRO_WAC_GLD100_DTM_79S79N_100m_v1.1.tif
LROC PDS node + USGS mirrors (RECOMMENDED route for the same products)	WAC Hapke 7-band colour mosaic (6.5 GB); WAC empirical 7-band (6.4 GB); WAC 643 nm (18.6 GB); GLD100 topography (5.3 GB); NAC pole mosaics	up to 189 GB	LROC/LRO data via PDS; credit NASA/GSFC/ASU	https://pds.lroc.im-ldi.com/ and https://astrogeology.usgs.gov/
LOLA PDS gridded data	the authoritative altimetry the SVS kit is derived from	PDS IMG	PD-US	https://pds-geosciences.wustl.edu/missions/lro/lola.htm
LROC NAC (optional close-up layer)	0.5 m/px panchromatic swaths; 1 m/px or better over most of the Moon; DTMs at landing sites	PDS, per-image	PD-US (credit NASA/GSFC/ASU)	https://www.lroc.asu.edu/ and PDS volumes

10.1b LICENCE WARNING ON THE LROC CURATED DOWNLOADS PAGE (CORRECTION)

The LROC team's own Terms of Use page states, verbatim:

- "Data products available through the LROC Data Node of the PDS or through NASA PDS interfaces are in the public domain."
- "Images available on the LROC website, including but not limited to Featured Images, Featured Sites, curated products, downloads and non-PDS interfaces listed above, are copyrighted."
- "All commercial use of LROC images not downloaded from the LROC PDS archive require prior permission."

Source: <https://lroc.im-ldi.com/about/terms> CONSEQUENCE FOR ZIG: the curated products on <https://lroc.im-ldi.com/images/downloads> (WAC Hapke 7-band 6.5 GB, WAC empirical 643 nm 18.6 GB, NAC pole mosaics 189 GB) are distributed from a NON-PDS interface and are therefore copyrighted for a commercial product. The same or equivalent data are public domain when obtained from the PDS LROC node (<https://pds.lroc.im-ldi.com/>, <https://data.lroc.im-ldi.com/lroc/search>) or from USGS Astropedia. Use lroc.im-ldi.com only as a product index and as the source of the citation list. Recommended citation for PDS-delivered LROC data: Robinson, M.S. et al. (2010), Space Science Reviews 150, 81-124.

The NASA SVS "CGI Moon Kit" (Section 10.1) is unaffected: it is a NASA SVS/GSFC product, not an LROC website image.

10.2 WHY THIS IS THE BEST COMBINATION

- The LROC WAC morphology mosaic + the SVS CGI Moon Kit's LOLA displacement map are, together, exactly the "colour + displacement" pair a PBR renderer needs, and both are direct derivatives of LRO instruments. This is better than any

"moon model" on any 3D marketplace, and it is public domain.

- The SVS kit's colour map is a Hapke-normalised, white-balanced, human-vision-matched composite of three LROC WAC bands (643 nm→R, 566 nm→G, 415 nm→B) – i.e. it corrects for the photometric function, which is exactly what a CGI texture needs (a raw image would bake in the illumination).
- The displacement map is available in float kilometres and 16-bit half-metres relative to radius 1737.4 km (or +20,000 half-metres offset for unsigned) – ideal for building both displacement and normal maps without re-deriving units.

10.3 RECOMMENDED CONSTRUCTION – TYPE C (sphere + displacement) + D

- Sphere radius 1,737.4 km, 512x256 to 1024x512 tessellation for displacement; ship a separate lower-poly sphere with a normal map for distant views.
- Base colour: SVS colour map downsampled to 8192x4096 (already high quality); add a subtle normal map built from the 64 px/deg LOLA height field so micro-relief (crater rims, mare wrinkle ridges) reads at grazing sun angles – this matters enormously for lunar realism because the Moon has no atmosphere to soften anything.
- Roughness/metallic: high roughness (0.85-1.0), metallic 0. Lunar regolith shows an opposition surge (shadow-hiding), which in practice means "the full moon is brighter than Lambertian"; a simple fix in a custom surface-shading function is a backscatter lobe (or use a fitted Hapke/Hapke-Holm-Warren-style function with published parameters). Mention it as an approximation and cite Hapke.
- No atmosphere, no clouds. Any "lunar haze" is wrong.
- Night side: none (except earthshine – Section 23).
- Optional premium layer: for a small number of "hero" landing sites (Apollo 11/15/17, Chang'e, Luna), add 1-5 m/px LROC NAC orthoimagery + a NAC-derived DTM as a high-resolution patch. This is a "pixel-peeping" feature no consumer app has, and the data are free.

10.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
USGS LROC WAC 100 m + SVS CGI Moon Kit (LOLA)	10	10	10	10	10	9	8	10	9.7
LROC Hapke 7-band + GLD100 DTM (via the PDS LROC node, not the LROC website)	10	9	10	9	10	7	8	10 (PDS route)	9.3
Solar System Scope 8K moon (CC BY 4.0)	4	8	9	8	2	9	9	9	7.0

#1 wins on every axis: it is purpose-built for CGI, it is the finest global colour product, it carries a real laser-altimetry displacement, and its licence is as clean as it gets.

10.5 INSPECTION SPEC (single-object viewer)

- DEPTH, THREE TIERS: (1) LROC WAC global morphology mosaic at 100 m/px, which is 109,152 x 54,576 px (about 5.96 Gpx; roughly 5.96 GB as 8-bit imagery and about 3.5 GB as an ASTC 6x6 pyramid with mips) - the deepest global mosaic of any body in the project; (2) LOLA ldem_64 height field at 474 m/px for global relief; (3) LROC NAC at 0.5 m/px for hero sites - at 0.5 m/px a 5 x 5 km patch is 10,000 x 10,000 px, about 44 MB as ASTC 6x6, so 20 curated patches fit in under 1 GB.
- HERO SITE LIST (all public domain via PDS or USGS): Apollo 11, 12, 14, 15, 16, 17 landing sites (descent stages and astronaut tracks are visible), Tycho central peak, Aristarchus plateau, Reiner Gamma swirls, Hadley Rille, South Pole-Aitken basin floor, Mare Tranquillitatis wrinkle ridges, and the Chang'e and Luna sites.
- CORRECTION TO REVISION 2: the previous Moon row printed "5.5 GB TIF; 23040x11520" as if both described the albedo mosaic. They do not. The WAC morphology mosaic at 100 m/px is 109,152 x 54,576 px; 23040 x 11520 (474

- m/px) is the LOLA ldem_64 HEIGHT field. The two must not be confused when budgeting memory.
- TOGGLES: WAC morphology (default, photometrically normalised) | NASA SVS CGI Moon Kit colour (Hapke-normalised, human-vision matched) | LROC Hapke 7-band from the PDS node (mineralogy, labelled enhanced) | LOLA height/normal overlay | NAC hero patches. Sun-angle slider is ESSENTIAL here: sweeping the terminator across a 0.5 m/px NAC patch is the single most convincing demonstration of real relief in the entire app.
 - RELIEF CEILING: 474 m/px global (ldem_64); NAC DTMs at roughly 1-2 m exist for selected sites and should be used where shipped.

11. MARS

11.1 THE BEST FREE MARS ASSETS

Asset	Content	Resolution/Size	Licence	URL
Caltech Murray Lab Global CTX Mosaic V01	5.0 m/px global mosaic, 86,571 CTX images, 2006-2020, 99.5% coverage 88°S-88°N, 5.7 terapixels	4°x4° tiles; full mosaic 5.6 TB compressed; streaming endpoint available	"The Murray Lab/Caltech grants free use of the V01 version of the mosaic for all purposes. Data credit should be to: NASA/JPL/MSSS/The Murray Lab"	https://murray-lab.caltech.edu/CTX/ ; tiles https://murray-lab.caltech.edu/CTX/V01/tiles/
USGS Mars MGS MOLA DEM 463 m	laser altimetry DEM, 128 px/deg, ±3 m total uncertainty	2 GB TIF, 16-bit	"Access Constraints: CC0 (public domain)"	https://planetarymaps.usgs.gov/mosaic/Mars_MGS_MOLA_DEM_mosaic_global_463m.tif
USGS Mars Odyssey THEMIS-IR Day Global Mosaic 100 m v12	daytime IR temperature mosaic, useful for albedo/thermal context	21 GB TIF	PD-US, credit authors	https://planetarymaps.usgs.gov/mosaic/Mars_M0_THEMIS-IR-Day_mosaic_global_100m_v12.tif
USGS Mars Viking Global Color Mosaic 925 m / MDIM 2.1 grayscale 232 m	Viking-era colour and controlled basemap	764 MB / 4 GB	PD-US	https://planetarymaps.usgs.gov/mosaic/Mars_Viking_ClrMosaic_global_925m.tif
HiRISE DTMs and orthoimages	1 m/px stereo topography at dozens of sites (Jezero etc.)	per-site 1-2 GB	"All of the images produced by HiRISE ... are within the public domain: there are no restrictions on their usage by anyone in the public, including news or science organizations" (credit requested: NASA/JPL/University of Arizona)	https://www.uahirise.org/dtm/ ; https://www.uahirise.org/media/usage.php
Mars 2020 TRN HiRISE DTM mosaic (Jezero)	1 m DTM + orthoimage mosaic for rover landing	1.8 GB + on AWS S3	PD-US	https://planetarymaps.usgs.gov/mosaic/Mars2020_trn/HiRISE/
MRO CTX PDS archive (original EDRs)	for building your own higher-quality subsets	TB-scale	PD-US (PDS)	https://pds-imaging.jpl.nasa.gov/volumes/mro.html

11.2 WHY CTX IS A GENUINE BREAKTHROUGH FOR ZIG

The Murray Lab mosaic's own grant language ("free use ... for all purposes") is the most permissive redistribution statement in this entire report, and the data are the highest-resolution global planetary product that exists for any body other than Earth: 5 m/px globally, traceable to individual CTX images with seam maps (so you can document provenance), and archived in PDS (<https://doi.org/10.17189/m23v-ae12>). EOSS does not have this. No free solar system app has this.

11.3 RECOMMENDED CONSTRUCTION – TYPE C + D (displacement + multi-layer material)

- Sphere radius 3,389.5 km; use the IAU/MOLA reference shape if you animate limb-accurate views.
- Colour base: the CTX mosaic is panchromatic (grayscale). Build the visible colour as: CTX luminance at high frequency + Viking/THEMIS colour at low frequency (a classic and defensible "panchromatic + colour" recipe, standard in planetary cartography). Document that the colour is low-frequency

- (Viking colour is 925 m/px; THEMIS IR-day is a temperature proxy, not visible colour – use it for geology emphasis in a separate mode).
- Displacement/normal: MOLA 463 m DEM → displacement + high-frequency normal from CTX-derived slope (or, better, from the HiRISE DTMs in the hero regions).
- Regional hero detail: Jezero/HiRISE DTM at 1 m/px, plus a handful of other HiRISE DTMs, dynamically loaded when the camera is within X km. This is the "premium close-up" feature that sells the app.
- Dust/atmosphere: Mars's atmosphere is thin but very visible: CO₂ Rayleigh (weak), dust Mie (dominant) with strong forward scattering ($g \sim 0.7-0.8$), an optical depth that varies seasonally, and dust storms as a rare dynamic event. Use the same scattering LUT code with Mars parameters; add a global "dust opacity" scalar driven by real data (e.g. published MRO/MEDA or THEMIS dust indices) so the app can show a real 2018-class global storm.
- Polar caps: CO₂/H₂O seasonal caps need seasonal texture variation – derive from the MOLA/albedo data or add a seasonal mask from published cap extents.
- Night side: honest darkness (no lights).

11.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
CTX Murray Lab + MOLA + HiRISE DTMs + scattering	10	10	10	10	10	7 (data volume)	7	10 (explicit all-purposes)	9.6
USGS Viking colour 925 m + MOLA 463 m	8	6	8	6	10	9	9	10	8.0
Solar System Scope 8K Mars (CC BY 4.0)	4	8	9	8	2	9	9	9	7.0

#1 wins: it is the highest-resolution global planetary dataset that exists and it comes with an explicit all-purposes grant. It is also the only option that lets ZIG do legitimate metre-scale close-ups.

11.5 INSPECTION SPEC (single-object viewer)

- DEPTH, THREE TIERS: (1) global context at 232 m/px (USGS MDIM 2.1) or a CTX-derived 32,768-wide overview (about 650 m/px); (2) CTX 5 m/px over 99.5% of the surface between 88 deg S and 88 deg N - but understand the volume: a full 4 x 4 deg CTX block at 5 m/px is roughly 47,000 px across, about 2.2 Gpx and about 1 GB as ASTC 6x6, so curate REGIONS rather than the planet; (3) HiRISE at 0.25-0.5 m/px for site-scale inspection.
- RECOMMENDED HERO REGIONS (CTX, 5 m/px): Jezero crater and the delta, Gale crater and Aeolis Mons, Valles Marineris wallrock, the Olympus Mons basal scarp, Hellas basin rim, the south polar layered deposits, Medusae Fossae, Nili Fossae, Mawrth Vallis. Each region at 1 x 1 deg is about 11,850 px across (about 62 MB as ASTC 6x6), so twenty regions fit in roughly 1.2 GB.
- HiRISE PATCHES: at 0.5 m/px a 1 x 1 km patch is 2,000 x 2,000 px (about 2 MB); ship 50-100 patches (rover traverses, fresh impact craters, recurring slope lineae, dune fields) for a few hundred MB. This is the deepest zoom in the app outside the black hole: individual boulders and rover wheel tracks.
- RELIEF CEILING (state it): albedo can reach 0.5 m/px while the global DEM (MOLA) is 463 m/px. Use HiRISE DTMs (about 1 m) at hero sites; elsewhere the normal map is interpolated from MOLA and the HUD should say so.
- TOGGLES: CTX luminance with Viking low-frequency colour (the standard panchromatic-plus-colour recipe, described as such) | THEMIS IR day mosaic (labelled thermal, not visible colour) | MOLA hypsometric tint | dust opacity (real, season-dependent) | polar cap season | atmosphere on/off.

12. JUPITER

12.1 THE PROBLEM WITH JUPITER (and every gas giant)

There is no "Jupiter surface". There is only a time-stamped cloud-top snapshot.

Physically defensible options, in order:

- (a) a real global cloud-top map from a dated observation, animated by differential zonal winds (the winds ARE real and well measured);
- (b) a real map plus procedural turbulence for sub-map-scale motion;
- (c) pure procedural (rejected: no scientific provenance).

Also: Jupiter's published "true colour" maps are filter composites. Example from the OPAL programme (Simon, Wong, Orton 2015, ApJ 812, 55): the global maps use R = F631N (631 nm), G = F502N (502 nm), B = F395N (395 nm), mapped between 79.9°N and 79.9°S at 0.1°/px. That is "approximately true colour", and the app should say so.

12.2 THE BEST FREE SOURCES

- #1 – Hubble OPAL annual global maps (highest quality, licence caveat)
 - OPAL (Outer Planet Atmospheres Legacy) produces yearly global maps of all four giant planets; data are immediately public in the MAST archive with no proprietary period (stated in the ApJ paper).
 - NASA-hosted release pages (verified):
 - <https://science.nasa.gov/asset/hubble/jupiter-global-map-from-hubble-opal-data-rotation-2/>
 - downloads: 3600x1598 TIFF (5.22 MB) + JPG; credit "NASA, ESA, A. Simon (GSFC), M. Wong (UC Berkeley), and G. Orton (JPL-Caltech)"
 - <https://science.nasa.gov/asset/hubble/multiwavelength-opal-jupiter/>
 - <https://science.nasa.gov/asset/hubble/opal-jupiter-observations/>
 - ESA mirror route (CC BY 4.0): <https://esahubble.org/> (see 29.4 for why this matters legally – ESA/Hubble is CC BY 4.0, AURA/STScI is not).
 - LICENCE WARNING: the same OPAL image can be governed by two different regimes depending on which site publishes it. AURA/STScI states plainly: "Hubble images and materials created, authored, and/or prepared by AURA and STScI are copyrighted... Commercial use: permission should be obtained from the copyright owner, AURA/STScI." NASA's own guidelines say NASA content generally is not copyrighted in the US. The safest route is to derive your map from the archived calibrated science data yourself (MAST/ESA archive), which is mission data rather than a copyrighted presentation image; otherwise use the ESA/Hubble (CC BY 4.0) publication of the same map.
- #2 – JPL Photojournal Cassini cylindrical map of Jupiter (clean, older, lower res)
 - PIA07782 "Cassini's Best Maps of Jupiter (Cylindrical Map)", NASA/JPL/Space Science Institute, image addition date 2006-03-27; downloads: JPEG (431 KB) and TIFF (18.56 MB); constructed from 36 Cassini NAC images in two colours (750 nm near-IR and 451 nm blue); the JPL text states "the map's colors are close to those the human eye would see"; smallest visible features ~120 km. <https://photojournal.jpl.nasa.gov/catalog/PIA07782>
 - <https://www.jpl.nasa.gov/images/pia07782-cassinis-best-maps-of-jupiter-cylindrical-map/>
 - Related: PIA07783/PIA07784 (polar stereographic), PIA00011 (Voyager 1 cylindrical map, 1979, 2.26 MB TIFF).
- #3 – Solar System Scope 8K Jupiter (CC BY 4.0): artistic, but the only 8K option with a clear commercial licence. Use as fallback/optimisation target.
 - Also evaluated and REJECTED as a primary source: the JPL Solar System Simulator texture page (<https://space.jpl.nasa.gov/tmaps/jupiter.html>) whose own description says "From Voyager images, badly processed", 720x360, 2 px/deg, and whose Uranus entry is literally a solid colour (#0087d5). It is honest but far below target quality.

12.3 RECOMMENDED CONSTRUCTION – TYPE D + E (multi-layer + shader-driven motion)

- Sphere radius 71,492 km (equatorial) with the correct 1/15.4 oblateness – this is visible to the naked eye in any real image and EOSS does model it (a flattened spheroid, not a ball). Do not skip it.
- Base: OPAL map resampled to 4096x2048 (0.1°/px native ≈ 3600x1598, so 4K is already an upsample; do not pretend otherwise) or PIA07782 for a

time-independent composite.

- Motion: implement real zonal wind shearing (latitude-dependent angular velocity from published wind profiles; the OPAL/Cassini wind profiles are in the literature) by sampling the base map with a latitude-dependent longitude offset, plus a curl-noise advection for sub-map detail. This produces the correct "bands shear and vortices roll" look with a single texture.
- Great Red Spot: it is a real feature; keep it from the map and add a subtle swirl normal map so it reads as a vortex rather than a sticker.
- Atmospheric limb: Jupiter's limb shows strong Rayleigh/Mie-plus-CH₄-absorption shading; use the scattering LUT with a methane absorption term to darken the limb correctly, plus a bright limb haze.
- Aurora and magnetosphere (both exist as separate EOSS models – you cannot copy those files, but you can build your own): Jupiter's UV aurora ovals are a famous, defensible premium feature; use published aurora oval positions.
- Night side: Jovian lightning flashes (real, detected by Juno) as an optional premium effect; do not add city-style lights.

12.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
OPAL map (derived from MAST/ESA data or ESA/Hubble route) + wind shader	10	10	9	9	9	9	9	7-9 (see 29.4)	9.0
JPL PIA07782 Cassini cylindrical map	8	7	8	7	10	9	9	10	8.5
Solar System Scope 8K Jupiter (CC BY 4.0)	4	8	9	8	2	9	9	9	7.0

#1 wins on scientific accuracy (dated, calibrated, filter-defined, 0.1°/px) and

on visual quality; #2 wins on licence cleanliness. If the legal review is conservative, ship #2 as the default and offer the OPAL map as an optional, separately-credited download.

12.5 INSPECTION SPEC (single-object viewer)

- DEPTH: the OPAL global map is 3600 x 1598 (0.1 deg/px), i.e. about 125 km/px at the equator; the Cassini-derived Photojournal map resolves features of about 120 km. Jupiter therefore has the SHALLOWEST real data of any planet here - a fact to design around rather than hide.
- MAX USEFUL ZOOM: full disk plus about 3-4x. Past that the user is looking at interpolation, and the data-limit indicator must fire. Do not add procedural cloud detail without labelling it.
- WHAT SAVES THIS OBJECT: motion and scale. Real zonal wind shear, real great red spot, real oblateness (1/15.4), the correct limb darkening and methane absorption, and a sun-angle control that makes the terminator sweep the bands. A rotating, shearing Jupiter at 4096 px reads as far more real than a static 8192 px one.
- TOGGLES: OPAL epoch (annual maps exist, so the user can compare years) | Cassini PIA07782 (2000-era, lower resolution, cleaner licence) | wind-shear animation speed | aurora ovals and lightning flashes (real phenomena from Juno, labelled and optional).

13. SATURN

13.1 PLANET BODY

- Same data situation as Jupiter: use the OPAL annual global maps (<https://science.nasa.gov/asset/hubble/multiwavelength-opal-saturn/> and the OPAL releases) or Cassini-era JPL products. Jupiter's PIA07782 has a Saturn analogue in the Cassini "Map of ..." series; search the PIA catalogue (<https://photojournal.jpl.nasa.gov/>) for the Saturn globe maps released during/after the Grand Finale.

- Physical accuracy requirements: Saturn's equatorial radius 60,268 km, polar radius 54,364 km → the most visually oblate planet ($\approx 1/10$). Anything that renders Saturn as a sphere looks wrong instantly.
- Axial tilt 26.73°, so the ring plane presents correctly over the 29.5-year orbit – animate it from real orbital elements so ring opening angle is right for any date (this is a big authenticity signal).
- Atmosphere: strong methane absorption in the blue/red (hence the muted yellow), plus a distinct limb haze; use the scattering LUT with Saturn parameters.

13.2 RINGS – BUILD THEM, DON'T BUY THEM (this is the highest-value upgrade in the whole project, because EOSS uses two PNG billboards and its own ring sprite is only 3,588 bytes)

Scientific basis (PDS Ring-Moon Systems Node, <https://pds-rings.seti.org/>):

- COUVIS_8001 (v2, 438 MB tar.gz): radial profiles of Saturn's rings at 1 km and 10 km resolution from more than 200 Cassini UVIS stellar occultations, plus the 8-volume Cassini UVIS_Ring_Occultation_Atlas; processing documented in Colwell et al. (2010).
Page: <https://pds-rings.seti.org/cassini/uvis/index.html>
- CORSS_8001 (v2, 6.14 GB): Cassini RSS occultation profiles, including diffraction-limited optical depth and phase-shift profiles.
Page: <https://pds-rings.seti.org/ringocc/>
- COVIMS_8001 (v2, 320 MB): VIMS stellar occultations, giving compositional (colour) information alongside optical depth.
- Ground-based: EBROCC_0001 (5.14 MB) – historical Earth-based occultations; Nicholson et al. 2002 (Icarus) provides optical depth profiles at 0.9, 2.1, 3.3, 3.9 μm .

Construction recipe (a preprocessing job, then a tiny runtime asset):

1. Compute a high-resolution normal optical depth profile $\tau(r)$ by combining UVIS (1 km) profiles (median-combined to remove azimuthal variation), cross-checked against RSS.
2. Convert $\tau(r)$ into a 1D RGBA gradient texture indexed by radius:

$$\alpha = 1 - \exp(-\tau / \mu) \quad (\mu = \text{ring-plane incidence cosine})$$
 (i.e. the physically correct transmission through an optically thin slab)
 colour = from the VIMS/Cassini visible-colour observations per ring region
 (C ring grey-brown, B ring bright with reddish tint, Cassini Division dark, A ring with the Encke and Keeler gaps)
3. Encode the profile as a 2D texture (radius $\times$ azimuth) or a 1D texture (width = 16384 texels is plenty for 74,500 km of ring span at ~ 4.5 km/texel; use 32768 if you insist: that is $\sim 9\times$ finer than any published ring image).
4. Ship also: ring thickness (~ 10 -100 m for the main rings – so render as a two-sided plane with a separate volumetric option), and per-region particle-size/roughness hints from the literature.

Runtime rendering:

- Double-sided, translucent, custom-material disc with: analytic radial transmission, forward scattering (the rings are strongly forward-scattering, which is why they look different from the opposite side), self-shadowing (rings cast a real shadow onto Saturn's cloud tops, and Saturn casts a shadow onto the rings – this is a must-have authenticity feature and is physically straightforward: compute the sun-ray intersection with the planet ellipsoid and the ring plane in the shader), plus a subtle "spokes" mask (a real, documented phenomenon).
- Because the ring is a plane, this is cheap: 1 draw call, a handful of textures, an analytic shadow test (~ 10 ALU ops). EOSS cannot match it.
- Uranus's 13 narrow rings and Neptune's Adams/Le Verrier rings: build from published radii/widths/optical depths (Uranus ring data are in PDS ring

node datasets and the Uranus ring atlas literature). They are narrow and faint – cheap but a strong authenticity signal (EOSS ships a single PNG each).

13.3 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
OPAL/Cassini body + PDS-derived procedural rings	10	10	10	9	10	9	8	9-10 (PDS = PD)	9.6
OPAL/Cassini body + Solar System Scope ring PNG (CC BY 4.0)	6	7	9	6	3	9	9	9	7.4
Any downloaded "Saturn ring texture" PNG	3	7	8	6	1-2	9	9	UNCERTAIN	UNUSABLE

#1 wins because the rings become a scientifically derived, physically transmissive, self-shadowing structure instead of a decal.

13.4 INSPECTION SPEC (single-object viewer)

- DEPTH: body about 105 km/px (OPAL 0.1 deg/px). Rings, by contrast, have 1 km radial resolution from Cassini UVIS occultations - the rings are roughly 100x sharper than the planet they circle. That asymmetry is real and worth saying out loud in the app.
- RING DETAIL: encode the radial optical-depth profile at 16,384-32,768 texels across the ring span (74,500 km for Saturn's main rings, so 32,768 texels is about 2.3 km/texel - matching the 1-10 km UVIS profile resolution). Exceeding 16,384 in one dimension may exceed the device texture limit, so store the profile as a 2D array or two halves; see 28.3.
- MAX USEFUL ZOOM: extreme on the rings (down to individual density waves and the Encke and Keeler gaps), modest on the cloud tops (about 100 km/px), and huge on the ring/planet shadow interaction, which is geometry rather than texture and therefore stays sharp at any zoom.
- TOGGLES: ring optical-depth rendering (transmission from tau) | ring thickness/volumetric E-ring haze | ring shadow on planet | planet shadow on rings | spoke overlay (labelled, transient phenomenon) | ring azimuthal variation | sun-angle control to change the ring opening angle, which is the object's most beautiful behaviour.
- OBLATENESS: 1/10.2 with a 26.73 deg tilt. Rendering Saturn as a sphere is the fastest way to look amateur; the flattening is visible in any real photograph.

14. URANUS

14.1 DATA REALITY

- Uranus has been visited once (Voyager 2, 1986) and is monitored by Hubble OPAL; there are also 2023-2024 JWST NIRCам images (near-infrared, not visible). There is no high-resolution global visible-light map beyond OPAL.
- COLOUR CAVEAT (must be in-app): the long-standing "deep cyan" look of Uranus and the "deep blue" look of Neptune come from vintage processing. Irwin et al. (2024, MNRAS) re-derived the true colours from HST/STIS spectra and multi-band imaging: Uranus and Neptune are both pale blue/cyan, and Neptune's famous deep blue is an artefact of contrast enhancement. Use the corrected colour and explain it – this is exactly the "scientific accuracy" the project wants. Sources: Irwin et al. 2024, MNRAS 527, 11521, "Modelling the near-infrared colours of Uranus and Neptune" (see Section 35 for the DOI style citation and NASA/ESA press coverage).
- Also physically important: Uranus's 97.77° obliquity (it rolls), its exceptionally low internal heat flux (hence a featureless disk), the bright polar cap that has been growing since equinox 2007 (documented in OPAL releases), and its faint narrow rings.

14.2 RECOMMENDED CONSTRUCTION

- Sphere radius 25,559 km (equatorial), 24,973 km (polar) – extremely oblate.
- Texture: OPAL Uranus global map (true colour); if unavailable at high resolution, use a conservative approach: a low-contrast cloud-deck texture with the correct pale-cyan albedo, the polar cap brightening, and subtle latitudinal bands. Add procedural zonal banding at low contrast (physically motivated: Uranus's atmosphere is nearly featureless).
- Atmosphere: methane-dominated absorption (very strong in the red/near-IR), Rayleigh scattering from H₂/He; the result is the pale blue and a soft limb. Use the scattering LUT with Uranus parameters and a methane absorption term.
- APPEARANCE TOGGLE (added in revision 2): offer "Corrected appearance (Irwin et al. 2024)" as the default and "Historic Voyager-era appearance" as a labelled alternative, so users can compare the pale-cyan 2024 result with the saturated deep-cyan archival look. Both states are labelled; neither is presented as the only truth.
- Rings: 13 narrow rings from published radii/widths; nearly edge-on to the observer for years (13 rings appear as thin lines) – a great authenticity detail.
- Aurora: Uranus has real UV aurora (HST/UVS detections); optional.
- Axial tilt: 97.77°; rotation period –17.24 h (retrograde); the IAU longitude convention for Uranus is unusual (rotation around the tilted axis) – get this from SPICE kernels rather than from blog posts.

14.3 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
OPAL map + Irwin et al. 2024 colour correction + scattering LUT	10	9	10	8	9	9	9	7-9	9.0
Voyager 2 frames assembled into a map	8	6	8	6	10	9	9	9	8.1
Solar System Scope 2K/8K Uranus (CC BY 4.0)	4	7	9	7	2	9	9	9	7.0
#1 wins because Uranus is a planet where <i>colour correctness</i> is the dominant accuracy story in 2026, and OPAL + Irwin 2024 is the only route that gets it right.									

14.4 INSPECTION SPEC (single-object viewer)

- DEPTH: about 45 km/px (OPAL 0.1 deg/px). There is very little to see and that is the scientifically correct outcome.
- MAX USEFUL ZOOM: modest. The honest presentation is a nearly featureless pale cyan disk with faint latitudinal banding, a bright polar cap growing since the 2007 equinox, a soft methane limb, and 13 narrow rings seen nearly edge-on.
- THE TILT IS THE SHOW: 97.77 deg obliquity means the ring plane is almost perpendicular to the orbital plane. A rotation-axis indicator, an obliquity readout and a "pole-on" camera preset are worth more here than any texture work.
- TOGGLES: corrected appearance (Irwin et al. 2024) | historic Voyager-era saturated appearance (labelled) | ring set | polar cap emphasis | aurora (real, HST-detected, optional).

15. NEPTUNE

15.1 DATA REALITY

- Same as Uranus: Voyager 2 (1989) + Hubble OPAL + JWST (NIR, not visible).
- Neptune is more visually active than Uranus: dark spots (Great Dark Spot in 1989, NDS-2018, etc.), bright companion clouds, and a strong zonal wind system. OPAL's decade of observation documents these (NASA release:

- <https://science.nasa.gov/missions/hubble/nasas-hubble-celebrates-decade-of-tracking-outer-planets/>).
- Colour correction applies (Irwin et al. 2024).
- Recommended: OPAL Neptune global map, with a dark-spot feature composited from the actual dates in which it existed, so the app can show a *time-travel* mode: move the date slider and the spots come and go. This is both more accurate and more compelling than a static texture.

15.2 CONSTRUCTION

- Sphere radius 24,764 km equatorial (very nearly spherical, oblateness ~1/58).
- Atmosphere: same scattering model with Neptune parameters; a stronger methane absorption signature than Uranus (which is why it reads bluer).
- APPEARANCE TOGGLE (added in revision 2): "Corrected appearance (Irwin et al. 2024)" as the default and the contrast-enhanced 1989 Voyager appearance as a clearly labelled historical option. Neptune's true colour is paler and less saturated than the iconic deep-blue press images; this toggle turns a correction into a feature.
- Rings: Adams ring + arcs (Liberté, Égalité, Fraternité, Courage) from published ring data; the arcs' confinement is a nice "did you know" feature.
- Winds: 2,100 km/h zonal winds – you can shear the texture realistically, and the cloud bands should visibly differ from Jupiter's sharper bands.
- Aurora: real (JWST 2025 detection of aurora + H₃⁺ emission); optional.

15.3 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
OPAL map + corrected colour + time-varying spots + scattering LUT	10	9	10	8	9	9	9	7-9	9.1
Voyager 2 mosaic (PIA00046-class)	8	6	8	6	10	9	9	10	8.2
Solar System Scope 2K/8K Neptune (CC BY 4.0)	4	7	9	7	2	9	9	9	7.0

#1 wins for the same reason as Uranus, plus it enables a documented date-driven feature set (spots, arcs, aurora).

15.4 INSPECTION SPEC (single-object viewer)

- DEPTH: about 43 km/px (OPAL 0.1 deg/px), plus Voyager 2 mosaics from 1989 for the spot epochs.
- MAX USEFUL ZOOM: modest, and epoch-bound. The interesting content is historical rather than spatial: the 1989 Great Dark Spot, NDS-2018 and later OPAL dark spots, each paired with bright companion clouds.
- DESIGN CONSEQUENCE: because ZIG has no time simulation, ship these as LABELLED EPOCHS the user selects ("Neptune, August 1989", "Neptune, 2018") rather than pretending to animate them.
- TOGGLES: corrected appearance (Irwin et al. 2024) | historic deep-blue appearance (labelled) | epoch selector | Adams ring with the Liberte, Egalite, Fraternite and Courage arcs | aurora (JWST 2025 detection, optional).

16. MILKY WAY

16.1 THE THREE COMPETING APPROACHES (and the recommended hybrid)

- Approach A – a single HDR equirectangular or cubemap texture.
Pro: cheapest, most photoreal for the unresolved glow. Con: a fixed image; bright stars blur or alias; no parallax-free correctness for camera motion (it is correct for camera rotation only, which is fine because stars are at infinity); memory cost is the problem at 8K+.
- Approach B – real star catalogue rendered as GPU point sprites.
Pro: scientifically correct positions/magnitudes/colours; can show named

stars, constellations, time travel (proper motion, precession); the sky changes with the camera's position in the galaxy (if you model depth).
 Con: unresolved Milky Way glow (the "cloud" of the galaxy) is not a point set; needs a very large catalogue to look dense.

Approach C – procedural galaxy (noise-based dust lanes).

Pro: arbitrarily detailed. Con: zero scientific provenance. Reject as a primary method; acceptable only as an added layer of fine dust structure on top of real data.

RECOMMENDED: A + B hybrid, i.e. the deep star map for the diffuse band/dust lanes, real catalogue stars as HDR sprites on top, procedural noise only for sub-textel grain.

16.2 THE BEST FREE ASSETS

- #1 – NASA SVS "Deep Star Maps 2020" (ID 4851), visualisation by Ernie Wright
<https://svs.gsfc.nasa.gov/4851/>
 Available files (verified on the page): starmap_2020_4k/8k/16k/32k EXR (16384x8192 = 422.9 MB), milkyway_2020_16k.exr (413.9 MB), 32k (1.4 GB), 64k (3.7 GB), hiptyc_2020_4k/8k/16k/32k/64k (bright-star foreground), galactic-coordinate variants, constellation boundaries/figures, celestial grid, all as TIFFs/EXRs.
 Content per the SVS page: plotted from 1.7 billion stars of Hipparcos-2, Tycho-2 and Gaia DR2, with help from the Yale Bright Star Catalog, UCAC3 and XHIP; star colours from B-V → T_{eff} with a corrected mapping; Gaia DR2's two data holes filled with UCAC3.
 Licence: NASA-produced visualisation (NASA-CONTENT) with an ESA/Gaia/DPAC credit line for the Gaia DR2 component. Practical rule: credit "NASA/Goddard Space Flight Center Scientific Visualization Studio; star data from Hipparcos-2, Tycho-2, Gaia DR2 (ESA/Gaia/DPAC), Yale BSC, UCAC3" on the credits screen. If legal counsel objects to the Gaia component, fall back to the hiptyc_2020 + milkyway_2020_gal variants (the page exposes the bright-star and Milky-Way layers separately, so you can drop the Gaia-derived layer).
 - #2 – ESO "The Milky Way panorama" (eso0932a, ESO/S. Brunier)
<https://www.eso.org/public/images/eso0932a/> – 6000x3000, CC BY 4.0 per ESO's copyright policy. CAVEAT: the page states that the full 800-megapixel original is not provided "for copyright reasons" (it belongs to Brunier), so only the 18-megapixel release is licensed. 6000x3000 equirect = ~1.7°/px, i.e. not sufficient alone; usable as a colour/dust-lane reference or a mid-distance skybox with credit.
 - #3 – Real catalogues for the point stars (chosen for licence cleanliness):
 - * Yale Bright Star Catalogue, 5th ed. (Hoffleit & Warren 1991) – the standard bright-star source (public domain US academic/NASA-funded publication; distributed via CDS/VizieR). Use for the ~9,000 brightest stars with names, spectral types, B-V colours.
 - * USNO catalogues (UCAC4, NOMAD, USNO-B1.0) – US Naval Observatory, US Government work → public domain; UCAC4 gives ~113 million stars to V≈16 with proper motions. This is the pragmatic way to get a genuinely dense, licence-clean starfield.
 - * 2MASS Point Source Catalog (NASA/IPAC) – for infrared cross-identification and for filling star colours (J-K), subject to IPAC's data-use terms; verify before shipping (LICENSE UNCERTAIN until checked).
- AVOID: Gaia (CC BY-NC 3.0 IGO), HYG/AT-HYG (CC BY-SA 4.0 – share-alike), Stellarium/Celestia catalogues (GPL).

16.3 RECOMMENDED CONSTRUCTION

- Star sprite system: one VBO of (position on unit sphere → actually store RA/Dec/parallax or a 3D unit vector, magnitude, B-V → colour) with the star as a point sprite or a 3-quad cross. Filament can render points? Use instanced quads (Filament supports instancing) with a custom material that

- computes the PSF-like profile in the fragment shader, and additive blending. Add magnitude-dependent size: a physical point source should be a fixed angular size, so the sprite size must be sub-pixel-aware: implement a minimum-size clamp plus flux-preserving energy so that AA + tone mapping behave (this is the classic "stars disappear when you resize" bug).
- Bright stars need a diffraction/optics PSF (4-8 spikes) to look photographic; that is an optical artefact, not a physical property, so label it as a camera model choice or omit it in a "scientific" mode.
- Deep map: use milkyway_2020_16k.exr downsampled to 8192x4096 for the diffuse band (a high-pass-free glow is low-frequency, so 8K is plenty), stored as an HDR-capable format if possible. Practical Android format decision:
 - * Option 1: RGB9_E5 / R11F_G11F_B10F KTX2 (no ASTC HDR on most devices) – a 8192x4096 R11F_G11F_B10F is 4 bytes/px = 134 MB. Too heavy to keep resident if you also want a 16K variant.
 - * Option 2 (recommended): store the skybox as ASTC LDR (sRGB) at 8K (~22 MB with ASTC 6x6 + mips) and carry the HDR information in the star sprite brightness (stars are what actually need range). Use Filament's ``iblLuminance``/exposure and tone mapping to keep the sky dark.
 - * Option 3 (best fidelity per byte): keep two layers – a 4K HDR skybox for the band and the sprite star system for the points (recommended default).
- Add real-time features that make this a "premium" Milky Way:
 - * proper motion and precession: with UCAC4 proper motions you can show the sky as it looked in 1000 AD or 10,000 AD;
 - * named stars, IAU names, Bayer/Flamsteed designations, constellation lines and boundaries (SVS provides IAU 1930 boundaries as TIFFs at 64K);
 - * the galactic-plane orientation must come from the IAU galactic pole definition, not eyeballed;
 - * spectral-class-based star colour (B-V → sRGB via a documented colour-temperature transform, and be honest that star "colour" beyond ~the Rayleigh-Jeans tail is eyesight-limited at low light levels.
 - * Optional deep-sky objects (Messier/NGC) as small sprites – but see the licensing caution: many amateur images are copyrighted; use DSS images (NASA/STScI) or ESA/Hubble (CC BY 4.0) if you want real imagery, and check terms.
- Performance: a million stars = a million point sprites = trivial vertex load (one vertex each) but the fragment cost is dominated by the size; keep sprites ≤ 3-4 px except for the ~100 brightest. This is how mobile planetariums do millions of stars at 60 fps.

16.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
SVS 4851 deep map + USNO/Yale sprite stars	10	10	n/a	10	10	9	8	9	9.5
SVS 4851 deep map alone (no catalogue stars)	8	9	n/a	10	10	9	9	9	8.9
ESO eso0932a panorama (CC BY 4.0) as the only sky	6	7	n/a	6	7	9	9	9	7.3
Gaia-derived own rendering	10	9	n/a	9	10	6 (volume)	6	2 (CC BY-NC)	6.0

#1 wins: the only combination that gives both the luminous band and catalogue-accurate stars, is NASA-produced, and avoids the non-commercial Gaia licence.

16.5 INSPECTION SPEC (the "look around" object)

- This is the only object the user stands INSIDE. Zoom does not reveal detail, because the Milky Way is an angular pattern: the 16,384 x 8,192 deep map is about 1.3 arcmin per texel, and every star beyond that is a point source at any magnification.
- WHAT ZOOM DOES: it magnifies the sprite stars (which should grow as optical point-spread functions, with flux preserved, not as blurry photographs), so

- zooming into the Pleiades or the Orion Nebula region shows resolved member stars and catalogue colours rather than a smeared texture.
- WHAT TO MAKE INTERACTIVE: tap a star for its catalogue identity - designation, magnitude, colour index, distance where known; constellation figures and IAU boundaries as an overlay; a magnitude-limit slider that dims to naked-eye, binocular or telescope views.
- HONESTY: a "this is a visualisation" note, because the deep map is a rendering of catalogue data rather than a photograph, and the star colours are derived from B-V through a stated transform.

17. ISS

17.1 THE BEST FREE ISS MODEL – MEASURED, NOT GUESSED

I downloaded and parsed the actual binary to make sure this is not a blog claim.

#1 – NASA VTAD `ISS\_stationary.glb`

URL: https://assets.science.nasa.gov/content/dam/science/psd/solar/2023/09/i/ISS_stationary.glb

Landing page: <https://science.nasa.gov/resource/international-space-station-3d-model/>

(also available as USDZ, 15.97 MB; release date 2023-09-09;

credit: "NASA Visualization Technology Applications and Development (VTAD)")

Measured facts (from the file itself):

- * 44,495,916 bytes (~42.4 MiB), valid glTF 2.0 binary (magic 0x46546C67, version 2)
- * generator: "Khronos Blender glTF 2.0 I/O" → i.e. a Blender-exported glTF
- * 131 meshes, 132 nodes, 28 materials, 26 images (all image/png)
- * 247,547 triangles (sum over index accessors)

No glTF extensions are required (extensionsUsed is absent), i.e. it loads in any conformant glTF 2.0 loader including Filament's gltfio.

#2 – NASA 3D Resources "International Space Station (ISS) (B)"

<https://science.nasa.gov/3d-resources/international-space-station-iss-b/>

file: [https://assets.science.nasa.gov/content/dam/science/cds/3d/resources/model/international-space-station-\(iss\)-\(b\)/International%20Space%20Station%20\(ISS\)%20\(B\).glb](https://assets.science.nasa.gov/content/dam/science/cds/3d/resources/model/international-space-station-(iss)-(b)/International%20Space%20Station%20(ISS)%20(B).glb)

source credit: "NASA/Michael D. Carbajal"

#3 – NASA 3D Resources "International Space Station (ISS) (C) (High Res)"

<https://science.nasa.gov/3d-resources/international-space-station-iss-c-high-res/>

35.53 MB ZIP in original Lightwave format, "very high resolution model of the ISS in many parts", February 2011 configuration – the CAD-derived option if you need more fidelity than the GLB. Requires a Blender/Lightwave conversion step by you. Note: it is a 2011 configuration (no later modules).

For reference (do NOT ship): the EOSS runtime asset

https://eyes.nasa.gov/assets/static/models/sc_iss/ISS_stationary.gltf

is 501,729 bytes + 22,186,880 bytes .bin, generator "Khronos glTF Blender I/O v4.2.60", 149 meshes / 150 nodes / 47 materials / 25 images, ~264,809

triangles, and it *does* use KHR_materials_clearcoat, KHR_materials_specular and KHR_materials_ior (i.e. it has richer material definition). It is tempting because it is slightly richer, but it is a runtime asset of a closed app with no redistribution grant, so prefer the NASA 3D Resources copies above.

If you want those extra material properties, port them by hand onto the #1 model (clearcoat on the MLI blankets, specular/ior tinting).

17.2 MODEL vs POSITION – THE SEPARATION YOU ASKED FOR (and the correct design)

Two independent concerns must not be coupled:

- VISUAL MODEL: a NASA glTF asset, loaded once, cached, at the correct geometric scale (109 m truss span, ~73 m with arrays; solar array wings 34 m long). Set the model unit scale so that 1 scene unit = 1 metre and verify the truss span measures ~109 m; if not, apply the correction factor

derived from measurement, not from guesswork.

- B. ORBITAL POSITION: keep ZIG's existing solution untouched. The renderer should expose a single API – "attach this model to body/instance X at simulation time T" – and never compute orbital mechanics itself. This is a hard architectural boundary; it guarantees no regression in ZIG's existing calculations.

Two techniques for placement:

1. Attach-to-body transform (recommended): ZIG's ephemeris produces a position (ECI/ECEF or your internal inertial frame) and a geodetic sub-point. The renderer converts that into the scene's floating-origin frame and sets the node transform. This is $O(1)$ and does not touch the orbit math.
2. For "future/backup" ephemeris: NASA JSC publishes ISS trajectory data in CCSDS OEM form (state vectors in the J2000 frame) at https://spotthestation.nasa.gov/trajectory_data.cfm – useful as a validation dataset for ZIG's existing solution and as a fallback data source for times beyond the current TLE validity. Note: this is data, not a model; do not let it replace ZIG's solution, use it to test it.

Orientation:

- The ISS is maintained in a near-nadir LVLH-ish attitude, but real attitude oscillates and changes (yaw manoeuvres, dockings, reboots). Two honest options: (a) "LVLH mode" – derive the local frame ($\hat{r}$, $\hat{v} \times \hat{r}$, $\hat{v}$) from the state vector and align the model's longitudinal (+X) axis with the velocity vector, with a documented $\pm$ few-degree error; (b) "attitude mode" – use real attitude telemetry/quaternions if you license/obtain them. Ship (a) as the default and label it "attitude approximated as LVLH".
- Calibrate the model's local axes ONCE by comparing a rendered frame with a public photograph (e.g. Expedition photographs from the Gateway to Astronaut Photography, <https://eol.jsc.nasa.gov/>) so the truss/solar-array orientation is right. Document the calibration in-app or in the repo.

Scale, lighting, and premium details worth doing:

- Sun-lit + Earth-lit shading: the ISS is lit by both; use Filament's directional light for the Sun (with physical Lux scaling) and either a second directional light or a low-res indirect (IBL) for earthshine. Real images of the ISS show the Earth facing side notably warmer.
- Earth limb in the background: when the camera is attached near the ISS, the premium look requires the Earth from orbit – reuse the Earth asset set with a high-resolution local patch (this is where virtual texturing or a camera-anchored detail patch becomes necessary; see Section 26/28).
- Surface materials: convert the 26 PNGs to KTX2/ASTC (Section 26) and give the MLI blankets a clearcoat and a slight anisotropy to reproduce the "crumpled foil" specular response; Kapton/gold foil can be metallic with a high specular tint. This is the difference between a plastic toy and a photographed station.
- Distance-dependent LOD: 247k triangles is fine for a hero view, but build 3-4 LODs (e.g. 32k/64k/128k/247k) so that ZIG can render the ISS as a speck in an Earth-orbital overview without wasting a single draw call.

17.3 LICENCE ASSESSMENT

- The models are published by NASA on NASA web properties, with explicit source credit lines; NASA's media guidelines cover "files used in the rendition of 3-dimensional models, such as texture maps and polygon data in any format". Practical position: redistribution inside a commercial app is acceptable with clear attribution ("International Space Station 3D model: NASA/NASA VTAD" / "NASA/Michael D. Carbajal") and with no NASA logo and no implied endorsement.

- Residual risk to record in your legal notes: the ISS is a multi-national programme artefact and NASA's guidelines are a policy statement, not a licence deed. If ZIG ever becomes high-profile, the conservative move is to write to NASA's Brand/Media office with a description of use (they explicitly handle merchandise and commercial enquiries: <https://www.nasa.gov/nasa-brand-center/merchandise-approvals/>).
- What you MUST NOT do: download the third-party GitHub re-uploads of EOSS spacecraft models (e.g. "matthewgiarra/spacecraft-models") – those have no licence and are not NASA's publication.

17.4 SCORES

Source	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
NASA VTAD ISS_stationary.glb (44.5 MB)	9	9	9	8	9	9	8	9	8.8
NASA 3D Resources ISS (B) glb	8	8	8	8	9	9	9	9	8.5
NASA 3D Resources ISS (C) High-Res Lightwave	9	9	10	8	8	6 (needs conversion)	6	8	8.0

#1 wins: it is NASA-published, glTF 2.0 with zero required extensions, big enough to look premium (247k tris, 26 textures), small enough to load in a second, and it is derived from the same lineage as the EOSS model.

17.5 INSPECTION SPEC (single-object viewer)

- DEPTH: the model is the limit. NASA VTAD ISS_stationary.glb carries 247,547 triangles, 28 materials and 26 PNG textures, so close inspection shows module and truss geometry well, and texture resampling starts to show on the MLI blankets and array blankets.
- UPGRADE PATH (recommended): re-encode the 26 textures to KTX2/ASTC, then add bespoke material work rather than more polygons - clearcoat on the MLI blankets, metallic gold Kapton foil with a high specular tint and slight anisotropy, and a roughened white paint for the truss. That contrast is what makes a spacecraft read as photographed rather than plastic.
- LODs: build 4 (about 32k / 64k / 128k / 247k triangles) so the inspection camera can pull back without wasting fill rate.
- INTERACTION: orbit, pinch zoom, tap-to-focus on a module, a scale bar in metres (truss span 109 m) and optional module labels.
- SUN ANGLE: here the control genuinely matters. The real station is lit by the Sun and by earthshine; expose "sun only", "sun plus earthshine" and "orbital dusk" presets so the user can reproduce the lighting of real photographs.

18. BLACK HOLE

This is the section where I refuse the lazy answer ("find an Interstellar model"). There is no Interstellar asset to license; DNGR is proprietary Double Negative code. What exists is a well-documented body of physics plus several open-source implementations with very different licences. Below: the science, the technique landscape, the ranked candidates, and a concrete real-time plan for mobile.

18.1 WHAT "REALISTIC" MEANS (the checklist ZIG must satisfy)

The visually correct features of a Kerr/Schwarzschild black hole seen against a star field with a thin accretion disk, all of which are observable consequences of GR, not art direction:

1. The shadow: a dark region of angular radius set by the critical impact parameter. For Schwarzschild, $b_c = 3\sqrt{3} M \approx 5.196 M$ (i.e. shadow radius $2.598 r_s$); for Kerr it is asymmetric and shifted by frame dragging.
2. The photon ring: an extremely thin, bright ring at the critical curve, produced by light that orbited the hole ($n = 1$ and higher). Measured for real: EHT/Broderick et al. 2022, "The Photon Ring in M87*", ApJ 935, 61 – the $n=1$ ring diameter is $\sim 4.15\text{--}4.22 \mu\text{as}$ radius (i.e. $\sim 8.4 \mu\text{as}$ across), constant across the 2017 campaign, consistent with theory. NASA/observations

- for the shadow itself: EHT 2019, ring diameter $42 \pm 3 \mu\text{s}$.
3. Multiple images of the disk: the far side of the disk appears *above* and *below* the shadow (the "split" that made Interstellar famous), and near the photon ring the disk appears multiple times, exponentially demagnified.
 4. Doppler beaming + gravitational redshift: the approaching side of a Keplerian disk is brighter (by up to a factor of ~ 10 - 100 at the inner edge depending on inclination and spin) and blueshifted; the receding side is dimmer and redder. The total intensity transform follows Liouville's theorem (I_ν/ν^3 invariant, hence $I \propto g^3 \cdot I_{\nu'}$, and $I_\lambda \propto g^5 \cdot I_{\lambda'}$ in wavelength form – the exact relation used by Bruneton and stated in the DNGR paper).
 5. Blackbody colour: a thin disk emits a blackbody at a temperature that rises inward ($T \propto r^{-3/4}$ for a standard thin disk), so the inner edge is blue-white and the outer regions are orange-red. The *observed* colour comes from a shifted blackbody (temperature shifted by the Doppler/redshift factor), convolved with the observer's colour matching functions.
 6. Lensing of the background sky: the starfield is distorted into the Einstein-ring pattern, with a secondary image inside the ring.
 7. Time-dependent effects: disk shear and orbital motion make the pattern rotate with a Keplerian angular velocity; for a real stellar-mass BH the inner disk orbits in milliseconds, so the app needs an explicit, labelled time-scaling control.
 8. Interstellar's own choices (documented, and different from the physics): James, von Tunzelmann, Franklin & Thorne 2015 (Class. Quantum Grav. 32 065001, open access) describe DNGR, ray-bundle propagation, caustics, camera lens flare and the disk model (constant-temperature blackbody with Doppler and gravitational frequency shifts). DNGR's Gargantua as visualised used a *moderate* spin $a/M = 0.6$ (stated in the paper's appendix), and the paper explicitly documents removing the Doppler-induced intensity change in one of its figures by dividing by a weighted mean, plus a 6500 K white balance. Press reporting at the time (New Scientist) quotes Oliver James saying Nolan vetoed the extreme asymmetry because audiences would misread it. Conclusion: the film look is a deliberately photometrically simplified version. ZIG can offer BOTH: a "Cinematic (Interstellar-style)" mode and a "Physically correct (beaming on)" mode, and it will be more honest than any competitor.

18.2 THE TECHNIQUE LANDSCAPE (with what is actually published)

Method class 1 – Full relativistic ray tracing per pixel (Kerr or Schwarzschild).

Integrate null geodesics (e.g. Hamiltonian form with RK4, or the planar

$u'' = -u + 3/2 \cdot (r_s) \cdot u^2$ ODE for Schwarzschild) per pixel per frame.

Pros: exactness, including photon ring and multiple images if step count is high. Cons: hundreds to thousands of integration steps per pixel.

Evidence of cost: Bruneton's paper compares against ray marching that needs "up to 1000 integration steps" for near-equal precision, which runs at ~ 45 fps at 1920x1080 on an NVIDIA GTX 960 (a 2015 desktop GPU). A 2026 mobile GPU is in that performance class, but at 1080p on battery with a full scene, brute force is not the design you want.

Method class 2 – Precomputed tables / LUT-based beam tracing (RECOMMENDED).

Bruneton 2020, "Real-time High-Quality Rendering of Non-Rotating Black Holes" (arXiv:2010.08735; code BSD-3-Clause at

https://github.com/ebruneton/black_hole_shader). Core idea: precompute, for each impact parameter e , the total deflection Δ and the relationship between the ray's angle from the hole and the radial coordinate $u = 1/r$, so that the fragment shader can find where a curved beam intersects the disk plane and where it exits to the sky in *constant time per pixel*.

The paper states the implementation uses two small tables –

D(e,u): 512x512 RG32F

U(e,φ): 64x32 RG32F

"which are very fast to precompute" – and reports ~150 fps at 1920x1080 on a GTX 960, versus ~65 fps (stars rendered with a different method) and ~45 fps (ray marching, 1000 steps). It also includes a custom texture filtering scheme so that point stars integrate correctly into a beam (avoiding the classic aliasing/disappearing-star problem), plus disk shading with gravitational lensing, relativistic Doppler and beaming.

This is, by a wide margin, the best published starting point for a mobile GPU: 5125128 bytes + 64328 bytes ≈ 2.1 MB of LUTs, and the per-pixel cost is a bounded number of texture fetches.

Method class 3 – Analytic strong-deflection approximations.

Use the closed-form strong-field deflection for the photon sphere (Bozza et al. 2002, Phys. Rev. D 66, 103001; astro-ph/0208086) to get the divergent bending near $b \rightarrow b_c$ analytically, and the weak-field $\alpha \approx 4M/b$ far away. Cheap and smooth, and useful as (a) a fallback tier on weak devices, (b) a way to keep the photon-ring region visually correct when a full LUT is not available. It is an approximation: label it.

Method class 4 – Ray bundles / beam tracing with filtering (research/offline).

This is the DNGR approach (and Bruneton's method is described as "based on beam tracing"). Full ray-bundle rendering with magnification-aware filtering is what makes the Interstellar image smooth and film-like. On mobile you cannot do it per-frame; you CAN do it offline to bake high-quality "hero" stills/cinematics, or to precompute a magnification LUT.

Method class 5 – GRMHD-informed disk appearance (research/offline).

For maximum scientific credibility (e.g. for M87-like or Sgr A-like modes), use published GRMHD simulation results to derive emission profiles; do not attempt GRMHD on device.

What is NOT acceptable as the primary technique: "a black sphere with an orange torus" and a fake ring (i.e. the majority of Game-jam/tutorial black holes). It will fail the "scientifically defensible" bar immediately.

18.3 RANKED CANDIDATES (code + licence, verified via the GitHub API on

2026-09-14 unless noted)

#1 – ebruneton/black_hole_shader (Eric Bruneton)

https://github.com/ebruneton/black_hole_shader

Licence: BSD-3-Clause (verified; LICENSE + per-file headers). 240 stars, last push 2020-11-11 (stable/inactive but complete and correct).

What it is: WebGL2 demo + a small generic GLSL library implementing Algorithm 1 of arXiv:2010.08735: precompute(), traceRay(), custom star texture filtering, accretion-disk shading, gravitational lensing, relativistic Doppler and beaming; plus reusable preprocessing code (black_hole/preprocess/), GLSL library (black_hole/*.glsl), and demo geometry for stars/color maps.

Can it be reused legally? YES, including commercially, with BSD-3 obligations: retain the copyright notice, include the licence text in your app's open-source notices, do not use the author's name to endorse ZIG. Mobile feasibility: excellent – LUT-based, constant cost/pixel, 2 MB of tables, no compute shaders required; the shader is WebGL2/GLSL ES 3.0-class and ports to Filament's material language or to a custom GLES3 pass with minimal changes (Filament material syntax differs, but the maths is GLSL-compatible).

What NOT to copy: his `gaia\_sky\_map` preprocessing pipeline pulls Gaia data (CC BY-NC) for the star texture, and his demo's `rocket` model and `color\_maps` assets have their own provenance. Use the science code, supply your own star map (NASA SVS 4851) and your own disk noise/colour textures. Production quality? Research-grade but real and complete; treat it as a reference implementation you adapt, not as a plug-in library.

- #2 – KleaSCM/tika (MIT) – <https://github.com/KleaSCM/tika>
 C++23 + GLSL, Kerr metric ray tracing with Hamiltonian geodesic integration (RK4), frame dragging, ergosphere, blackbody accretion disk with Liouville flux transport, gravitational redshift. Licence MIT (verified). New/small (0 stars at time of writing), so label it "educational, not battle-tested"; its value is the Kerr shader maths to lift into an offline reference / "hero" mode, and to sanity-check your Schwarzschild results.
- #3 – hydrogendeuteride/BlackHoleRayTracer (MIT, 41 stars, pushed 2026-01-15) <https://github.com/hydrogendeuteride/BlackHoleRayTracer>
 OpenGL ray marching of a Schwarzschild hole, compute-shader based (requires OpenGL 4.3 – i.e. NOT mobile-friendly as-is). Use as a visual ground-truth reference and for validating your LUT accuracy; do not port the compute path to GLES 3.1 unless you have a specific need.
- #4 – zongzhengli/black-hole (CC-BY-4.0, 3 stars, GLSL/JS) <https://github.com/zongzhengli/black-hole>
 Real-time WebGL ray tracer with dynamic step sizes based on distance to the hole over an Andromeda background. Licence CC-BY-4.0 (permissive, attribution required). Useful as a compact reference for adaptive stepping.
- #5 – sirxemic/Interstellar (MIT, 375 stars, 2022) <https://github.com/sirxemic/Interstellar>
 Interactive wormhole + black hole "similar to those in the movie"; MIT code, but the demo textures are separately credited (many are NASA; verify individually). Useful for camera UX/interaction ideas.
- #6 – ImanetorX98/KerrTraceCpp2 (MIT, pushed 2026-09-09) <https://github.com/ImanetorX98/KerrTraceCpp2>
 Kerr/Kerr-Newman/Kerr-de Sitter ray tracer, Boyer-Lindquist and Kerr-Schild charts, RK4 + DOPRI5, ray-bundle mode, CPU/Metal/CUDA. Excellent offline "ground truth generator" to validate a mobile approximation; MIT licence.
- VERIFIED BUT UNUSABLE IN A PROPRIETARY APP:
- rantonels/starless – GPL-3.0 (714 stars). Excellent CPU reference (Schwarzschild lensing with real geodesics), but GPL-3.0 code cannot be incorporated into a closed-source commercial app. Cite it, do not copy it.
 - vlvkaos/threejs-blackhole – derives from starless and is stated GPL-3.0.
 - LGhassen/Singularity (KSP mod) – no SPDX-recognised licence (NOASSERTION) and self-described as "not a physically correct simulation"; also derived from sirxemic. Not usable.
 - rossning92/Blackhole (509 stars) – no licence file (verified: license=None) → ALL RIGHTS RESERVED. GitHub availability is not a licence. Do not use.
 - oseiskar/black-hole (276 stars) – repository has no LICENSE file (verified: NOASSERTION; the LICENSE path 404s) → ALL RIGHTS RESERVED. Do not use.
 - Any Shadertoy shader without an explicit licence: default is CC BY-NC-SA 3.0 (Shadertoy's own terms page) → non-commercial and share-alike. Unusable in ZIG. Algorithms/equations are not copyrightable, so you may read them, understand them, and reimplement the physics in your own code – but do not copy or translate the source line-by-line.
- #7+ – For completeness, datasets and papers (not code) to cite:
- * Luminet 1979, "Image of a spherical black hole with thin accretion disk", *A&A* 75, 228 – the first correct rendering of a Schwarzschild lensed disk.
 - * James et al. 2015, *CQG* 32, 065001 (open access) – DNGR.
 - * Bruneton 2020, *arXiv:2010.08735* – LUT method (BSD-3 code).
 - * Bozza 2002, *PRD* 66, 103001 – analytic strong-deflection.
 - * *EHT Collaboration 2019, *ApJL* 875, L1/L6 – M87 shadow, $42 \pm 3 \mu\text{s}$.*
 - * *Broderick et al. 2022, *ApJ* 935, 61 – the photon ring in M87.*
 - * Gralla, Lupsasca & Strominger 2020 (*PRD* 102, 124004) – photon ring observables and the "n=1" hierarchy.

18.4 THE CONCRETE MOBILE PLAN (what ZIG should build)

Tier A – Flagship (Adreno 740+/Mali-G715+/Xclipse-class, 2023+):

1. Render the black hole pass into a half-resolution R11F_G11F_B10F or RGBA16F RenderTarget (Filament RenderTarget), then upsample with a sharpening/cubic filter and composite into the HDR scene. This halves the per-pixel geodesic cost with almost no visible loss (the lensed field is smooth).
2. Use Bruneton-style LUTs: $D(e,u)$ 512x512 RG32F + $U(e,\phi)$ 64x32 RG32F for the Schwarzschild case. These are static assets shipped in the APK (2.1 MB).
3. Disk: analytic thin-disk annulus between $r_{in} = ISCO$ and r_{out} (default: $ISCO$ to ~ 20 M). Sample a 2D noise/procedural texture for clumping and turbulence, and a temperature profile $T(r) \propto r^{-3/4}$ (Shakura-Sunyaev-like). Compute the observed spectrum as a blackbody at $T_{obs} = g \cdot T_{emit}$, with g the total redshift factor (gravitational + Doppler). Precompute a blackbody-temperature $\rightarrow$ CIE RGB table (Bruneton does exactly this with a $C(xy,D)$ precomputed map) so the shader is a lookup, not an integral. Colour management must then use the same transfer as the rest of the scene.
4. Apply the observed intensity transform with the g^3 (I_v/v^3) or g^5 (I_λ) relation – pick one, state which, and be consistent. This is what produces the bright approaching side and the dark receding side.
5. Photon-ring region: because the LUT handles the full deflection, the higher-order images come out automatically at the correct radii; add a slight extra exposure inside a narrow annulus around the critical curve to reproduce the natural brightness pile-up (physically justified as the demagnified images' cumulative flux).
6. Post: Filament bloom with a threshold set above the disk's mean luminance, ACES/AgX tone mapping (the disk is a genuine HDR source: use it), and a conservative lens-flare/ghosting layer for the "cinematic" preset. Note explicitly in-app that lens flare is a *camera* artefact, not an astrophysical one (the DNGR paper models this too).
7. Modes to ship: (i) "Physically correct" (beaming/redshift on, spin parameters honest), (ii) "Cinematic" (Doppler asymmetry reduced – the Interstellar-style look, documented as such, including the $a/M \approx 0.6$ choice and the 6500 K white balance from James et al. 2015), (iii) "M87*-like" (low-inclination, photon-ring-dominated, radio-wavelength emission, GRMHD-informed profile, with the EHT measurement cited), (iv) "Sgr A*-like" (face-on-ish, high-inclination variability).

USER-FACING QUALITY CONTROL (added in revision 2): expose the quality tier as a user-visible control with three named presets and one advanced slider -

"Cinematic (physically correct)" | "Balanced" | "Battery saver",
plus an advanced "ray steps / table resolution" slider (default 2 tiers below the device maximum) and an FPS readout. This makes the performance/quality trade-off legible instead of invisible, and it is also the cleanest way to pass a device test matrix without shipping 30 material variants.

Tier B – Mid-range: half-resolution + LUTs with a reduced disk sampling count and single-scatter disk (no self-illumination), analytic weak-field deflection outside a radius threshold, no photon-ring over-brightening pass.

Tier C – Low-end: replace the lensing with the analytic Bozza deflection + a precomputed 512x512 "deflection field" cubemap baked for the current camera radius (re-bake only when the camera radius changes by $> 20\%$), rendering at quarter resolution.

Everything above can run inside Filament: the lensing pass is a material with ``shadingModel : unlit`` rendered into a RenderTarget (or, for a full-screen background, as a custom skybox material with ``eyeDirection``), which keeps Filament's tone mapping/bloom/TAA downstream.

18.5 WHAT TO AVOID

- Do not call any of this "the Interstellar black hole". You cannot use the film's imagery, and the film's physics is deliberately altered.
- Do not present the cinematic mode as physically accurate; the DNGR paper

itself documents the alterations.

- Do not use physics-illiterate shaders that draw a glowing ring outside the photon sphere without lensing; the giveaway is a disk that never appears above/below the shadow.
- Do not use a 3D "black hole model" (a torus mesh). There is no such thing as a black hole mesh; the appearance is a lensing phenomenon.
- Do not copy from a repository that has NO LICENCE FILE. Verified example: rossning92/Blackhole (509 stars, last push 2024-12-02) has no LICENSE file (the repository root contains only .gitignore, CMakeLists.txt, README.md, assets/, docs/, shader/, src/) and its README contains no permission statement; GitHub reports license: None. Its README also credits oseiskar.github.io/black-hole for its physics, and that project is likewise unlicensed - so copying it would stack two unlicensed claims. Absence of a licence file means all rights reserved (Section 29.1, rule R0).

18.7 THE THREE LEGITIMATE ROUTES TO A LICENCE-FREE REFERENCE

1. ASK. For small research repositories, opening an issue or sending a short email that describes the commercial product and requests a specific licence (MIT/BSD-3/Apache-2.0) frequently succeeds, and costs one message. Written permission by email is itself a licence; keep it in the provenance manifest.
2. REIMPLEMENT FROM THE PUBLISHED PHYSICS. Copyright protects expression, not ideas, methods or mathematical formulae. You may read a paper's equations and write your own shader. What you may not do is translate that specific source file line by line. The published basis for a clean reimplementation is Luminet 1979 (A&A 75, 228), Bozza 2002 (PRD 66, 103001), James et al. 2015 (CQG 32, 065001) and Bruneton 2020 (arXiv:2010.08735) - and Bruneton's own implementation is BSD-3-Clause, so for the Schwarzschild case you do not even need to reimplement anything.
3. USE A LICENSED ALTERNATIVE. KleaSCM/tika (MIT, Kerr), hydrogendeuteride/BlackHoleRayTracer (MIT), ImanetorX98/KerrTraceCpp2 (MIT), zongzhengli/black-hole (CC-BY-4.0) are all licensed for reuse with attribution.

18.6 SCORES

Candidate	SciAcc	Visual	Geom	Tex	Provenance	Android	Perf	Licence	Overall
Own implementation on Bruneton BSD-3 LUT method + own assets	10	10	n/a	n/a	10	9	9	10	9.7
KleaSCM/tika (MIT) Kerr reference, adapted	9	9	n/a	n/a	8	6 (needs port)	4	10	8.0
hydrogendeuteride (MIT) compute raymarch reference	9	8	n/a	n/a	8	3	3	10	7.0
starless (GPL-3.0)	9	8	n/a	n/a	8	5	4	1 (cannot ship)	REJECT
Shadertoy black-hole shaders (default CC BY-NC-SA)	3-6	6-8	n/a	n/a	2	8	8	1 (cannot ship)	REJECT
Any "Interstellar black hole model" asset	0	5	n/a	n/a	0	8	9	UNCERTAIN/INFRINGING	REJECT

#1 wins because it is the only candidate that is simultaneously (a) peer-reviewed technique, (b) constant cost per pixel, (c) commercially licensed, and (d) already proven at 150 fps on a 2015 desktop GPU.

18.8 INSPECTION SPEC (the only object with unlimited meaningful zoom)

- WHY IT IS DIFFERENT: every other object is limited by the resolution of a texture or a mesh. A black hole is limited by nothing but the renderer, because the image is computed from geodesics: the photon ring is a mathematical curve and the higher-order disk images are exact solutions. The user can zoom indefinitely and the structure remains correct, which makes this the flagship object of the inspector concept.
- CAMERA: orbit at fixed radius, logarithmic zoom, an inclination control (a near-edge-on view for the classic image, a more face-on view for the photon ring), and a field-of-view control that mimics changing the focal length.
- LIVE CONTROLS THAT MAP DIRECTLY TO PHYSICS: disk inner radius (ISCO), disk outer radius, disk temperature scale, beaming and redshift on/off (the cinematic versus physical toggle), spin parameter where the approximation

permits, and the user-facing ray-step slider from 18.4.

- WHAT THE USER WILL ACTUALLY SEE WHEN THEY ZOOM: the shadow growing, the photon ring separating from the shadow edge, the disk's far side arcing over and under the shadow, the approaching side brightening and blueshifting, and the background starfield compressing into the Einstein-ring pattern. Every one of those is a general-relativity prediction, not an art choice.

19. STARFIELD

19.1 THE DESIGN (one system, three layers)

Layer 1 – Deep background: NASA SVS 4851 milkyway/starmap EXR, downsampled to 8K equirect, ASTC-compressed (see 16.3 for the HDR/format decision).

Layer 2 – Catalogue stars: GPU point sprites from public-domain catalogues (Yale BSC5 + UCAC4), with magnitude, colour index, proper motion, and optionally parallax for nearby stars. Rendered additively with a physically motivated PSF; sprites clamped to ≥ 1 pixel with flux compensation.

Layer 3 – Constellations and labels: IAU 1930 boundaries (SVS 4851 provides 16K/32K/64K TIFFs of the boundaries, plus constellation figures); star names from the IAU Working Group on Star Names catalogue (freely published), rendered as line primitives + text (Filament LabelComponent equivalent = your own text-in-3D via ViewNode or a screen-space pass).

19.2 WHY NOT A SINGLE BIG TEXTURE OR A SINGLE HUGE CATALOGUE

- A single texture cannot resolve bright stars (aliasing on camera rotation) and cannot support proper motion, spectral colours, or named-star picking.
- A single catalogue cannot produce the unresolved galactic glow without hundreds of millions of faint sources (Gaia would, but it is non-commercial).
- The hybrid gives you the best visual *and* the best science per byte, and it is exactly what the mobile planetarium apps that render millions of stars do.

19.3 PERFORMANCE (the part that bites)

- Vertex load: 1 vertex/star; 1M stars = 1M vertices/frame – fine on modern mobile GPUs (they move $\sim 100\text{M}+$ vertices/s) but the draw must be a single instanced/point draw with no per-star CPU work. Use a static VB0, a magnitude-sorted layout, and GPU frustum/limit culling by magnitude.
- Fragment load: total screen area of sprites is the budget. A 6 px average sprite at 1080p covers $\sim 5\%$ of the screen $\rightarrow$ cheap. The killer is bloomy overdraw; keep the bright-star count low and let HDR bloom do the work.
- Precision: star directions should be computed in double precision offline and stored as fp32 unit vectors (angular resolution $\sim 1\text{e-}7$ rad, plenty).
- Streaming a 100M-star UCAC4-derived set is possible (a few GB) but pointless on a phone screen; 2-10M stars is the sweet spot for a "premium" look with a $\sim 50\text{-}150$ MB binary.

19.4 REVISION 3 NOTE: THE STARFIELD IS NOW AN OBJECT, NOT A BACKDROP

In the multi-object design the starfield was scenery behind the planets. In the inspector it is one of the thirteen objects: the Milky Way button is the only object the user stands inside, and the star layer is what makes it interactive.

Consequences:

- The whole star system (deep map, sprite VB0, constellation overlay, labels) loads only when that object is selected, so it costs nothing in the other twelve packs. Its total footprint is small: an 8192-wide ASTC skybox (about 15 MB with mips), a sprite vertex buffer of 32-64 MB for 2-4 million stars, and under 1 MB of constellation and naming data.
- The sprite system must be drawn with flux-preserving sizing at every zoom level, because in this object zoom is the primary interaction (Section 16.5).

- No other object shows stars except the black hole, which lenses them: there, the sky is an input to the lensing shader rather than a background layer, so the black-hole pack carries its own low-resolution copy of the same sky.

20. LIGHTING, HDR AND PBR

20.1 THE LIGHTING MODEL ZIG SHOULD USE

- Sun = a single directional light with a physically scaled intensity in Lux (Filament supports physical light units and a configurable exposure; the solar illuminance at 1 AU is $E = 1361 \text{ W/m}^2$, and in photometric terms the Sun's illuminance at Earth is $\approx 128,000 \text{ lx}$, i.e. $\sim 1.3\text{e}5 \text{ lux}$). At Mercury it is $\sim 6.7\text{x}$ that; at Jupiter $1/27\text{th}$; at Neptune $1/900\text{th}$. This is the single most under-appreciated realism feature: outer planets must be *dark*, and a well-exposed Earth must be *blown out* at Mercury unless you expose per target. Provide an explicit "exposure: auto/manual (EV)" control and an on-screen readout – that is NASA-Eyes-plus behaviour.
- Secondary light: a low-resolution irradiance field for the dominant reflected light in the scene (e.g. a captured IBL or an analytic earthshine/moonshine approximation, see Section 23). For most views the correct magnitude is: nothing (space is black). Resist the urge to add fill light; the blackness of space shadows is a defining trait.
- IBL/environment: use a procedural sky environment for each scene (Filament builds IBLs from cubemaps via cmgen with SH). For interplanetary scenes the environment is the deep star map + the local dominant body's bounce.
- Emissive: Earth night lights, the Sun, the accretion disk, aurorae. All must go through HDR, not be clamped LDR.

20.2 HDR PIPELINE

- Render target: RGBA16F (Filament supports half-float targets; on Vulkan backends this is efficient).
- Bloom: Filament's HDR bloom with a threshold above the scene's mean; for the Sun/disk use a low threshold (they are genuinely 2-4 orders of magnitude above the sky), for planets a high threshold.
- Tone mapping: offer ACES (filmic, the movie look), AgX (modern, handles saturated emissives gracefully – best for the Sun and the accretion disk), PBR Neutral (Khronos, colour-accurate, best for "scientific" mode), and Filmic (legacy). Filament supports all of these as built-ins/easily built. Default recommendation: AgX for cinematic modes, PBR Neutral for "science" mode.
- Colour grading: exposure, white balance (matching the DNGR 6500 K choice for cinematic BH mode), contrast, saturation. Filament ships grading controls.
- Dithering: enable – a dark starfield on a phone screen will band without it.

20.3 PBR MATERIAL CONSTANTS FOR PLANETARY SURFACES (documented values)

- Roughness/albedo anchors (use these rather than random values):
 - * Lunar/Mercurial regolith: albedo 0.07-0.13 (Moon geometric albedo 0.12; Mercury 0.09-0.14), roughness 0.85-1.0, metallic 0.
 - * Mars: bright dust albedo ~ 0.30 , dark basaltic regions ~ 0.10 -0.15; roughness 0.9. Iron-oxide reds are *not* uniform; there are grey basalt areas (Syrtis Major) – keep them.
 - * Earth: vegetation 0.10-0.25, deserts 0.35-0.45, snow/ice 0.7-0.9, ocean: high specular, roughness ~ 0.02 -0.1, IOR 1.33 (use Filament's specular/Fresnel; a rough-water microfacet model with sun glint is the correct treatment).

- * Venus cloud tops: albedo ~0.7-0.8 (that is why it is so bright), roughness high, Mie-dominated.
- * Icy moons (Europa/Enceladus): albedo 0.6-1.0, roughness 0.2-0.5, some subsurface scattering is physically appropriate (Filament has a `subsurface` shading model – use it for Europa's ice at close range).
- Normal/height: always derive normal maps with proper tangent-space conventions (OpenGL-style, +Y up) and verify against a known feature; a flipped green channel is the classic silent error.
- Anisotropy: useful for Jupiter/Saturn's banded sheen (a subtle anisotropic specular along the bands) – Filament supports anisotropy.

20.4 WHAT "PREMIUM" LOOKS LIKE (concrete checklist)

- Correct relative brightness across the solar system (Section 20.1).
- Correct terminator: a hard terminator for airless bodies, a soft one for atmospheric bodies (atmosphere scattering must bend light past the geometric terminator – this is a hallmark of good rendering and a place where EOSS is visibly approximate).
- Limb darkening on the Sun; limb brightening/haze on gas giants.
- Ring shadows and cross-shadows when the geometry is right.
- A visible photon ring, and the accretion disk's far side arcing over the shadow.
- No banding, no z-fighting (use a logarithmic or reversed-Z depth buffer and a floating origin – Section 24/28), no popping from LOD.
- Camera: per-target framing presets, inertia-free cinematic moves, focal length control (a real photographic parameter), and floating origin so that surface-level views do not collapse in float precision.

21. ATMOSPHERES

21.1 THE REFERENCE TECHNIQUE (and its freely licensed implementation)

The standard for real-time physically-based skies is Bruneton & Neyret (2008), "Precomputed Atmospheric Scattering" (Computer Graphics Forum 27(4)) and its successor work by Hillaire (2020), "A Scalable and Production Ready Sky and Atmosphere Rendering Technique" (Computer Graphics Forum 39(4)) used in

Unreal's SkyAtmosphere. The original reference implementation is open source:

https://github.com/ebuneton/precomputed_atmospheric_scattering (BSD-3-Clause, ~1,065 stars; the code ships a C++/OpenGL demo and GLSL/HLSL/WGSL shaders).

Licence verified BSD-3-Clause → commercially usable with notice retention.

What it computes: single scattering via a 4D LUT (transmittance x scattering), then a 4D multiple-scattering LUT (Hillaire's improvement uses a smaller 3D "multi-scattering" LUT + a 2D transmittance LUT and is much friendlier to mobile GPUs).

For ZIG: the 2020-style LUT set (2D transmittance LUT, 3D multi-scattering LUT, plus a small sky-view LUT) is the pragmatic mobile target. Cost: 2-3 texture lookups + a handful of exponentials per sky/fog pixel. The LUTs are computed offline (heavy preprocessing is fine – it is a genuine advantage here): you can bake higher-order scattering offline into a bigger LUT than any runtime could afford.

21.2 PER-BODY PARAMETERS (use published values, don't invent them)

Body	Surface/cloud-top pressure	Rayleigh	Mie	Key absorbers	Notes
Earth	1013 hPa	$\sigma_R(\lambda) \propto \lambda^{-4}$, H ≈ 8 km	H ≈ 1.2 km, g ≈ 0.76	O ₃ (Chappuis band – gives the blue sky), H ₂ O	The reference case; ozone is essential for a believable Earth limb/twilight
Venus	92 bar	CO ₂ , denser → stronger	H ₂ SO ₄ droplets, g ≈ 0.7-0.75	SO ₂ , UV absorber	The atmosphere IS the visible surface. Multiple scattering dominates; model as a thick cloud deck, not a clear sky
Mars	~6 hPa	CO ₂ (weak)	dust, g ≈ 0.7-0.8, τ varies 0.2-2+	O ₃ (thin)	Dust storms; blue-ish sunsets are real (dust extinction)
Jupiter/Saturn	H ₂ /He	very weak	NH ₃ /NH ₄ SH cloud haze, g ≈ 0.8	CH ₄ (strong bands)	CH ₄ absorption is what makes Jupiter's belts dark and the limb brown-ish

Body	Surface/cloud-top pressure	Rayleigh	Mie	Key absorbers	Notes
Uranus/Neptune	H ₂ /He	weak	methane haze	CH ₄ (very strong)	Gives the pale cyan/methane blue; Uranus is hazier/flatter
Titan (if ever added)	1.45 bar	N ₂	thick organic haze, $g \approx 0.9$	CH ₄	Orange haze; a classic "wow" moon
Moon/Mercury	~0	none	none	none	Do not add an atmosphere. Optionally add a very faint exospheric glow with a citation
Sun	–	–	–	–	Use a corona/emission model instead (Section 6)

21.3 TWO IMPLEMENTATION ROUTES IN FILAMENT

- (a) Analytic sky LUT shading inside `customSurfaceShading` on the planet shader → produces atmospheric shading of the planet's own surface (aerial perspective toward the limb, terminator softening).
- (b) A separate "atmosphere shell" mesh (a slightly larger sphere, back-face rendered, additive/translucent) sampling the same LUTs → produces the limb glow visible against space (this is mandatory: the atmosphere must be visible *outside* the planet's silhouette, which (a) alone cannot do).

Best practice: do both (a) and (b) with the same parameter set.

Quality tiers: (1) full LUT + shell + Hillaire multi-scattering; (2) single-scattering LUT only; (3) analytic approximation (Rayleigh with a fake horizon gradient) – the last is what looks "cheap" and is exactly what to avoid in the default tier.

21.4 REVISION 3: WHAT THE SINGLE-OBJECT BUDGET UNLOCKS

- FULL-RESOLUTION RAY-MARCHED ATMOSPHERE becomes affordable on flagships. In a multi-body scene the sky pass competes with everything else for fragment throughput; with one object on screen it does not. Recommended tiers:
 - (1) flagship: analytic integration with a 2D transmittance LUT and a 3D multi-scattering LUT sampled per pixel at full resolution, plus a separately rendered atmosphere shell;
 - (2) mid-range: same LUTs evaluated at half resolution and upsampled;
 - (3) low-end: single-scattering only.
- The multiple-scattering LUT can be baked OFFLINE at a resolution that no runtime could afford (heavy preprocessing is explicitly acceptable), which improves the twilight and terminator colour without touching frame time.
- The atmosphere shell should be rendered twice: once behind the planet (so the limb glow is visible against black space) and once in front (so the haze fogs the surface near the limb). That ordering is only cheap to get right when there is a single object to sort against.
- Because there is no other body in the scene, the atmospheric scattering can include the light of the object's own surface bounce if desired: a single extra ambient term driven by the surface albedo, which measurably improves Venus and Titan-class haze realism.

22. PLANETARY RINGS

See Section 13.2 for the Saturn construction. Summary of the full requirements:

- Geometry: a flat annulus (or a slightly displaced multi-layer stack for the main rings) with correct inner/outer radii from published values: Saturn D ring from ~66,900 km, C/B/A out to ~136,780 km (F ring at 140,180 km; G and E much further out and very faint). Uranus's ϵ ring at 51,149 km, Neptune's Adams ring at 62,933 km (with the arcs).
- Texture: derived from PDS optical-depth profiles (COUVIS_8001 / CORSS_8001), converted to a 1D/2D $\tau(r)$ map; $\alpha = 1 - \exp(-\tau/\cos \theta)$. Colour from VIMS/Cassini visible colour. Ringlet-level structure (the B ring has hundreds of density waves and gaplets) is a genuine "premium" detail that a 3,588-byte sprite can never have.

- Shading: two-sided; forward vs back scattering asymmetry (particles are much brighter forward-scattered); the opposition surge/"opposition effect" is a real property of the B ring's particles.
- Shadows: (1) planet casting onto rings (an ellipse projected along the sun direction – analytic), (2) rings casting onto the planet (banded shadows – computed by transforming the sun-ray intersection into ring radius and sampling τ ; this is cheap and looks spectacular), (3) ring-on-ring shading (optional, expensive; skip).
- Physical thickness: the main rings are ~10-100 m thick over 10^5 km diameter – i.e. flatter than paper; never render them as a "torus". Radial "warping" is real but sub-pixel at true scale; the only reason to add thickness is the E ring/G ring.
- E-ring/G-ring: very faint, large, and visible in real Cassini imagery as a forward-scattered haze – optional Layer 2.
- Uranus/Neptune: narrow rings with small optical depths; render as thin annuli with the correct width and the arc structure (Neptune).

22.4 REVISION 3: RINGS AT MAXIMUM QUALITY (no longer a side feature)

With Saturn as the only object on screen, the ring budget can be spent freely.

- RADIAL RESOLUTION: encode the optical-depth profile at 16,384-32,768 texels across the main ring span (74,500 km for D through A). At 32,768 texels that is 2.3 km/texel, which matches the 1-10 km resolution of the Cassini UVIS stellar-occultation profiles - i.e. the texture can be as sharp as the data. Because 32,768 exceeds the guaranteed maximum texture dimension (Section 28.3), store the profile as a 2D array of two 16,384-wide slices, or as an RG16F buffer sampled with a manual index.
- AZIMUTHAL VARIATION: sampling the UVIS profiles per-azimuth rather than collapsing them to a median is what separates a real ring from a gradient. Keep at least a low-resolution azimuthal dimension (for example 256 columns) to carry the B ring's density waves and the transient spokes.
- OPTICAL DEPTH RENDERING: transmission $\alpha = 1 - \exp(-\tau / \mu)$ with μ the incidence cosine, evaluated per pixel. This is one line of shader code and it is what makes the thin C ring visibly translucent and the B ring opaque.
- MULTIPLE SCATTERING WITHIN THE RING: a single extra isotropic term, driven by the same τ , reproduces the observed brightening of optically thick regions and is cheap at this object's polycount.
- SELF-SHADOWING: planet shadow on the rings (analytic ellipse), ring shadow on the planet (transform the sun ray into a ring radius and sample τ), and ring-on-ring occlusion for the densest regions if the budget allows. These three are the highest-value effects on this object because they are pure geometry and therefore stay pixel-sharp at any zoom.
- FORWARD VERSUS BACK SCATTERING: the rings look materially different when viewed from the sunlit versus the shadowed face. Implementing the asymmetry (a phase-dependent multiply on the α and colour) is what makes the object feel observed rather than painted.
- E RING AND G RING: large, faint and forward-scattering; a second thin annulus with a much lower optical depth and no sharp radial structure.

23. SELF-SHADOWING AND LIGHT GEOMETRY

Revision 3 note: this section replaced "Shadows and eclipses". Inter-body eclipses, transits and umbra/penumbra cones required two bodies in one scene and are no longer features of ZIG. What remains is everything that happens WITHIN a single object - which is, conveniently, where most of the visual credibility lives.

23.1 THE FIVE EFFECTS THAT MATTER, IN ORDER OF VALUE

1. TERMINATOR AND LIMB: the day/night boundary. Free from N.L for airless

bodies; for atmospheric bodies the terminator must be softened by the scattering pass so light wraps past the geometric terminator. Getting the terminator right is the single strongest realism cue on any planet.

2. TERRAIN SELF-SHADOWING at grazing sun. On the Moon, Mercury and Mars this is decisive: at a 5-10 degree sun elevation, crater interiors go black and only the rims catch light. Without it a displaced Moon looks like a bumpy ball.
3. RING SHADOWS (Saturn, Uranus, Neptune): the planet's shadow on the rings and the rings' shadow banded across the planet. Both are analytic, both are pixel-sharp at any zoom, and both are unmistakable in real photographs.
4. LIMB DARKENING on the Sun, and limb brightening/haze on the gas giants.
5. LOCAL OCCLUSION on the ISS (modules shadowing each other) and inside NAC hero patches.

23.2 TECHNIQUE: ANALYTIC FIRST, SHADOW MAPS LAST

Every astronomical body here is a sphere or an ellipsoid and every ring is a plane, so shadowing does not need shadow maps at all: a ray/sphere or ray/plane intersection in the fragment shader is cheaper and sharper than any shadow map, and it never suffers from cascade artefacts across zoom levels.

- Sun rays are parallel at these distances, so the maths reduces to a direction vector plus two intersections.
- Filament's shadow-map options (PCF, PCSS, DPCF, VSM) are therefore reserved for the one object that is genuine geometry: the ISS, and for hero terrain patches where a local shadow map is denser than an analytic test.

23.3 TERRAIN SELF-SHADOWING WITHOUT A SHADOW MAP (the high-value trick)

Because the height field is known offline and preprocessing is unlimited, bake a HORIZON MAP per body: for each texel, the maximum terrain elevation angle in N azimuthal directions (N = 8 or 16 is plenty). At runtime the shader compares the current sun elevation against the horizon profile in the sun's azimuth and shadows the texel if the sun is below it.

- Cost: 8 bytes per texel for an 8-direction R8 horizon map. At 4096 x 2048 that is 8 MB; at 8192 x 4096, 32 MB - trivial in a single-object budget.
- Benefit: correct, stable, zoom-independent shadows at any sun angle with no shadow-map resolution problem, which is exactly the failure mode users notice when they sweep the sun across a crater field.
- The same map drives ambient occlusion (sky visibility), so craters get darker floors for free.
- Bake it from the same DEM used for displacement, so shadows and relief agree.

23.4 THE SUN-DIRECTION CONTROL (the feature that replaces orbits)

With no ephemeris there is no "correct" sun position, so make the sun an explicit user control - and make it physical:

- Controls: azimuth 0-360 deg, elevation -90 to +90 deg, and an "exposure (EV)" slider. Presets: high noon, quarter phase, crescent, grazing sun (2-5 deg), north-pole-on, south-pole-on.
- Relative brightness is still physical. Each object pack stores its real distance from the Sun, and the directional light is scaled by $1/r^2$, so Mercury is blown out and Neptune is nearly dark at the same exposure. This is the honest way to show that space is not evenly lit. Relative solar irradiance (Earth = 1): Mercury 6.7, Venus 1.9, Earth 1.0, Moon 1.0, Mars 0.43, Jupiter 0.037, Saturn 0.011, Uranus 0.0027, Neptune 0.0011.
- Presets to ship per object: "as photographed" (a sun angle chosen to match a famous real image of that body, cited), "noon", "grazing", "night side".
- EARTHSHINE, retained in a reduced and labelled form: for the Moon, an optional secondary fill light representing sunlight reflected from a full Earth, at the correct order of magnitude (Earth's albedo 0.30). Since there is no orbit simulation, the app offers it as a labelled approximation rather than as a computed phase.

23.5 WHAT WAS DELETED, AND WHY IT DOES NOT COST REALISM

- Inter-body eclipses and transits: required two bodies; a solar eclipse is a relationship, not a property of the Moon.
- Umbra and penumbra cones: same reason.
- Earth shadowing the Moon: same reason; the Moon's appearance in eclipse is instead offered as a labelled lighting preset, "eclipse-like illumination (deep red)", with the colour driven by refraction through Earth's atmosphere rather than by a second object.
- Nothing about a single object's own appearance is lost by these removals.

24. SCIENTIFIC DATA SOURCES (intrinsic object data)

Revision 3 note: this section previously described an ephemeris backbone (SPICE kernels, Horizons validation, baked trajectory tables). ZIG has no orbits, so all of that is out of scope. What remains is the data that describes an object's own appearance - which is exactly what an inspector needs.

24.1 THE INTRINSIC CONSTANTS (authoritative, small, and worth getting exactly right)

Category	What it covers	Authoritative source
Physical dimensions	radius, equatorial/polar radii, oblateness	IAU Working Group on Cartographic Coordinates and Rotational Elements (WGCCRE) reports; NASA planetary fact sheets
Rotation	sidereal rotation period, direction (retrograde flags), rotation-axis orientation	IAU WGCCRE rotation elements (alpha0, delta0, W rate + terms)
Orientation	axial tilt and pole direction in inertial space	IAU WGCCRE
Atmosphere	composition, scale heights, scattering coefficients	published per-body values (see 21.2)
Rings	inner/outer radii, width, normal optical depth, particle properties	PDS Ring-Moon Systems Node; Cassini UVIS/RSS profiles; published ring atlases
Surface reflectance	geometric albedo, roughness anchors	published photometry (see 20.3)
Imagery and relief	the maps and DEMs per object	USGS Astrogeology, NASA SVS, PDS, Murray Lab, NASA Visible Earth, Landsat

Implementation note: the rotation period is the one ephemeris-adjacent quantity that stays, because it drives the spin animation (Section 26.4). Store it as a single scalar per object from the IAU report - no runtime kernel, no date maths.

24.2 WHAT ZIG DELIBERATELY DOES NOT NEED

- SPICE kernels and the SPICE toolkit: not required at runtime. They remain useful only as a reference for the rotation elements above.
- JPL Horizons and any orbital propagation: out of scope by design.
- Any per-date imagery pipeline: the app ships fixed, labelled epochs rather than a clock. This removes an entire class of maintenance (and an entire class of ambiguity about "what date am I looking at").

24.3 COLOUR AND PROVENANCE DISCIPLINE (unchanged, and now more important)

Because ZIG claims fidelity, every map shipped must be classified in the data manifest and shown to the user in the same terms:

- "True colour": RGB channels map to stated visible wavelengths combined with a documented transform (for example JPL's statement that the Cassini Jupiter map's colours are close to human-eye appearance).
- "Approximately true colour": filter composites, for example OPAL's 631 / 502 / 395 nm channels.
- "Enhanced / false colour": Mercury's enhanced-colour mosaic, any UV or EUV image (including the SDO channels and the Carrington map), Venus's synthetic C3-MDIR colourisation, any infrared day-temperature mosaic.
- "Radar": all Magellan surface imagery and any Titan radar product.
- "Simulated / visualisation": Solar System Scope textures, the CGI Moon Kit colour map (a Hapke-normalised composite, not a photograph), the SVS deep star maps, all procedural layers (granulation, ring turbulence, cloud banding).

- "Procedural beyond data": any detail the app invents below the published resolution of the source (Section 2.5 makes this a visible HUD state).

24.4 DATA WEIGHTS, STATED PER OBJECT

For each shipped texture the manifest records: source product name, publisher, licence, the pixel resolution, the real ground resolution it encodes (m/px or km/px), the wavelength or band meaning, the processing applied, and the URL. Appendix B lists this per object button.

25. OPEN-SOURCE RENDERING CODE (per-project assessment)

Format per project: name · URL · licence (verified) · stars/forks · activity · language/API · what it does · legal reuse · Android portability · useful modules/files · what NOT to copy · production vs educational.

- 1) Google Filament – <https://github.com/google/filament>
 Apache-2.0 (verified) · 20,489 stars · pushed 2026-09-13 (very active) ·
 C++ core with Java/Kotlin/JS/Swift bindings · OpenGL ES 3.0+, Vulkan 1.0,
 Metal, GL 4.1+, WebGL2/WebGPU.
 What it does: production real-time PBR renderer: clustered forward lighting,
 IBL (cmgen), physical light units, HDR bloom, DOF, SSAO, shadows (PCF/PCSS/
 DPCF/VSM), TAA/FXAA/MSAA, tone mappers (ACES/AgX/PBR Neutral/filmic),
 grading, gltfio loader with KTX2/Basis + Draco/meshopt providers, matc
 offline material compiler.
 Legal reuse: full, including in a proprietary app, with notice retention and
 no trademark use. Android portability: it IS the Android choice.
 Useful modules: libs/filament (engine), libs/gltfio (glTF + KTX2/Draco),
 libs/math, tools/matc, tools/cmgen, tools/mipgen, docs/Materials.md.
 What NOT to copy: nothing is off-limits, but avoid vendoring the whole repo
 into your own source tree (use the Maven artifacts and only fork what you
 must patch).
 Production/educational: production.
- 2) SceneView – <https://github.com/SceneView/scenview> (canonical; verified
 Apache-2.0, 1,316 stars, pushed 2026-09-12; Maven io.github.scenview:
 scenview 2.3.x)
 What it does: Compose-native node/composable API over Filament; model,
 material, environment and IBL loaders; gestures; geometry generators;
 post-processing (bloom/DOF/SSAO/vignette/colour grading/tone mapping) per
 its README.
 Legal reuse: full (Apache-2.0). Android portability: it is Android/JVM.
 Useful modules: scenview-core (Filament bindings, node types),
 the Compose wrappers, the loaders, the camera/gesture delegates.
 What NOT to copy: its higher-level assumptions (single-scene/single-context,
 simplified camera) if they conflict with a floating-origin multi-scale
 astronomy scene – and do not let it lock you out of custom .filamat
 materials or custom RenderTargets. Educational/demo grade in places:
 verify every feature you rely on. Production: production-leaning library,
 younger than Filament.
- 3) Eric Bruneton, black_hole_shader – see Section 18.3 #1.
 BSD-3-Clause · 240 stars · last push 2020-11-11 · GLSL/JS.
 Reuse: yes (notices). Portability: high (GLSL ES 3.0 class maths).
 Useful files: the precompute stage and the GLSL deflection/disk shading.
 Do not copy: the gaia_sky_map data pipeline (Gaia is non-commercial) or
 demo assets. Reference implementation, not a library.
- 4) Eric Bruneton, precomputed_atmospheric_scattering –
https://github.com/ebuneton/precomputed_atmospheric_scattering
 BSD-3-Clause (verified) · ~1,065 stars · GLSL/HLSL/WGSL/C++.

- Reuse: yes. Portability: high. Useful: the LUT generation + the shader sampling maths; the newer Hillaire-style 3D multi-scattering LUT is the mobile-friendly target. Do not copy: the demo's terrain/water. Production: research reference of exceptional quality.
- 5) KhronosGroup/KTX-Software – <https://github.com/KhronosGroup/KTX-Software>
 Licence per its GitHub API: NOASSERTION (the API returned no SPDX id; the project's own documentation describes Apache-2.0, and there is an explicit patent grant discussion in its docs). ACTION REQUIRED: read the repository's LICENSE file and confirm the exact text and any bundled third-party licences (basis_universal, Zstd, etc.) BEFORE relying on it. Mark this "VERIFY BEFORE SHIPPING" rather than assuming Apache-2.0 here.
 What it does: toktex/toktx (create/transcode KTX2 + Basis Universal + UASTC + supercompression), critical for the texture pipeline. Android portability: it is a CLI tool, run it in your build pipeline (no Android runtime needed; the runtime part is the KTX2 loader inside Filament).
 - 6) google/draco – <https://github.com/google/draco> – Apache-2.0 (verified by convention and by Filament's bundled third-party notices). What it does: mesh compression used by the glTF KHR_draco_mesh_compression extension. Note: Draco can hurt GPU cache locality at runtime (decode cost + non-optimised index order). Prefer meshopt (below) for most ZIG content.
 - 7) zeux/meshoptimizer – <https://github.com/zeux/meshoptimizer> – MIT.
 What it does: mesh simplification (good LOD generation), vertex/index cache optimisation (real FPS win on mobile), overdraw reduction, and EXT_meshopt_compression. Recommended for ZIG's model pipeline: simplify → optimise → compress.
 - 8) Blender – <https://www.blender.org/> – GPL-2.0-or-later for the *application*. Important legal clarity: the GPL applies to the Blender program, NOT to your assets or your app. You may use Blender to produce commercial assets without any obligation. (This is a common misunderstanding; state it so nobody in the team worries.)
 - 9) Stellarium – <https://github.com/Stellarium/stellarium> – GPL-2.0 (verified, 9,981 stars). Reference only: star catalogues and constellation data are GPL-encumbered as distributed; do not ship its files. You may study its algorithms. Educational/reference for ZIG.
 - 10) Celestia – <https://github.com/CelestiaProject/Celestia> – GPL-2.0 (verified, 2,360 stars). Reference only: its algorithms (planetary texture generation, orbital mechanics) and its asset pipeline are instructive; its textures are third-party with mixed licences. Do not copy code or assets.
 - 11) Gaia Sky – <https://github.com/langurmonkey/gaiasky> – MPL-2.0. Reference only for architecture (it is a Java/libGDX astronomy engine with a proven catalogue pipeline). MPL-2.0 is file-level copyleft: copying a file obliges you to publish that file's source. Simplify: study, don't copy.
 - 12) rantonels/starless – <https://github.com/rantonels/starless> – GPL-3.0 (verified, 714 stars). Excellent black-hole physics reference; GPL-3.0 forbids proprietary incorporation. Cite as literature.
 - 13) NASA 3D Resources – <https://science.nasa.gov/3d-resources/> – not code, but the asset source (see Section 17). Individual models have credit lines.
 - 14) three.js – <https://github.com/mrdoob/three.js> – MIT. Reference only for ZIG (it is what EOSS uses). Useful if you need to cross-check visual results, or for an internal web-based tool.
 - 15) s0xDk/ghostty-blackhole – MIT (verified) · 1,668 stars · single-file GLSL Schwarzschild shader with exact-Binet integration and an analytic weak-field blend (used for a terminal background effect). Reuse is permitted under MIT, but it is a *stylised* shader, not a physically curated renderer; its main value is showing a very cheap step budget (N_STEPS) with an analytic tail.
 - 16) sirxemic/Interstellar – MIT (verified) · 375 stars · WebGL wormhole/black

- hole. Reuse permitted for the code; verify the texture credits individually.
- 17) zongzhengli/black-hole – CC-BY-4.0 · 3 stars · adaptive-step Schwarzschild ray tracer. Reuse permitted with attribution.
 - 18) KleaSCM/tika – MIT · C++/GLSL Kerr ray tracer · use for validation/hero renders.
 - 19) hydrogendeuteride/BlackHoleRayTracer – MIT · 41 stars · OpenGL 4.3 compute ray marcher · desktop reference.
 - 20) ImanetorX98/KerrTraceCpp2 – MIT · Kerr/Kerr-Newman ray tracer with ray-bundle mode · offline ground truth.

LEGAL REUSE SUMMARY RULE (revision 2): for a proprietary ZIG, permit MIT/BSD/Apache-2.0/CC0 and CC BY where the attribution burden is acceptable; treat MPL-2.0 as "file-level copyleft" (avoid copying files); treat GPL-2.0/-3.0 code as unusable in the binary; treat CC BY-NC / CC BY-NC-SA / CC BY-SA assets as unusable unless you accept the corresponding obligations; and treat a repository with NO LICENCE FILE as ALL RIGHTS RESERVED, not as public domain. The common belief that "no licence file means no copyright" is incorrect: copyright is automatic on creation, and no registration, notice or licence file is required for it to subsist. See Section 29.1 rule R0 for the authoritative statements, and Section 18.7 for the three lawful routes to a licence-free reference.

26. ASSET CONVERSION AND PREPROCESSING PIPELINE (A-J)

PHASE A – ASSET ACQUISITION (offline, on a workstation, not on the phone)

- A1. Planetary rasters: download from USGS Astropedia / planetarymaps.usgs.gov (all URLs in Sections 6-15). Use `curl -L` (302 → S3) and verify with `file`/`gdalinfo` before trusting (I verified Venus GTDR this way: TIFF LE, 8192x4096, 16 bps, uncompressed).
- A2. Models: NASA 3D Resources GLBs (ISS). Verify with a parser (I verified ISS_stationary.glb: glTF 2.0, 131 meshes, 247,547 tris) and record the hash + credit line in a provenance manifest.
- A3. Imagery for heroes: HiRISE DTMs, LROC NAC, OPAL maps (record observation IDs, filters, dates).
- A4. Every asset gets an entry in a machine-readable provenance manifest (JSON) with: name, source URL, download date, licence, credit string, calibration notes (colour space, bands, processing level). Ship the manifest inside the APK and show it in-app. This is your legal and scientific paper trail.

PHASE B – GEOMETRIC / PHOTOMETRIC PREPROCESSING

- B1. Reproject to simple cylindrical (equirectangular) if not already; unify on longitude convention per the IAU WGCCRE report (note the 0-360 vs -180/180 differences and the planetographic/planetocentric latitude split – Jupiter's longitude convention, for example, is a classic trap).
- B2. Radiometric normalisation: linearise (most PNG/TIFF are sRGB or gamma-encoded; many scientific products are linear DN). Build a documented transfer curve per asset; for SVS CGI Moon Kit use their stated Hapke/colour normalisation rather than re-inventing it.
- B3. Mosaicking/registration for multi-product composites (e.g. CTX luminance + Viking colour). Do the registration at high resolution and downsample once (avoid repeated resampling).
- B4. Seam removal: for global mosaics, fix the 0°/360° seam and pole distortions (polar rows compress 360° into a few pixels; either oversample the polar caps or use a cube-sphere mapping).
- B5. Altimetry → height/normal/displacement: convert DEM to metres relative to the reference radius, then generate (i) a 16-bit height map for displacement, (ii) a tangent-space normal map via Sobel on the *metric* height field (not pixel units), (iii) an optional low-frequency ambient occlusion bake.

PHASE C – TEXTURE CONVERSION (the quality-determining step)

- C1. Downsample with a high-quality filter (Lanczos-3 or better) in linear light to the target dimension: 8192x4096 for hero bodies, 4096x2048 for mid-tier, 2048x1024 for small bodies (choose by the body's angular size on a phone screen, not by the source size).
- C2. Build the full mip chain in linear space (this is where ordinary pipelines go wrong; mips of an sRGB texture must be generated in linear then re-encoded).
- C3. Choose the container/format:
 - * Base colour: KTX2 + Basis/UAStC (transcoded to ASTC at install/load) or direct ASTC (6x6 or 8x8) + KTX2 wrapper.
 - * Normal maps: ASTC (6x6 is usually enough; 4x4 for hero close-ups) or UAStC for maximum fidelity, then transcode.
 - * Height/masks: R8/R16 or ASTC (mask channels compress poorly in block codecs; if you see artefacts, use ETC2 RGBA or a separate R8 with mips and accept the size).
 - * HDR (star map, disk colour LUTs): RGB9_E5 or R11F_G11F_B10F in KTX2 (ASTC HDR is not broadly supported); for LUTs use RG32F/RG16F.
- C4. Record the colour space in the manifest (sRGB vs linear) – mismatched colour space is the most common cause of "why does my planet look washed out".

PHASE D – MODEL OPTIMISATION (for the few true models: ISS, rovers, spacecraft)

- D1. Import the GLB into Blender; verify units and scale against known dimensions (ISS truss 109 m) and fix the transform ONCE.
- D2. Split into LODs with meshoptimizer (e.g. 100%, 50%, 25%, 10%, plus an impostor billboard for the smallest tier).
- D3. Optimise vertex cache/index order (meshopt), then optionally apply EXT_meshopt_compression or Draco.
- D4. Convert all textures to KTX2 (see Phase C) and rewrite materials to a small number of Filament-compatible material variants (a single PBR variant with a texture atlas beats 28 unique materials for draw-call reasons).
- D5. Bake a low-res irradiance/ambient occlusion where it helps; keep the original high-poly as a build-time asset only.

PHASE E – PACKAGING (Android)

- E1. Ship large assets via Play Asset Delivery install-time asset packs (up to 1.5 GB per pack; 4 GB cumulative for install-time packs; Play requires the base module ≤ 200 MB compressed – with APK size limits this means the base app should stay ~100-200 MB and everything heavy goes in packs). Verify the current limits in the Play Console docs at build time, since they change. No compression of already-compressed KTX2 (set noCompress for ktx2/glb/astc inside the pack) – double compression wastes time and CPU.
- E2. Structure the packs by scene so a user can download the solar system and the deep-sky pack separately.
- E3. Keep an assets manifest with SHA-256 hashes; verify on first launch.
- E4. Total realistic budget: base 150 MB + planetary pack 1.2-2.0 GB + extras. Because the phone is the target and quality is the priority, this is fine.

PHASE F – RUNTIME LOADING

- F1. Load KTX2 (Basis-transcoded or ASTC direct) with Filament's Ktx2Provider / gltfio; load GLBs with AssetLoader; do all of this off the main thread (SceneView's async loaders or your own coroutine dispatchers).
- F2. Virtual texturing (Filament supports virtual texturing via Texture::Builder + .stp/.ktxst streaming textures generated by `cmgen`/`filamat`-adjacent tooling): use it for the Earth surface, Mars CTX, Moon NAC patches and any other gigapixel source, so that only visible tiles are resident. This is THE technique that makes "5 m/px Mars, locally" possible.
- F3. Eviction policy: LRU by distance/visibility with a hard VRAM budget (e.g.

1.5-2.5 GB on a 12 GB device) and a GPUMEM threshold that triggers downsampling of the LOD bias rather than a crash.

- F4. Pre-warm: on the loading screen, upload the shader variants and the first-frame textures to avoid hitching (Filament's material compilation is offline, so the remaining hitch source is texture upload – use mip upload ordering and, if needed, a small number of synchronous uploads at load).

PHASE G – RENDERING ARCHITECTURE

- G1. Floating origin: keep the camera near the world origin and express every body's position relative to it in double precision on the CPU, uploading fp32 offsets. Without this, surface-level views jitter.
- G2. Draw-call discipline: instanced star sprites (1 draw), one draw per planet (plus atmosphere shell, clouds, rings), a handful for models – a total of 10-30 draws is a good target.
- G3. Pass order: sky/deep-map background → star sprites → planets (opaque) → emissive layers → translucent (clouds, atmosphere shells, rings) → the relativistic pass (if in a black-hole scene, ideally as the sky background, since it is the background) → post (bloom → tone map → grade → TAA/dither → UI overlay).
- G4. Precision: keep the far plane large but do not rely on depth for interplanetary distances; render bodies as separate "local scenes" with their own near/far when close, or use a logarithmic depth buffer.

PHASE H – LIGHTING

- H1. Sun as a physical directional light (Section 20.1) with per-scene exposure.
- H2. Analytic secondary lights where physically justified (earthshine, planet-shine on moons; ring-shine on Saturn; the accretion disk's own light in the black-hole scene).
- H3. Atmosphere and sky LUTs computed offline (Phase A/B) and shipped as KTX2.
- H4. Emissive: Earth night, the Sun, aurorae, the accretion disk – all in HDR.

PHASE I – CAMERA

- I1. Modes: overview (solar-system-scale), approach/target (body-centred), orbit (body-relative), surface (camera at altitude with a terrain-tangent frame), chase (spacecraft/ISS), and free flight.
- I2. Controls: orbit (drag), zoom (pinch/FOV), bank/roll for cinematic, focal length presets, and a "cinematic" auto-camera for showcases (documented cinematic, not "simulation").
- I3. Precision/UX: a true "distance to target" HUD, the body's angular size, the local time, and the current exposure value. This is the "premium presentation" layer EOSS does well and it will make ZIG feel authoritative.

PHASE J – INTERACTION

- J1. Selection: ray-pick bodies by their *rendered* disc (including the atmosphere/ring silhouette), not by a bounding sphere, so tapping a ring works.
- J2. Labels: screen-space labels with occlusion tests against body/ring silhouettes, distance-based decluttering, and a stable sort (labels are one of the hardest things to get right; budget UI time).
- J3. Time control: a scrubber with real date/time, "now", and speed presets; every dynamic element (rotation phases, moon positions, ring shadows, ISS position, eclipse geometry, BH disk rotation) must be a pure function of the simulation clock – never of wall-clock animation. This single rule is what separates a simulation from an animation.
- J4. Deep links/share: "show me Mars on 2033-04-11" as a shareable state; cheap to build, high perceived value.

26.4 REVISION 3 DELTAS TO PHASES A-J (single-object inspector)

The A-J phases remain the spine of the pipeline, with four changes and one new phase.

SUPERSEDED: PHASE G1 (floating origin). Floating-origin camera maths existed to keep precision across solar-system distances. With one object per screen there is no long-range translation, so it is replaced by a simpler and stricter scheme: every object is normalised to UNIT RADIUS at load (planet radius = 1.0), the camera works in object radii, and near/far planes are set from the current zoom (near = 0.001 radii when inspecting a 0.5 m/px NAC patch, far = 8 radii when framing the whole body). Double precision is used only during the load-time normalisation, never in the frame loop. This removes an entire class of precision bugs and makes zoom-to-surface numerically boring.

SUPERSEDED: PHASE J3 (simulation clock). Replace the date scrubber with two object-local controls: (a) ROTATION - spin the object at its real sidereal period, with presets 1x, 60x, 3600x and "hold", and (b) SUN DIRECTION - the azimuth/elevation control from Section 23.4. Every dynamic effect in the viewer (cloud drift, ring shadow position, terminator, specular glint) must be a pure function of these two controls and nothing else. There is no wall-clock animation anywhere in the app.

AMENDED: PHASE E (packaging) - the pack unit is now the BUTTON. One object = one asset pack, install-time for the core bodies and on-demand for the heavy ones (Section 27.4). The base module ships a low-resolution version of every object so that all thirteen buttons work immediately with no download.

AMENDED: PHASE F (runtime loading) - because only one object is resident, the loader can be aggressive: load the object's base mips, show it immediately, then stream the tile pyramid in the background while the user is already orbiting. Time-to-first-pixel and time-to-full-detail are now separate, measurable targets (suggested: under 400 ms and under 6 s respectively for a hero object on a flagship).

NEW: PHASE K - THE LOCAL TILE STORE (the core of the inspector)

- K1. Offline, build a mip-like pyramid of ASTC tiles for each object from the deepest source data (for example 512 x 512 texel tiles with a 4-texel apron to prevent seams, at every level down to the native resolution of the source).
- K2. Store the pyramid on device - inside the asset pack, or in app storage for on-demand packs - never fetched over the network.
- K3. At runtime, maintain a tile cache keyed by (object, level, tile) with an LRU eviction policy and a hard VRAM ceiling (Section 28.2).
- K4. Choose tile level per frame from the projected texel density of each screen region, so a 0.5 m/px NAC patch at maximum zoom pulls only the tiles it renders, and the far side of the object pulls nothing at all.
- K5. Prefetch along the camera's motion vector and the zoom direction, so inspection feels continuous rather than tiled.
- K6. Decode on a worker pool: KTX2 tiles are decoded/transcoded off the render thread and uploaded asynchronously; never block the frame on I/O.
- K7. Budget: for a hero object, plan for 200-800 MB of resident tiles and 0.5-3.5 GB on disk (Appendix B gives the per-object numbers).

27. ANDROID LOCAL PACKAGING STRATEGY

27.1 THE HARD CONSTRAINTS (verify current numbers at build time)

- APK/AAB base module: keep well under 200 MB compressed (Play's per-module limit; the historical limit was 150 MB for AAB base, currently Play allows up to 200 MB per module with 4 GB total install-time size).
- Play Asset Delivery: install-time/fast-follow/on-demand packs, up to 1.5 GB each; install-time packs count toward the "max installed size" (4 GB at time of writing). This is exactly the mechanism a 2-4 GB astronomy app needs.
- If Google Play distribution is not essential, you can also ship an APK + OBB or a direct-download APK, but Play Asset Delivery is the well-behaved route.

27.2 RECOMMENDED PACK SPLIT

- Pack 0 (base, install-time, ≤ 200 MB): code, materials (.filamat), LUTs, low/mid-resolution textures for every body (so the app is fully functional offline immediately), font/UI, provenance manifest.
- Pack 1 (install-time, ~600-900 MB): high-resolution planetary surfaces (Earth 8K, Moon 8K + normal, Mars 8K + MOLA-derived normal, Mercury 8K,

Venus 4-8K, giants 4K), atmosphere/scattering LUTs.

Pack 2 (install-time, ~400-800 MB): gigapixel virtual-texturing pyramids – Earth VIIRS-derived high-res; Mars CTX selection; Moon LROC selection; hero-site HiRISE/LROC NAC patches; the .ktxtst/.stp page sets.

Pack 3 (on-demand, ~200-600 MB): spacecraft/ISS high-res + rovers + historic probes (3D Resources), plus extra deep-sky imagery.

Pack 4 (on-demand, ~100-300 MB): the black-hole scene's high-order LUTs, disk textures, and any cinematic extras.

27.3 THE RULE THAT MAKES THIS WORK

The app must be fully functional with Pack 0 only (e.g. when a user installs on a device with limited storage), gracefully upgrading when the packs are present. Never make a core feature depend on an on-demand pack.

27.4 THE BUTTON-TO-PACK MAP (the packaging plan for the inspector)

Each of the thirteen buttons maps to one asset pack. The base module must ship a usable low-resolution version of EVERY object, so the app is complete and offline the moment it is installed; the packs then upgrade specific objects to maximum fidelity. Play Asset Delivery supports install-time packs (up to about 1.5 GB each, counted against roughly a 4 GB installed-size ceiling) and on-demand packs of the same size, which is exactly the right primitive for this product.

Button	Base module (always present)	Install-time pack	On-demand pack
Sun	2K procedural base + AIA 304 disk	8K procedural params + 10 AIA frames	SDO frame set, corona
Mercury	8K BDR downsample	16K BDR + 8K DEM	full 166 m/px pyramid (about 2.5 GB)
Venus	8K radar + cloud shader	8K topography, Fresnel, slope	17% stereo patch set
Earth	8K Blue Marble, 4K night	12 monthly BMNG, 8K night, clouds	Landsat hero tiles (0.6-1.4 GB)
Moon	8K WAC + 4K LOLA	16K WAC + 8K LOLA + normals	100 m/px global pyramid (about 3.5 GB) and NAC patches
Mars	8K MDIM	16K CTX-derived + 8K MOLA	CTX hero regions (1-3 GB) and HiRISE patches
Jupiter	2K OPAL	4K OPAL + wind LUT	per-epoch OPAL maps
Saturn	2K body + coarse rings	4K body + 16K ring profile	32K ring profile + E-ring haze
Uranus	2K map	4K map + ring set	historic and corrected epoch pair
Neptune	2K map	4K map + ring set	Voyager 2 epoch mosaics
Milky Way	4K skybox + 200k stars	8K skybox + 2M stars + constellations	16K skybox, deep catalogue
ISS	64k-triangle LOD	full 247k-triangle model + KTX2 textures	hero module textures
Black hole	LUTs + star sky + disk	full-quality LUTs and textures	cinematic presets and extras

Rules that make this work:

- The base module stays under about 200 MB compressed and must render all thirteen objects at a presentable quality, offline, with no packs installed.
- No object's core appearance may depend on an on-demand pack.
- The credits screen lists which sources are actually installed, so the attribution grows with the content rather than promising data the user does not have.
- Pack contents are hashed (SHA-256) and verified at first launch.

28. PERFORMANCE AND MEMORY ANALYSIS (single-object budgets)

Revision 3 note: this section is rewritten. Wide multi-object scenes forced conservative budgets; a one-object inspector does not, and the numbers below reflect that. It also corrects an arithmetic error carried in revision 1 and 2 (see 28.2).

28.1 THE HARD LIMITS ON ANDROID

- Texture format support: ETC2 is available on over 95% of active devices and ASTC on over 80% per Google's own guidance. Ship ASTC as primary.
- Maximum texture dimension: OpenGL ES 3.0 guarantees only 2048. In practice most modern flagships report 8192 or 16384, but it is NOT safe to ship a single 16384-wide texture as a hard requirement. Rule: treat 8192 as the

guaranteed single-texture width, and express anything wider as tiles, a sampler2DArray slice set, or virtual texturing.

- Memory: mobile GPUs share system RAM. Budget the whole app (Java/Kotlin heap, native, GPU textures) within about 1.5-3 GB on a flagship, and keep resident GPU textures under roughly 1.5-2 GB to stay clear of the low-memory killer.
- Draw calls: this product needs 5-30 per frame. There is no excuse for more.
- Fill rate, not triangle count, is the currency of an inspector: the expensive pixels are the atmosphere, the rings, the displacement shading and the black hole. Triangles are cheap (see 28.4).
- Thermals and battery: sustained 60 fps for ten minutes is a different target from a 3-second benchmark. Use Android's thermal status API to step down effect tiers automatically.

28.2 CORRECTED MEMORY ARITHMETIC (this table is authoritative)

ASTC payload: 4x4 block = 128 bits per 16 texels = 8 bits/texel = 1.0 B/texel;
6x6 = 128/36 = 3.56 bits/texel = 0.444 B/texel; 8x8 = 2 bits/texel = 0.25 B/texel.

Mip chain adds about 33%. A spherical 2:1 equirectangular texture is what ZIG uses everywhere.

Resolution | Texels | ASTC 4x4 (+mips) | ASTC 6x6 (+mips) | ASTC 8x8 (+mips)

4096 x 2048	8.4 M	8.4 MB (11.2)	3.7 MB (5.0)	2.1 MB (2.8)
8192 x 4096	33.6 M	33.6 MB (44.7)	14.9 MB (19.9)	8.4 MB (11.2)
16384 x 8192	134.2 M	134.2 MB (178.6)	59.6 MB (79.5)	33.6 MB (44.7)

CORRECTION: revision 1 and 2 stated "8K ASTC 6x6 about 22.3 MB" for the Earth day map. The correct figure is 14.9 MB (19.9 MB with mips); the earlier number was an arithmetic slip. Every other budget statement in this report used the same method and should be recomputed from the table above rather than from that figure.

28.3 THE 16K PROBLEM, AND HOW TO SHIP 16K ANYWAY

A single 16,384 x 8,192 texture is 134 MB of texels - acceptable in a one-object budget - but it may exceed the device's maximum texture dimension.

Three supported solutions, in order of preference:

1. TILE IT: split the equirect into 4 or 8 ASTC tiles of at most 8192 wide and treat them as a mip level of the pyramid. This is also exactly what the tile store wants (Phase K), so it costs nothing extra.
2. ARRAY IT: sampler2DArray with 4-8 slices, indexed by a computed slice in the shader. Filament supports sampler2DArray parameters.
3. VIRTUAL TEXTURE IT: keep tiles resident only where the camera looks; this is what makes the 92,160-wide Mercury BDR and the 109,152-wide lunar WAC mosaic inspectable at all.

The rule for the shader author: never assume a texture dimension above 8192, and never assume a texture is resident.

28.4 WHAT ONE OBJECT BUYS (multi-body budget versus inspector budget)

Resource	Multi-body scene (rev 2 plan)	Inspector (rev 3)	Why
Resident albedo	8192 x 4096, ASTC 6x6 (20 MB)	16384 x 8192, ASTC 4x4 (179 MB)	one object, four times the linear resolution, four times the quality per texel
Resident normal	4096 x 2048 (5 MB)	8192 x 4096 ASTC 6x6 (20 MB)	grazing-sun relief is now a headline feature
Height field	4K, streamed	8-16K plus offline horizon map (32 MB)	self-shadowing at grazing sun
Triangles	under 150k per body	0.5-4M for the focused body	20-25x the geometry, spent only where the user is looking
Atmosphere	half-resolution LUT pass	full-resolution, plus offline-baked multi-scatter LUT	fill rate is no longer shared
Post-processing	bloom + tone map	bloom, tone map, TAA, optional 1.25-1.5x supersample	one object, one pass
Draw calls	20-40	5-30	less sorting, fewer state changes
Resident total	about 250-400 MB	300 MB - 1.5 GB (streamed tiles)	comfortably inside the VRAM ceiling

28.5 QUALITY TIERS (automatic, never changing scientific content)

Tier 0 (flagship): 16K albedo, 4M-triangle displacement, full-res atmosphere and rings, TAA plus optional supersampling, 60 fps.

Tier 1: 8K albedo, 1M triangles, half-res atmosphere upsample, no supersampling, 60 fps.

Tier 2: 4K albedo, normal-mapped sphere without displacement, single-scattering atmosphere, 30-60 fps.

Tier 3 (fallback): 2K albedo, static lit sphere with normal map, no atmosphere shell, 30 fps.

Across all tiers the ephemeris-free facts are identical: radii, oblateness, tilt, rotation, colours, labels and credits. Quality degrades; the science does not.

28.6 TARGETS TO HOLD THE TEAM TO

- Time to first pixel after a button tap: under 400 ms (base mips only).
- Time to full detail for a hero object: under 6 s on a flagship.
- Steady-state frame time: 16.6 ms at Tier 0, with no frame over 33 ms while tiles stream (prefetch, do not hitch).
- Total resident GPU memory: under 1.5 GB at Tier 0, under 700 MB at Tier 0 for any object whose pack is smaller than 150 MB.
- Zoom range: the camera must be able to reach the object's data ceiling (2.4) and no further without a labelled procedural step - verified per object in the test matrix.

29. LICENSING AND LEGAL SAFETY

29.1 THE DECISION RULES (apply mechanically, in this order)

R0. NO LICENCE FILE AT ALL. This rule is placed first because it is the most commonly misunderstood. A repository, dataset or asset with no licence file is NOT public domain and is NOT free to use. Copyright subsists automatically from the moment a work is fixed in a tangible medium; no registration, no copyright notice and no licence file is required. Absence of a licence means the author retains all exclusive rights (reproduction, derivative works, distribution) and you have no permission.

Authoritative statements, quoted:

- * GitHub Docs, "Licensing a repository": "You're under no obligation to choose a license. However, without a license, the default copyright laws apply, meaning that you retain all rights to your source code and no one may reproduce, distribute, or create derivative works from your work." The same page notes that publishing in a public repository gives other GitHub users "the right to view and fork your repository" - a right that exists on the host, and which does not extend to shipping the code inside your product.

<https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/licensing-a-repository>

- * Choose a License, "No License": "When you make a creative work (which includes code), the work is under exclusive copyright by default... If you find software that doesn't have a license, that generally means you have no permission from the creators of the software to use, modify, or share the software. Although a code host such as GitHub may allow you to view and fork the code, this does not imply that you are permitted to use, modify, or share the software for any purpose."

<https://choosealicense.com/no-permission/>

- * U.S. Copyright Office, Circular 1 (Copyright Basics): protection is secured automatically upon creation; neither registration nor a copyright notice is required. <https://www.copyright.gov/circs/circ01.pdf>
- * Berne Convention: protection is automatic, without formality, in all member states.

THE TWO GENUINE EXCEPTIONS (neither applies to a random GitHub repo):

- (i) Works of the U.S. Government are not subject to copyright in the U.S. (17 U.S.C. 105) - which is why NASA/USGS/USNO material is public domain. That is a statutory exemption for a class of author, not an inference from a missing file.
- (ii) An author may explicitly dedicate a work to the public domain (CC0, Unlicense, or plain wording). Explicit is the operative word.

IMPLIED LICENCES: courts can in narrow, fact-specific circumstances find an implied licence (for example where an author publicly invites reuse of code written for exactly that purpose). Do not build a commercial product on this doctrine. It is a litigation defence, not a rights clearance.

REMEDIES, in order of preference: ask the author for a written licence (one email, frequently successful for small research repos); use a licensed alternative; reimplement from the published mathematics (equations, methods and algorithms are not copyrightable - only the specific expression is); negotiate a licence. Never "just copy it because there is no licence".

- R1. Is the asset a work of the US Government (NASA/USGS/NOAA/NIST/DOE/NSF contractor deliverables with a PD statement)? → PUBLIC DOMAIN IN THE US. You may use/modify/redistribute/sell, with courtesy credit. NASA/USGS still ask you to credit.
- R2. Is the object a NASA logo/insignia/identifier? → CANNOT be used, ever.
- R3. Is it a NASA publication (SVS, Visible Earth, Photojournal) that embeds third-party or institutional material (ESA, university, contractor)? → check the credit line and treat that component separately.
- R4. Is it ESA/Hubble or ESO imagery? → CC BY 4.0 for images/videos/music (credit + licence link + no endorsement; ESA/Hubble and ESO both state this), BUT there are exclusions: no commercial use of imagery dominated by an identifiable person without permission; logos excluded; scientific papers/code excluded. Read the exact policy text and reproduce the credit exactly as specified.
- R5. Is it AURA/STScI imagery? → copyright AURA/STScI; educational/non-commercial use OK, commercial use requires permission. Do not ship in a paid app without permission.
- R6. Is it ESA Space Science Archives data (including Gaia)? → CC BY-NC 3.0 IGO; commercial use requires written authorisation from ESA (data.licences@esa.int). DO NOT ship commercial products derived from Gaia.
- R7. Is it from a data repository that states CC0 (e.g. SDO/JSOC)? → safest category in the project, with the requested courtesy credit.
- R8. Is it a GitHub project? → READ THE LICENSE FILE. GitHub showing no licence means ALL RIGHTS RESERVED (this is exactly the case for rossning92/Blackhole and oseiskar/black-hole). Do not use "it's on GitHub" as a licence.
- R9. Is it a Shadertoy shader with no explicit licence statement? → default CC BY-NC-SA 3.0 (Shadertoy terms). Reimplement the maths; do not copy code.
- R10. Is it a CC BY-SA / GPL asset or code? → share-alike/copyleft. Do not incorporate into a proprietary product unless you intend to comply fully.
- R11. Is the licence unclear, or does the page say "free" without a licence? → mark exactly: LICENSE UNCERTAIN – DO NOT USE UNTIL VERIFIED.
- R12. When several routes exist, prefer the route where YOU derive the asset from archive data yourself (mission data) over the route where you redistribute somebody's prepared picture (a presentation product). This is why ZIG should build its own maps from PDS/USGS/MAST products.

29.2 PER-SOURCE LICENCE TABLE (exact URLs to cite)

Source	Licence	Commercial	Modify	Ship in APK	Attribution	Exact page
NASA media (general policy)	US PD generally; logos excluded; third-party items excluded	Yes (no endorsement)	Yes	Yes	"NASA" as source; logo prohibited	https://www.nasa.gov/nasa-brand-center/images-and-media/
NASA Brand Center / merchandise	Special approvals for commercial merchandise	Case-by-case	—	—	—	https://www.nasa.gov/nasa-brand-center/merchandise-approvals/
Visible Earth / NASA Earth Observatory	PD-US (NASA content)	Yes	Yes	Yes	Credit line per image, e.g. "NASA Earth Observatory images by Joshua Stevens, using VIIRS data from NASA EOSDIS/LANCE/GIBS"	https://visibleearth.nasa.gov/
SDO / JSOC data	CC0 waiver + requested credit	Yes	Yes	Yes	"Courtesy of NASA/SDO and the AIA and HMI science teams"	http://jsoc.stanford.edu/JSOC_Data_Policy.html and https://sdo.gsfc.nasa.gov/data/rules.php
NASA SVS (SVS items, e.g. 4851, 4720, 30362)	NASA content; individual items state their own credits including non-NASA sources	Yes, subject to the credited components	Yes	Yes	Per-item credit; for 4851 include the Gaia/Hipparcos/Tycho/UCAC3 credits	https://svs.gsfc.nasa.gov/4851/
USGS Astrogeology (Astropedia) products	Public domain (many noted CC0)	Yes	Yes	Yes	"Please cite the authors" per product page	https://astrogeology.usgs.gov/ and https://planetarymaps.usgs.gov/
USGS Astrogeology DEMs (e.g. MOLA 463 m)	CC0 1.0	Yes	Yes	Yes	cite authors	https://astrogeology.usgs.gov/search/map/mars_mgs_mola_dem_mosaic_global_463m
HiRISE images & DTMs	PD ("no restrictions on their usage by anyone in the public")	Yes	Yes	Yes	"credit NASA/JPL/University of Arizona"	https://www.uahirise.org/media/usage.php
Caltech Murray Lab CTX mosaic V01	"grants free use ... for all purposes"	Yes	Yes	Yes	"NASA/JPL/MSSS/The Murray Lab"	https://murray-lab.caltech.edu/CTX/
NASA 3D Resources models	NASA content (credit lines per model)	Yes for editorial; obtain approval for merchandise/promotional	Yes	Yes	e.g. "NASA/Michael D. Carbajal", "NASA VTAD"	https://science.nasa.gov/3d-resources/
NASA/JPL Photojournal images	NASA content (JPL images generally PD; credit NASA/JPL-Caltech or the instrument team)	Yes	Yes	Yes	"Image credit: NASA/JPL-Caltech/SSI" (per PIA page)	https://photojournal.jpl.nasa.gov/
ESO images (eso0932a etc.)	CC BY 4.0	Yes (except identifiable-person limitation)	Yes	Yes	credit ESO/S. Brunier + licence link	https://www.eso.org/public/copyright/
ESA/Hubble images	CC BY 4.0 (same structure as ESO)	Yes (same limitation)	Yes	Yes	"ESA/Hubble" + licence link	https://esahubble.org/copyright/
Hubble/OPAL images via STScI	© AURA/STScI	Non-commercial only; commercial needs permission	with permission	with permission	—	https://hubblesite.org/copyright
ESA Space Science Archives / Gaia	CC BY-NC 3.0 IGO	NO (written authorization required: data.licences@esa.int)	NC only	NO for commercial	—	https://www.cosmos.esa.int/web/esdc/terms-and-conditions
NAIF/SPICE	Free, commercial use permitted	Yes	Yes	Yes	acknowledgement per NAIF rules	https://naif.jpl.nasa.gov/naif/rules.html

Source	Licence	Commercial	Modify	Ship in APK	Attribution	Exact page
PDS datasets (rings, LOLA, etc.)	PD-US (per dataset)	Yes	Yes	Yes	cite dataset DOI	https://pds.nasa.gov/ and https://pds-rings.seti.org/
HYG / AT-HYG star catalogue	CC BY-SA 4.0	Yes but share-alike	Yes with SA	Yes with SA	credit + licence	https://github.com/astronexus/HYG-Database (CDS/VizieR, USNO)
Yale BSC5 / USNO UCAC	US Government/academic PD	Yes	Yes	Yes	cite catalogue	
Gaia DR3	CC BY-NC 3.0 IGO	NO	NC only	NO for commercial	—	https://www.cosmos.esa.int/web/gaia/data-licensing
Any repository with NO licence file (e.g. rossning92/Blackhole, oseiskar/black-hole, matthewgiarra/spacecraft-models)	No licence granted – all rights reserved by default	NO	NO	NO	none exists	https://choosealicense.com/no-permission/ and https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/licensing-a-repository
LROC curated website products (non-PDS interface)	Copyrighted (LROC terms of use)	Requires prior permission	Requires permission	NO without permission	per permission	https://lroc.im-ldi.com/about/terms
LROC data via the PDS LROC node / USGS mirrors	Public domain	Yes	Yes	Yes	NASA/GSFC/ASU + the Robinson et al. 2010 citation	https://pds.lroc.im-ldi.com/
Solar System Scope textures	CC BY 4.0	Yes	Yes	Yes	"Solar System Scope"	https://www.solarsystemscope.com/textures/
Filament	Apache-2.0	Yes	Yes	Yes	NOTICE retention	https://github.com/google/filament/blob/main/LICENSE
SceneView	Apache-2.0	Yes	Yes	Yes	notice retention	https://github.com/SceneView/sceneview
Bruneton atmosphere / black hole	BSD-3-Clause	Yes	Yes	Yes	keep copyright notice	https://github.com/ebruneton/black_hole_shader
KTX-Software	API reports NOASSERTION – VERIFY the LICENSE file	VERIFY	VERIFY	VERIFY	VERIFY	https://github.com/KhronosGroup/KTX-Software
Shadertoy shaders (no explicit licence)	CC BY-NC-SA 3.0 (site default)	NO	SA	NO	—	https://www.shadertoy.com/terms
Stellarium / Celestia starless	GPL-2.0 GPL-3.0	Refer only Refer only	— —	NO NO	— —	repo LICENSE files https://github.com/rantonels/starless
Gaia Sky	MPL-2.0	File-level copyleft	—	avoid copying files	—	https://github.com/langurmonkey/gaia-sky
oseiskar/black-hole, rossning92/Blackhole, matthewgiarra/spacecraft-models	NO LICENCE FILE → ALL RIGHTS RESERVED	NO	NO	NO	—	repos (verified absent)

29.3 ATTRIBUTION IMPLEMENTATION IN ZIG (do it properly, it is cheap)

- A "Data & Credits" screen reachable from the main menu and from every object screen ("i" button), listing for each asset: source agency, dataset/product name, publisher URL, licence name with a link, the exact credit string, and the transformation ZIG applied ("downsampled to 8K, converted to sRGB, reprojected to equirectangular").
- A machine-readable credits.json shipped in the APK (also satisfies the "state your modifications" obligations of CC BY).
- An open-source notices screen for Apache-2.0/BSD/MIT components (Filament, SceneView, Bruneton code, meshoptimizer, etc.). This is required, not optional.
- Never place a NASA, ESA, ESO or JPL logo in ZIG. Never write "in partnership with NASA", "NASA-approved", "official".

- Where a licence requires a link (CC BY 4.0), the link must be live in-app.

29.4 THE THREE GENUINE LEGAL RISKS IN THIS PROJECT (state them plainly)

- Risk 1 – Gaia/ESA archive non-commercial licence. Mitigation: never ship Gaia-derived pixels or star catalogues in the commercial build; use NASA SVS 4851's non-Gaia layers (highlighted in its file list) and USNO/Yale catalogues. If you want the Gaia-quality starfield, contact ESA for commercial authorisation (which is a real, documented route).
- Risk 2 – Hubble OPAL imagery under AURA/STScI copyright vs NASA-publication status. Mitigation: derive your Jupiter/Saturn/Uranus/Neptune maps from the archived calibrated data in MAST (mission data), or use the NASA-published release pages and/or the ESA/Hubble CC BY 4.0 publication of the same image; or use JPL Photojournal products and accept lower resolution. Document the route you chose in credits.json.
- Risk 3 – "CC BY-SA" star catalogues (HYG) smuggling share-alike into your data assets. Mitigation: do not use HYG; use Yale BSC5 + USNO (PD/US-Government).

29.5 WHAT MUST NEVER BE SHIPPED

- EOSS runtime assets (models, cmts textures, star binaries, ring sprites) or EOSS code.
- Any GPL/MPL code compiled into the app.
- Any Shadertoy/CC BY-NC-SA shader copied rather than reimplemented.
- Any asset from a repository with no licence file, on the mistaken theory that "no licence means no copyright" (Section 29.1 rule R0). This includes rossning92/Blackhole, oseiskar/black-hole, LGassen/Singularity and the third-party re-uploads of NASA Eyes spacecraft models.
- LROC curated products fetched from the LROC website rather than the PDS node or USGS mirrors (Section 10.1b).
- Any asset with no licence statement, including "free model" sites (Sketchfab default is "no license", TurboSquid/3D Warehouse have their own EULAs) – verify or do not use.
- NASA/ESA/ESO logos, mission patches (the ISS patch etc.), and any identifiable-person imagery used commercially.

30. NASA EYES vs ZIG – GAP ANALYSIS (17 items, reframed for the inspector)

The comparison is now between "NASA's solar-system simulator" and "ZIG's single-object inspector". Four of the original items were about simulation features that ZIG deliberately does not have; they have been replaced with the axes that actually decide this product.

1. ASSET QUALITY. Eyes streams modest LOD textures and hides part of its asset tree. ZIG ships the deepest public datasets - Mercury at 166 m/px, the Moon at 100 m/px global and 0.5 m/px at sites, Mars at 5 m/px over nearly the whole surface - locally. VERDICT: AHEAD, decisively.
2. GEOMETRY. Eyes uses spheroids plus spacecraft models. ZIG uses ellipsoids with real oblateness and DEM displacement up to several million triangles. VERDICT: AHEAD.
3. TEXTURES. Eyes is LDR and streaming-limited. ZIG ships 16K ASTC tile pyramids on device. VERDICT: AHEAD.
4. PBR. Eyes uses three.js-class PBR; ZIG uses Filament with physical light units, anisotropy, clearcoat and subsurface. VERDICT: LEVEL to AHEAD.
5. ATMOSPHERES. Eyes approximates with a Rayleigh tint and shadow entities; ZIG runs LUT-based multiple scattering, full resolution, with offline-baked higher orders. VERDICT: AHEAD.
6. LIGHTING. Eyes has a directional sun and ambient. ZIG has a physical sun scaled by real distance, user-controllable direction, and honest darkness. VERDICT: AHEAD.
7. SHADOWS. Eyes does self-shadowing and ring shadows. ZIG does those plus

- offline horizon-map terrain shadows at grazing sun. VERDICT: AHEAD.
8. HDR. Eyes is LDR with a tone map. ZIG is RGBA16F end to end. VERDICT: AHEAD.
 9. TONE MAPPING. Eyes uses ACES. ZIG offers AgX, PBR Neutral and ACES, chosen per mode, with a science mode that is colour-faithful. VERDICT: AHEAD.
 10. POST-PROCESSING. Eyes has bloom. ZIG has bloom, DOF, SSAO, lens effects, colour grading and TAA - all affordable because one object owns the frame. VERDICT: AHEAD.
 11. CAMERA. Eyes has an excellent orbit and target camera for a simulation. ZIG needs the same feel plus surface-level precision and tap-to-focus. VERDICT: LEVEL to AHEAD (match the feel; add the depth).
 12. SCALE. Eyes works in real solar-system units with dynamic scaling. ZIG normalises each object to unit radius, which removes precision problems entirely but cannot show inter-object scale. VERDICT: EVEN - a deliberate trade of one capability for another.
 13. SCIENTIFIC DATA. Eyes draws on JPL mission data and is authoritative on positions. ZIG draws on the same public archives for appearance and does not attempt positions at all. VERDICT: EVEN on appearance provenance; NOT APPLICABLE on dynamics.
 14. INSPECTION DEPTH (replaces "orbital visualisation"). This is ZIG's home ground: the ability to descend to 166 m/px on Mercury, 0.5 m/px on the Moon, 15 m/px on Earth and effectively unlimited on the black hole, with the data limit made visible. Eyes optimises for breadth of objects and mission context, not depth per object. VERDICT: AHEAD, and it is the reason the product exists.
 15. STARFIELD. Eyes ships six small point-sprite bins. ZIG ships a 16K deep map with millions of catalogue stars, names and constellation overlays, and tap-for-identity. VERDICT: AHEAD.
 16. PERFORMANCE. Eyes is browser-bound. ZIG is native, but it spends its budget on quality rather than scene count, so it must manage thermals honestly. VERDICT: AHEAD on capability, EVEN on sustained load (both need tiering).
 17. ANDROID GPU PIPELINE. Eyes is WebGL in a browser: no ASTC texture arrays, no compute, no virtual texturing, limited formats. ZIG is Filament on Vulkan or GLES3 with KTX2/ASTC, tile streaming and half-float targets. VERDICT: AHEAD - this is the structural advantage.

SUMMARY: ZIG trades breadth for depth. It cannot show a mission timeline or a planet transiting another planet, and it should never claim to. What it can do - and what no free product does well - is let a user press "Moon", zoom to a half-metre of Apollo 17, check the data credit, and see exactly what is real.

=====

31. FINAL RECOMMENDED STACK (single-object inspector)

=====

Language and UI: Kotlin + Jetpack Compose. A launcher grid of thirteen object buttons; one Filament view for the selected object.

Rendering: Google Filament (Apache-2.0), filament-android 1.71.x, GLES3 by default with Vulkan where it wins. Offline-compiled .filamat materials (matc), one bespoke material per object family: solar surface + corona, rocky regolith with displacement and horizon-map shadows, gas-giant banded flow, ring transmission, cloud deck, ice giants, star sprites, relativistic lensing.

Compose and loading: SceneView (Apache-2.0) where it accelerates (async model and environment loading, gesture delegates, node composables); bypassed wherever custom materials or custom RenderTargets are required.

SCENE MODEL: one object at unit radius, camera in object radii, near/far set from zoom. No floating origin, no double precision in the frame loop.

INTERACTION: orbit, pinch zoom, tap-to-focus, double-tap reset; ROTATION control (real sidereal period, 1x / 60x / 3600x / hold) and SUN DIRECTION control (azimuth, elevation, exposure in EV, with per-object presets), plus a data-

resolution HUD and a procedural-detail indicator.

TEXTURES: KTX2 (ASTC or Basis/UASTC), full linear-space mip chains, alpha-aware masks, 8192 guaranteed per texture with wider data expressed as tiles or sampler2DArray slices; 16K albedo targets for hero bodies.

TILE STORE: offline-built ASTC tile pyramids (512x512 with a 4-texel apron) for every object whose source data exceeds one texture; LRU cache with a hard VRAM ceiling; prefetch along camera motion; decode on worker threads.

RELIEF AND SHADOWS: DEM-driven adaptive displacement; normal maps derived from metric height; offline horizon maps (8-16 directions) for grazing-sun terrain self-shadowing and ambient occlusion; analytic sphere/plane shadowing for rings and bodies; Filament shadow maps only for the ISS and local hero patches.

ATMOSPHERES: LUT-based single plus multiple scattering (Bruneton/Hillaire, BSD-3-Clause reference), per-body parameters, offline-baked higher orders, full resolution on flagships, rendered as both a surface term and a shell.

RINGS: optical-depth profiles from PDS (COUVIS 8001, CORSS 8001), transmission $\alpha = 1 - \exp(-\tau/\mu)$, azimuthal variation retained, forward/back-scatter asymmetry, analytic planet and ring shadows.

SKY: NASA SVS 4851 deep map (non-Gaia layers) plus USNO UCAC4 and Yale BSC5 sprite stars with proper flux-preserving sizing; constellation and naming overlays; the same sky feeds the black-hole lensing shader.

BLACK HOLE: own implementation of the Bruneton BSD-3 precomputed-table beam tracing method, blackbody disk with Doppler and gravitational shift, photon ring, quality tiers plus a user-facing ray-step slider, and a documented cinematic preset after James et al. 2015.

HDR AND POST: RGBA16F, HDR bloom, AgX (cinematic) or PBR Neutral (science) tone mapping, TAA, dithering, colour grading, optional supersample.

PACKAGING: Play Asset Delivery; base under about 200 MB with all thirteen objects usable offline; install-time packs for core bodies; on-demand packs for the deep pyramids (Moon 100 m/px, Mercury 166 m/px, CTX and HiRISE regions, Landsat tiles).

TOOLING: Blender (GPL applies to Blender, not to assets you create), GDAL and OpenImageIO for rasters, meshoptimizer for LODs and cache order, KTX-Software for KTX2 (licence to be verified), Filament's matc/cmgen/mipgen, and a custom horizon-map and tile-pyramid baker.

WHAT IS DELIBERATELY ABSENT: SPICE at runtime, Horizons validation, orbital propagation, trajectory rendering, a simulation clock, network access in the renderer, and any WebView.

WHY NOT THE ALTERNATIVES (unchanged reasoning, and now tested against the single-object scope in Section 4.5): raw GLES or Vulkan would be a smaller project than before but would re-implement materials, KTX2 loading, HDR and post-processing for no quality gain; Unity and Unreal add cost, weight and Compose friction; WebView breaks the offline requirement outright.

32. OBJECT-BY-OBJECT FINAL WINNERS (with inspection ceiling)

Object	#1 BEST	#2	#3	Inspection ceiling (real data)	Overall /10
Sun	SDO/AIA+HMI (CC0) + procedural photosphere + corona	SVS 30362 Carrington EUV map (labelled false colour)	Solar System Scope 8K (CC BY 4.0)	about 340 km/px (SDO disk)	9.4
Mercury	MESSENGER MDIS BDR/LOI 166 m/px + USGS DEM 665 m	MDIS colour mosaic 665 m	Solar System Scope 8K	166 m/px albedo; 665 m relief	9.5
Venus	Magellan C3-MDIR + GTDR + Fresnel + slope (radar, labelled)	Magellan topography with procedural clouds	Solar System Scope cloud texture	75 m/px radar on 17%; 4.6 km/px global	9.3
Earth	Blue Marble NG (12 months) + Black Marble + Landsat hero tiles	solar system scope day/night/cloud set	GIBS VIIRS baked mosaic	15 m/px pan over land; 460 m/px night	9.6
Moon	LROC WAC 100 m/px global + LOLA lDEM_64 + NAC 0.5 m/px patches	SVS CGI Moon Kit (colour + displacement)	LROC Hapke 7-band via PDS node	0.5 m/px at sites; 100 m/px global; 474 m relief	9.7

Object	#1 BEST	#2	#3	Inspection ceiling (real data)	Overall /10
Mars	Murray Lab CTX 5 m/px regions + MOLA + HiRISE patches	USGS Viking colour 925 m + MOLA	Solar System Scope 8K	0.25-0.5 m/px at sites; 5 m/px regional	9.6
Jupiter	OPAL annual maps (derived or ESA route) + zonal-wind shader	JPL PIA07782 Cassini cylindrical map	Solar System Scope 8K	about 125 km/px (the tightest of all)	9.0
Saturn	OPAL/Cassini body + PDS tau(r) rings + analytic shadows	Cassini products + simplified ring profile	Solar System Scope body and ring	about 105 km/px body; 1-10 km radial in the rings	9.6
Uranus	OPAL + Irwin et al. 2024 corrected colour + scattering LUT	Voyager 2 assembled map	Solar System Scope 8K	about 45 km/px; near-featureless by nature	9.0
Neptune	OPAL + corrected colour + labelled spot epochs	Voyager 2 mosaics	Solar System Scope 8K	about 43 km/px, epoch-bound	9.1
Milky Way	SVS 4851 deep map (non-Gaia) + USNO/Yale sprite stars	SVS 4851 alone	ESO eso0932a (CC BY 4.0)	angular: 1.3 arcmin/texel; stars are points	9.5
ISS	NASA VTAD ISS_stationary.glb (247547 tris)	NASA 3D Resources ISS (B) glb	ISS (C) high-res after conversion	its own texture resolution	8.8
Black hole	own Bruneton-method (BSD-3) lensing + blackbody disk	KleaSCM/tika (MIT) Kerr reference	hydrogendeuteride (MIT) desktop reference	UNLIMITED (analytic geodesics)	9.7

WHY THE TOP SIX WIN, IN ONE LINE EACH

Moon: the only body where the world's best global mosaic (100 m/px), the best laser altimetry (LOLA) and half-metre close-ups (NAC) all exist, all public domain, all via PDS.

Mars: 5 m/px over 99.5% of the surface under an explicit all-purposes grant, with HiRISE for the deepest zoom in the app.

Earth: Landsat gives 15-30 m/px of real optical detail over all land, free of restrictions, on top of Blue Marble and Black Marble for the global look.

Mercury: a 166 m/px global mosaic with a matching DEM, at the finest global coverage of any rocky body after the Moon.

Saturn: the rings are better resolved (1-10 km) than the planet (about 105 km), so the object rewards inspection in a way no texture pack can fake.

Black hole: the only object with no data wall at all.

===== 33. IMPLEMENTATION ROADMAP (single-object inspector) =====

Nine to twelve months for a small team, in seven phases. The ordering reflects that the tile store and the materials are the real engineering, not the engine integration.

PHASE 0 – LEGAL AND DATA FOUNDATION (3-4 weeks)

- Build the provenance manifest and credits.json; record licence, credit string, resolution and processing for every asset.
- Close the open legal items: the Hubble/OPAL route, KTX-Software's licence file, and permission (or replacement) for the licence-free black-hole references.
- Verify the Landsat tile selection and collect per-tile credit strings.

PHASE 1 – INSPECTOR SHELL (4-6 weeks)

- Filament + Compose app; the thirteen-button launcher; one object at unit radius; orbit, pinch zoom, tap-to-focus; the rotation and sun-direction controls; HDR RenderTarget with tone mapping, bloom and TAA; the data-resolution HUD.
- One object end to end (Earth at 8K) proving KTX2 loading, colour space, lighting and 60 fps. Milestone: a boring but correct Earth.

PHASE 2 – TILE STORE AND PIPELINE (6-10 weeks, the critical path)

- Implement Phase K: offline pyramid builder (GDAL + KTX2 + tiling with apron),

on-device LRU cache, prefetch, worker-thread decode, VRAM ceiling.

- Prove it on the two hardest sources: the 109,152-px lunar WAC mosaic and the 92,160-px Mercury BDR. Milestone: 0.5 m/px patch inspected smoothly on a phone with no hitch.

PHASE 3 – MATERIALS AND LIGHT GEOMETRY (6-8 weeks)

- Offline horizon-map baker; regolith material with displacement, horizon shadows and ambient occlusion; ring material with tau transmission and analytic shadows; atmosphere LUTs and shell for four parameter sets; gas-giant flow shader with real zonal winds.
- Milestone: a lunar terminator sweep that looks like a photograph, and a Saturn whose rings shadow the planet correctly at any sun angle.

PHASE 4 – THE THIRTEEN OBJECTS (6-8 weeks, parallelisable by object)

- Each object: source download, calibration, pack build, material tuning, inspection-spec verification (Section 2.4), credit strings.
- Milestone: all thirteen buttons working offline with maximum-fidelity packs installed for at least the Moon, Mars, Earth and Mercury.

PHASE 5 – MILKY WAY AND BLACK HOLE (8-12 weeks)

- Catalogue ingestion and sprite rendering; constellation and naming overlays.
- Black-hole implementation: LUT generation, lensing shader, blackbody disk, Doppler and redshift, photon ring, quality tiers, ray-step slider, cinematic preset validated against a CPU reference.
- Milestone: the photon ring within a few percent of a reference render, running at 30 fps on a mid-range device at Tier 2.

PHASE 6 – POLISH, PERFORMANCE, DEVICE MATRIX (6-8 weeks)

- Degradation tiers and automatic thermal stepping; load-time and no-hitch targets from 28.6; memory profiling; Adreno, Mali and Xclipse test matrix; accessibility; localisation; the Data & Credits surface; shareable state ("show me Apollo 17" as a deep link).
- Milestone: sustained 60 fps at Tier 0 for ten minutes on a flagship, and a stable 30 fps floor on a three-year-old mid-range device.

34. RISKS AND LIMITATIONS

THE DATA WALL (the defining risk of this product)

- Users will zoom past the real data and will be looking at interpolation. This is unavoidable; what is avoidable is pretending otherwise.
- Mitigation: the resolution HUD, the procedural-detail indicator, the ability to switch procedural layers off, and honest per-object ceiling documentation (2.4). Treat the wall as a feature of the product's integrity.
- Specific exposures: Jupiter (about 125 km/px) and the ice giants (about 45 km/px) hit the wall almost immediately; Venus has no visible-light surface data at all; Earth has no optical sea floor. Each of these needs a designed presentation, listed in its section, rather than a texture rescue.

SCIENTIFIC AND PRESENTATION

- Colour is never absolute: every map is a filter composite with a transfer. Mitigation: the labelling taxonomy (24.3) shown in-app, not buried in a file.
- Epoch-bound content (Neptune's spots, Jupiter's belts) must be dated in the UI. Mitigation: labelled epochs everywhere; never imply "current".
- Procedural detail can drift into decoration. Mitigation: a hard rule that procedural layers are visibly separable and labelled, and that they never

drive scientific claims.

TECHNICAL

- Tile streaming is the hardest engineering in the project and sits on the critical path (Phase 2). Mitigation: build it early, on the two largest sources, with a fallback to single-texture LOD per object if a deadline forces it - the fallback still beats every competitor.
- Texture dimension limits: never assume more than 8192 in one dimension (28.3). Mitigation: tile, array or virtualise, all three supported by the design.
- Thermal throttling during long inspection sessions. Mitigation: tiers, frame pacing, the Android thermal API, and honest FPS targets.
- Driver variance across Mali, Adreno and Xclipse, especially in long fragment shaders and half-float blending. Mitigation: Filament's abstractions, a maintained device matrix, and a shader-complexity tier system.
- Black-hole shader is the highest-risk feature per line of code. Mitigation: prototype it as a skybox material in Phase 1 to de-risk early, then expand.
- Asset preprocessing volume: the Moon pack alone touches a 6 Gpx source. Budget for a real workstation, reproducible build scripts from day one, and checksummed outputs.

LEGAL

- Rule R0 (no licence file means all rights reserved) must be enforced at the build step: a CI check that refuses to package an asset without a licence entry in the manifest.
- The three standing risks remain: Gaia/ESA-archive non-commercial terms, the AURA/STScI versus NASA fork on Hubble imagery, and share-alike licences (CC BY-SA, HYG).
- Newly recorded: LROC website products (curated downloads) are copyrighted for commercial use and must be sourced from the PDS node or USGS instead; and Copernicus Sentinel ESA-processed IMAGES are CC BY-SA 3.0 IGO, whereas the underlying Sentinel DATA is free for commercial use under the Copernicus legal notice. Prefer Landsat (USGS public domain, no restrictions) for Earth hero imagery and avoid this distinction entirely.

PRODUCT

- Breadth envy: users may ask for orbits and mission timelines. The answer is that the trade bought 4x resolution and inspection depth; if those features are ever added, they must be a separate mode, never a compromise on this one.
- The library is finite (thirteen objects). Mitigation: the depth of each pack is the content - hero regions, appearance modes, epochs and labelled data provenance are what keep users exploring.

35. COMPLETE SOURCE LIST

NASA EYES / PIONEER (primary and observed)

1. <https://eyes.nasa.gov/apps/solar-system/> (Observed bundle; 3,071-byte HTML shell)
2. <https://eyes.nasa.gov/apps/solar-system/config.js> (asset roots; 128 bytes)
3. <https://eyes.nasa.gov/apps/solar-system/vendors.js> (2,529,206 bytes; three.js + loaders)
4. <https://eyes.nasa.gov/apps/solar-system/app.js> (1,740,432 bytes; components/managers)
5. <https://eyes.nasa.gov/apps/solar-system/preload.js> (133,068 bytes)
6. <https://eyes.nasa.gov/assets/static/stars/stars.0.bin> ... [stars.5.bin](https://eyes.nasa.gov/assets/static/stars/stars.5.bin) (116,545-173,171 B)
7. <https://eyes.nasa.gov/assets/static/stars/galaxies.0.bin> (27,857 B); [constellations.bin](https://eyes.nasa.gov/assets/static/stars/constellations.bin) (10,952 B)
8. https://eyes.nasa.gov/assets/static/sprites/saturn_rings_top.png (3,588 B)
9. https://eyes.nasa.gov/assets/dynamic/earth/data/visibleEarth/dataset_manifest.json (533,061 B; VIIRS WMTS 2016-2026)

10. [https://eyes.nasa.gov/assets/static/models/sc_iss/ISS_stationary.glTF \(+ .bin 22,186,880 B\)](https://eyes.nasa.gov/assets/static/models/sc_iss/ISS_stationary.glTF (+ .bin 22,186,880 B))
- NASA USAGE POLICY / LEGAL
11. <https://www.nasa.gov/nasa-brand-center/images-and-media/> (NASA media guidelines; 3D models/textures sentence; logo exclusion)
12. <https://www.nasa.gov/nasa-brand-center/merchandise-approvals/> (commercial/merchandise approvals)
13. <https://www.nasa.gov/nasa-brand-center/multimedia-guidelines/> (brand usage)
14. <https://www.eso.org/public/copyright/> (ESO = CC BY 4.0, with exclusions)
15. <https://esahubble.org/copyright/> (ESA/Hubble = CC BY 4.0)
16. <https://hubblesite.org/copyright> (AURA/STScI ©; commercial permission)
17. <https://www.cosmos.esa.int/web/esdc/terms-and-conditions> (ESA Space Science Archives CC BY-NC 3.0 IGO)
18. <https://www.cosmos.esa.int/web/gaia/data-licensing> (Gaia CC BY-NC 3.0 IGO)
19. http://jsoc.stanford.edu/JSOC_Data_Policy.html (SDO data CC0 waiver text)
20. <https://sdo.gsfc.nasa.gov/data/rules.php> (SDO credit request)
21. <https://naif.jpl.nasa.gov/naif/rules.html> (SPICE free incl. commercial)
22. <https://www.shadertoy.com/terms> (default CC BY-NC-SA 3.0)
23. <https://developer.android.com/guide/playcore/asset-delivery/texture-compression> (ASTC/ETC2 coverage)
24. <https://developer.android.com/guide/playcore/asset-delivery> (Play Asset Delivery)
- SUN
25. <https://svs.gsfc.nasa.gov/30362> (Full Map of the Sun's Surface; euvi_aia304_2012_carrington.tif 4104x2304)
26. <https://svs.gsfc.nasa.gov/12613/> (SDO 4k content; 4096x4096)
27. <https://sdo.gsfc.nasa.gov/data/> (AIA/HMI data access)
- MERCURY
28. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_global_mosaic_250m
29. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_global_color_mosaic_665m
30. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_basemap_md3_color_mosaic_global_665m
31. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_basemap_enhanced_color_mosaic_global_665m
32. https://astrogeology.usgs.gov/search/map/mercury_messenger_global_dem_665m
- VENUS
33. https://astrogeology.usgs.gov/search/map/venus_magellan_global_topography_4641m
34. https://astrogeology.usgs.gov/search/map/venus_magellan_global_c3_mdir_synthetic_color_mosaic_4641m
35. https://astrogeology.usgs.gov/search/map/venus_magellan_global_fresnel_reflectivity_4641m
36. https://astrogeology.usgs.gov/search/map/venus_magellan_global_meter_scale_slope_4641m
37. https://astrogeology.usgs.gov/search/map/venus_magellan_sar_fmap_stereo_look_global_mosaic_75m
- EARTH
38. <https://visibleearth.nasa.gov/collection/1484/blue-marble>
39. <https://earthobservatory.nasa.gov/features/BlueMarble>
40. <https://visibleearth.nasa.gov/images/144898> (Black Marble 2016 colour maps)
41. <https://svs.gsfc.nasa.gov/30919> (BlackMarble_2016.tif 86400x43200, 868.6 MB)
42. <https://svs.gsfc.nasa.gov/12573> (Earth at Night flat map)
43. <https://nasa-gibs.github.io/gibs-api-docs/> (GIBS WMTS + acknowledgment)
44. <https://www.ncei.noaa.gov/products/etopo-global-relief-model> (ETOPO 2022)
- MOON
45. <https://svs.gsfc.nasa.gov/4720> (CGI Moon Kit: lroc_color 2k/8k/16k; ldem_4/16/64; ref sphere 1737.4 km)
46. https://astrogeology.usgs.gov/search/map/moon_lro_lroc_wac_global_morphology_mosaic_100m
47. https://planetarymaps.usgs.gov/mosaic/Lunar_LR0_LR0C-WAC_Mosaic_global_100m_June2013.tif (5.5 GB)
48. https://astrogeology.usgs.gov/search/map/moon_lro_lroc_wac_dtm_gld100_118m (10 GB)
49. <https://lroc.im-ldi.com/images/downloads> (LROC curated products; WAC Hapke, 643 nm mosaic)
50. <https://pds-geosciences.wustl.edu/missions/lro/lola.htm> (LOLA gridded data)
- MARS
51. <https://murray-lab.caltech.edu/CTX/> (CTX mosaic V01; grant text)
52. <https://murray-lab.caltech.edu/CTX/V01/tiles/> (tile download)
53. <https://doi.org/10.17189/m23v-ae12> (CTX mosaic PDS archive DOI)
54. https://astrogeology.usgs.gov/search/map/mars_mgs_mola_dem_mosaic_global_463m (CC0)
55. https://astrogeology.usgs.gov/search/map/mars_odyssey_themisd_global_mosaic_100m (THEMIS-IR day, 21 GB)
56. https://astrogeology.usgs.gov/search/map/mars_viking_color_global_mosaic_925m
57. <https://www.uahirise.org/dtm/> (HiRISE DTMs)
58. <https://www.uahirise.org/media/usage.php> (HiRISE PD statement)

59. https://planetarymaps.usgs.gov/mosaic/mars2020_trn/HiRISE/ (Jezero 1 m DTM/ortho)

GIANTS

60. <https://science.nasa.gov/asset/hubble/jupiter-global-map-from-hubble-opal-data-rotation-2/> (3600x1598 TIFF + JPG)

61. <https://science.nasa.gov/asset/hubble/multiwavelength-opal-jupiter/>

62. <https://archive.stsci.edu/prepds/opal/> (OPAL data products)

63. <https://photojournal.jpl.nasa.gov/catalog/PIA07782> (Cassini cylindrical map of Jupiter)

64. <https://photojournal.jpl.nasa.gov/catalog/PIA00011> (Voyager 1 cylindrical map of Jupiter)

65. <https://space.jpl.nasa.gov/tmaps/jupiter.html> (720x360, 2 px/deg, "badly processed")

66. <https://space.jpl.nasa.gov/tmaps/uranus.html> (fictional solid-colour texture, #0087d5)

67. Simon, Wong & Orton 2015, ApJ 812, 55 (OPAL global-map filter/wavelength scheme, 0.1°/px)

68. Irwin et al. 2024, MNRAS 527, 11521 (Uranus/Neptune true-colour re-derivation)

69. <https://science.nasa.gov/missions/hubble/nasas-hubble-celebrates-decade-of-tracking-outer-planets/> (OPAL decade; Neptune spots)

RINGS

70. <https://pds-rings.seti.org/cassini/uvis/index.html> – COUVIS_8001 (438 MB, 1 & 10 km τ profiles)

71. <https://pds-rings.seti.org/ringocc/> – CORSS_8001 (6.14 GB), COVIMS_8001 (320 MB)

72. <https://pds-rings.seti.org/> (Ring-Moon Systems Node home)

73. Colwell et al. 2010 (UVIS occultation processing, AJ 140, 1569)

BLACK HOLE

74. https://github.com/ebruneton/black_hole_shader (BSD-3-Clause, 240★) + https://ebruneton.github.io/black_hole_shader/

75. <https://arxiv.org/abs/2010.08735> (Bruneton 2020; LUT sizes; fps table)

76. https://github.com/ebruneton/precomputed_atmospheric_scattering (BSD-3-Clause, ~1,065★)

77. <https://github.com/KleaSCM/tika> (MIT; Kerr ray tracer)

78. <https://github.com/hydrogendeuteride/BlackHoleRayTracer> (MIT, 41★; OpenGL 4.3 compute)

79. <https://github.com/ImanetorX98/KerrTraceCpp2> (MIT; Kerr/ray-bundle, offline reference)

80. <https://github.com/zongzhengli/black-hole> (CC-BY-4.0; adaptive-step GLSL)

81. <https://github.com/sirxemic/Interstellar> (MIT, 375★)

82. <https://github.com/rantonels/starless> (GPL-3.0, 714★ – REFERENCE ONLY)

83. <https://arxiv.org/abs/1906.11238> ; <https://iopscience.iop.org/article/10.3847/2041-8213/ab0ec7> (EHT M87* shadow, 42±3 μ as)

84. Broderick et al. 2022, ApJ 935, 61 – "The Photon Ring in M87*" (Bibcode 2022ApJ...935...61B)

85. Luminet 1979, A&A 75, 228 – first black-hole-with-disk image

86. James, von Tunzelmann, Franklin & Thorne 2015, Class. Quantum Grav. 32, 065001 – DNGR; a/M=0.6; 6500 K; Doppler asymmetry discussion

87. Bozza 2002, Phys. Rev. D 66, 103001 – analytic strong-deflection

88. Gralla, Lupsasca & Strominger 2020, PRD 102, 124004 – photon-ring observables

STARFIELD / MILKY WAY

89. <https://svs.gsfc.nasa.gov/4851/> (Deep Star Maps 2020; 4k-64k EXR/TIFF; content credits)

90. <https://www.eso.org/public/images/eso0932a/> (ESO Milky Way panorama 6000x3000, CC BY 4.0 via ESO copyright page)

91. <https://github.com/astronexus/HYG-Database> (CC BY-SA 4.0 – AVOID for a proprietary build)

92. Yale Bright Star Catalogue 5th ed. (Hoffleit & Warren 1991) via CDS/VizieR

93. USNO UCAC4 / NOMAD (USNO, US Government public domain)

ISS / MODELS

94. <https://science.nasa.gov/resource/international-space-station-3d-model/> (ISS_stationary.glb 44,495,916 B; USDZ 15.97 MB; credit NASA VTAD)

95. https://assets.science.nasa.gov/content/dam/science/psd/solar/2023/09/i/ISS_stationary.glb (verified download)

96. <https://science.nasa.gov/3d-resources/international-space-station-iss-b/> (ISS (B), credit NASA/Michael D. Carbajal)

97. <https://science.nasa.gov/3d-resources/international-space-station-iss-c-high-res/> (ISS (C) High-Res, Lightwave, 35.53 MB)

98. <https://science.nasa.gov/3d-resources/> (NASA 3D Resources index)

99. https://spotthestation.nasa.gov/trajectory_data.cfm (ISS OEM state vectors, J2000, km/km-s – validation/fallback DATA for ZIG's own orbit solution)

100. <https://eol.jsc.nasa.gov/> (NASA astronaut photography – model orientation calibration reference)

ENGINES / TOOLS / CODE LICENCES

101. <https://github.com/google/filament> (Apache-2.0; 20,489★; pushed 2026-09-13)

102. <https://github.com/google/filament/blob/main/docs/Materials.md> (parameter types; samplerCubemap; customSurfaceShading; transmission; skybox material example)

103. <https://github.com/SceneView/sceneview> (Apache-2.0; 1,316★; Maven io.github.sceneview:sceneview 2.3.x)

104. <https://github.com/ChronosGroup/KTX-Software> (GitHub API licence = NOASSERTION → VERIFY LICENSE FILE)

105. <https://github.com/google/draco> (Apache-2.0)

106. <https://github.com/zeux/meshoptimizer> (MIT; LOD + cache optimisation + EXT_meshopt_compression)

107. <https://github.com/Stellarium/stellarium> (GPL-2.0; reference only)

108. <https://github.com/CelestiaProject/Celestia> (GPL-2.0; reference only)

109. <https://github.com/langurmonkey/gaiasky> (MPL-2.0; reference only)
110. <https://www.blender.org/> (GPL-2.0 applies to Blender, not to the assets you create with it)
- EPHEMERIS / COORDINATES
111. Archinal et al. 2018, Celest Mech Dyn Astr 130:22 – IAU WGCCRE 2015 report (rotation elements, longitude conventions)
112. <https://ssd.jpl.nasa.gov/horizons/> (JPL Horizons – validation authority)
113. <https://naif.jpl.nasa.gov/naif/data.html> (SPK/PCK/FK/LSK kernel access; DE440/DE441)
- ADDED IN REVISION 2
114. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_global_basemap_bdr_166m (Mercury BDR 166 m; 4 GB; cite authors)
115. https://astrogeology.usgs.gov/search/map/mercury_messenger_mdis_basemap_loi_global_mosaic_166m (Mercury LOI 166 m; 4 GB)
116. <https://lroc.im-ldi.com/about/terms> (LROC terms: PDS-delivered = public domain; website images and curated products = copyrighted; commercial use requires permission)
117. <https://pds.lroc.im-ldi.com/> and <https://data.lroc.im-ldi.com/lroc/search> (PDS LROC node: the public-domain distribution channel for LROC products)
118. <https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/licensing-a-repository> (no licence = default copyright, all rights retained)
119. <https://choosealicense.com/no-permission/> (what "no licence" means for users; the three lawful options)
120. <https://www.copyright.gov/circs/circ01.pdf> (U.S. Copyright Office Circular 1: copyright is automatic; no registration or notice required)
121. Irwin et al. 2024, MNRAS 527, 11521 (Uranus/Neptune corrected colours - basis of the appearance toggles in Sections 14 and 15)

APPENDIX A – SECOND-PASS REVIEW (not part of the 35-section structure)

A.1 WHAT IT WAS An independent research pass on the same brief (search-based, not artefact-based) was reviewed against this report. It reached the same headline conclusions - Filament plus SceneView, locally generated spheres rather than downloaded planet models, NASA 3D Resources for the ISS, SD0-derived Sun, the same Moon/Mars strategy, the same packaging approach - so the core recommendation is now independently corroborated.

A.2 ADOPTED FROM THAT PASS (changes now incorporated)

- A-1. Mercury 166 m/px MESSENGER BDR and LOI global mosaics (Section 7). This was a genuine improvement on the 250 m mosaic specified in revision 1; the products were verified on Astropedia and the files confirmed live.
- A-2. The LROC terms-of-use issue (Section 10.1b). The second pass flagged that the LROC downloads page has a citation policy; reading it revealed that its curated products are copyrighted for commercial use. Revision 1 cited that page without the caveat, which was an error; the PDS / USGS route is now specified instead.
- A-3. A labelled "historical vs corrected appearance" toggle for Uranus and Neptune (Sections 14, 15). Good product thinking: it converts a mandatory colour correction into a feature and keeps both states honestly labelled.
- A-4. A user-facing quality/iteration control for the black-hole pass (Section 18.4), which makes the performance trade-off legible.
- A-5. GEBCO as an additional bathymetry option for Earth, alongside ETOPO 2022.
- A-6. Reported (not verified) vendor figures for the engine section: SceneView's documented node types and its stated ~3 MB of APK overhead, and the community characterisation of raw Filament as "about 1000 lines to draw a cube". These are indicative, not contractual.

A.3 CONSIDERED AND REJECTED, WITH REASONS

- R-1. "Gaia/Hipparcos catalogues are freely usable with attribution." Rejected: Gaia and the ESA Space Science Archives are CC BY-NC 3.0 IGO; commercial use requires written authorisation (data.licences@esa.int). A Gaia-derived skybox or catalogue therefore cannot ship in a commercial build. The substitutes are the non-Gaia layers of NASA SVS item 4851 plus USNO UCAC4 and Yale BSC5.
- R-2. Promoting Solar System Scope art maps to #1 for Mercury, Venus and the four giants. Rejected: it inverts the stated priority order. For every one of those bodies a higher-resolution, better-provenanced, public-domain NASA/USGS product exists (Magellan + GTDR + Fresnel for Venus; OPAL or JPL PIA for the giants; MESSENGER BDR 166 m for Mercury). CC BY 4.0 makes an asset usable, not better.

- R-3. Brute-force Schwarzschild ray marching with 50-2000 steps as the primary method. Rejected as the primary path: the precomputed-table beam-tracing method (Bruneton 2020, BSD-3-Clause) gives constant cost per pixel and was measured at ~150 fps at 1080p on a 2015 desktop GPU, versus ~45 fps for 1000-step marching. Retained only as a user-selectable high-quality tier.
- R-4. Screenshot-space "distortion" lensing as a mid-tier. Rejected as anything above the lowest tier: a screen-space warp produces neither the photon ring nor the disk images above and below the shadow, which are the visible signatures of general relativity. If used at all, label it "artistic approximation".
- R-5. Sketchfab "CC-BY model (verified)" as an ISS backup, and rossning92/Blackhole as "the best native-portable reference". Rejected: Sketchfab's default is no licence, and rossning92/Blackhole has no licence file at all (verified: repository root contains no LICENSE; GitHub API reports license: None) - all rights reserved - and its README credits another unlicensed project for its physics.
- R-6. "Pioneer" and "Dynamo" treated as unconfirmed. Incorrect, and the difference matters: this report read the deployed application, where the 3D canvas root is literally `<div id="pioneer"></div>` and mission-data URLs are built as `$DYNAMIC_ASSETS_URL/dynamo/{mission-url}`. Search-only research cannot see inside a running application; artefact-based research can.

A.4 WHAT THE SECOND PASS MISSED The Murray Lab Global CTX Mosaic (5 m/px, 5.7 terapixels, "free use for all purposes") - the best Mars asset available anywhere. The PDS ring optical-depth datasets and the tau-to-alpha transmission relation. The AURA/STScI versus NASA copyright fork on Hubble/OPAL imagery. The HYG CC BY-SA share-alike risk. The Shadertoy CC BY-NC-SA default. The KTX-Software NOASSERTION licence flag. The critical impact parameter $b_c = 3\sqrt{3}M$, the Liouville intensity transform (I proportional to g^3), the EHT photon-ring measurement and the DNGR parameters ($a/M = 0.6$, 6500 K) that define the interstellar-style look. And the observed contents of the Eyes bundle: six star-sprite binaries, a 3,588-byte Saturn ring sprite, the CMTS colour/height/normal/specular/night/cloud channels and the GIBS VIIRS tile manifest.

A.5 EFFECT ON THE FINAL RECOMMENDATION None of the above changes the engine, the stack, the ranking of any object, or the licensing posture. It improves one asset specification (Mercury), fixes one licence error (LROC), strengthens the black-hole plan, and adds three product ideas. The recommendation below stands as revised.

APPENDIX B – OBJECT BUTTON MANIFEST (per-object asset budget)

Thirteen buttons. For each: the deepest real data, what ships in the base module, what the install-time pack adds, what the on-demand pack adds, and the disk estimate using ASTC 6x6 with mip chains (28.2). All figures are budgets to build against, not measurements.

B.1 SUN

Data ceiling: SDO AIA/HMI 4096x4096 (about 340 km/px).
 Base: 2K procedural photosphere parameters + one AIA 304 disk, labelled false colour. About 8 MB.
 Install-time: 8K procedural base, HMI continuum, magnetogram 1024x512, 10 AIA frames across the channel set, corona parameters. About 120 MB.
 On-demand: 100-frame SDO time series at 4096 per channel, for a "one solar rotation" playback. About 900 MB.
 Modes: photosphere | AIA 171 / 193 / 304 (labelled) | magnetogram | far-side context (helioseismic, labelled low-resolution) | corona intensity.

B.2 MERCURY

Data ceiling: MDIS BDR/LOI 166 m/px (92160x46080, 4.25 Gpx); DEM 665 m/px.
 Base: 8K downsample of BDR (about 2.2 km/px) + MD3 colour at 4K. About 25 MB.
 Install-time: 16K BDR (about 940 m/px), 8K DEM as 16-bit height, 8K normal, MD3 and enhanced-colour 4K. About 260 MB.
 On-demand: the FULL 166 m/px tile pyramid (about 2.5 GB) plus a regional stereo DTM set where higher-resolution elevation exists.
 Modes: BDR (74 deg) | LOI (45 deg) | MD3 colour | enhanced colour (labelled) |

DEM and normal overlay | sun angle.

B.3 VENUS

Data ceiling: 4.6 km/px radar global; 75 m/px radar over about 17%.
Base: 4K C3-MDIR radar + procedural cloud-deck shader. About 10 MB.
Install-time: 8K radar, 8K topography (16-bit), 4K Fresnel reflectivity, 4K meter-scale slope, atmosphere parameter set. About 180 MB.
On-demand: the 17% stereo-look mosaic patches at 75 m/px. About 0.6 GB.
Modes: cloud deck (default, labelled) | radar surface (labelled) | topography | Fresnel | slope | atmosphere on/off | peel animation between cloud and radar.

B.4 EARTH

Data ceiling: Landsat 15 m/px (pan) and 30 m/px over all land since 2013;
Blue Marble 1.85 km/px global; Black Marble about 460 m/px night; ETOPO 2022 bathymetry about 450 m.
Base: 8K Blue Marble (one month), 4K Black Marble, 4K cloud layer, 2K ocean mask. About 45 MB.
Install-time: 12 monthly BMNG at 8K, 8K Black Marble, 8K normal from SRTM/ETOPO, cloud layer at 4K with opacity control. About 420 MB.
On-demand: 64-128 curated Landsat tiles at 15-30 m/px (cities, deltas, volcanoes, glaciers, deserts, reefs, megaprojects). About 0.6-1.4 GB.
Modes: month selector | day/night crossfade | clouds | ocean glint | bathymetry overlay | Landsat hero tiles | atmosphere | sun angle.

B.5 MOON

Data ceiling: NAC 0.5 m/px over large areas; WAC 100 m/px global (109152 x 54576); LOLA ldem_64 at 474 m/px global; NAC DTMs about 1-2 m at sites.
Base: 8K WAC morphology + 4K LOLA-derived normal. About 40 MB.
Install-time: 16K WAC (about 400 m/px), 8K LOLA height and normal, SVS CGI Moon Kit colour, 8K horizon map for terrain shadows. About 520 MB.
On-demand: the 100 m/px global tile pyramid (about 3.5 GB) and 20 NAC hero patches at 0.5 m/px (Apollo 11/12/14/15/16/17, Tycho, Aristarchus, Reiner Gamma, Hadley Rille, SPA floor). About 1.0 GB.
Modes: WAC morphology | SVS colour | Hapke 7-band via PDS (labelled) | LOLA overlay | NAC patches | horizon-map shadows on/off | sun angle (essential).

B.6 MARS

Data ceiling: HiRISE 0.25-0.5 m/px at sites; CTX 5 m/px over 99.5%; MOLA 463 m/px global; HiRISE DTM about 1 m at sites.
Base: 8K MDIM 2.1 (about 2.7 km/px) + MOLA-derived normal at 4K. About 35 MB.
Install-time: 16K CTX-derived overview, 8K Viking colour blended with CTX luminance, 8K MOLA height, 8K horizon map, dust-opacity parameter set. About 480 MB.
On-demand: CTX hero regions at 5 m/px (Jezero, Gale, Valles Marineris, Olympus flank, Hellas rim, south polar layers, Medusae Fossae, Nili Fossae) about 1.2 GB, plus 50-100 HiRISE patches at 0.25-0.5 m/px about 0.3 GB.
Modes: CTX plus Viking colour | THEMIS IR day (labelled thermal) | MOLA hypsometric tint | dust opacity | polar cap season | horizon-map shadows | sun angle.

B.7 JUPITER

Data ceiling: about 125 km/px (OPAL 0.1 deg/px, 3600 x 1598).
Base: 2K OPAL + band shader. About 6 MB.
Install-time: 4K OPAL, wind-profile LUT, cloud-turbulence params, oblateness model. About 30 MB.
On-demand: per-epoch OPAL maps (annual) for a decade, plus PIA07782. About

120 MB.

Modes: OPAL epoch selector | Cassini map | wind shear speed | aurora ovals | lightning (labelled, optional) | sun angle.

B.8 SATURN

Data ceiling: about 105 km/px body; rings 1-10 km radial (Cassini UVIS).

Base: 2K body + coarse ring annulus. About 8 MB.

Install-time: 4K body, 16K radial ring profile with azimuthal variation, analytic shadow setup, oblateness and 26.73 deg tilt. About 90 MB.

On-demand: 32K ring profile (as a 2D array), E-ring haze, spoke overlay. About 150 MB.

Modes: tau transmission rings | forward/back-scatter asymmetry | ring shadow on planet | planet shadow on rings | spokes | E ring | sun angle (changes the ring opening angle, the object's best behaviour).

B.9 URANUS

Data ceiling: about 45 km/px; near-featureless.

Base: 2K appearance + ring set. About 6 MB.

Install-time: 4K map, polar-cap emphasis, corrected and historic appearance pair. About 40 MB.

On-demand: none required.

Modes: corrected (Irwin et al. 2024) | historic Voyager-era (labelled) | rings | polar cap | obliquity/pole-on camera preset | aurora (optional).

B.10 NEPTUNE

Data ceiling: about 43 km/px; epoch-bound features.

Base: 2K appearance + ring set. About 6 MB.

Install-time: 4K map, Adams ring with arcs, epoch spot composite. About 45 MB.

On-demand: Voyager 2 mosaic epoch pack. About 80 MB.

Modes: corrected | historic (labelled) | epoch selector (1989, 2018 and later OPAL epochs) | rings and arcs | aurora (optional).

B.11 MILKY WAY

Data ceiling: 16384 x 8192 map (1.3 arcmin/texel); catalogue stars are point sources at any zoom.

Base: 4K skybox + 200k stars. About 25 MB.

Install-time: 8K skybox, 2M stars with magnitudes and colours, constellation figures and IAU boundaries, named stars. About 150 MB.

On-demand: 16K skybox and a deeper catalogue (to about 10M stars). About 600 MB.

Modes: magnitude-limit slider | constellation overlay | labels | deep-map on/off | "visualisation" provenance note.

B.12 INTERNATIONAL SPACE STATION

Data ceiling: the model's own textures (247547 triangles, 28 materials, 26 PNG textures).

Base: 64k-triangle LOD + KTX2 textures at half resolution. About 30 MB.

Install-time: full model, KTX2/ASTC re-encode, clearcoat MLI, Kapton foil materials, scale bar, module labels. About 150 MB.

On-demand: hero module textures and additional 3D Resources spacecraft as a later expansion. About 300 MB.

Modes: sun only | sun plus earthshine | orbital dusk | LOD auto/manual | rotation hold | module labels.

B.13 BLACK HOLE

Data ceiling: none (analytic).

Base: LUTs + low-resolution sky + disk texture. About 10 MB.
Install-time: full-quality LUTs (512x512 and 64x32 RG32F tables), disk turbulence textures, blackbody colour table. About 25 MB.
On-demand: cinematic presets, higher-order disk imagery, extra sky variants. About 60 MB.
Modes: physically correct (beaming and redshift on) | cinematic (documented after James et al. 2015) | M87-like | Sgr A-like | inclination, ISCO, disk temperature, ray-step slider.

B.14 TOTAL BUDGET

Base module, all thirteen objects usable offline: about 280 MB installed (about 200 MB compressed in the Play bundle).
Install-time packs for all objects: about 2.5 GB.
On-demand maximum-fidelity packs: about 8 GB if every option is installed.
Recommendation: ship install-time packs for Moon, Mars, Earth, Mercury and Saturn; make the lunar 100 m/px, Mercury 166 m/px, CTX and Landsat pyramids on-demand, so a typical user reaches roughly 3 GB installed and the completionist tops out around 11 GB.

=====

END OF REPORT Compiled 2026-09-14. Research and verification only; no implementation code. Every URL above was fetched or queried during this research except where a citation is given for a published paper (fetched via its publisher/abstract page). Items explicitly flagged NOASSERTION/UNCERTAIN must be re-verified before any shipping decision.

ADDED IN REVISION 3 (single-object inspector)

- 122. <https://www.usgs.gov/faqs/are-there-any-restrictions-use-or-redistribution-landsat-data> - "There are no restrictions on Landsat data downloaded from the USGS; it can be used or redistributed as desired"
- 123. <https://www.usgs.gov/faqs/are-landsat-data-cloud-still-considered-be-within-public-domain> - Landsat data are an official USGS record in the public domain; permission not required
- 124. <https://www.usgs.gov/faqs/are-usgs-reports-publications-copyrighted> - USGS-authored data and information are in the US public domain
- 125. <https://open.esa.int/copernicus-sentinel-satellite-imagery-under-open-licence/> - ESA-processed Sentinel images are CC BY-SA 3.0 IGO (share-alike)
- 126. https://sentinels.copernicus.eu/documents/247904/690755/Sentinel_Data_Legal_Notice - the legal notice governing Copernicus Sentinel DATA use (free and open, including commercial), distinct from portal content
- 127. <https://science.nasa.gov/asset/hubble/jupiter-global-map-from-hubble-opal-data-rotation-2/> - OPAL Jupiter global map release page (retained for the giant-planet routes)
- 128. <https://pds-rings.seti.org/cassini/uvis/index.html> - COUVIS_8001 optical-depth profiles, 1 km and 10 km (retained; now load-bearing for the ring material)
- 129. <https://github.com/google/filament/blob/main/docs/Materials.md> - Filament material reference: sampler2d / sampler2dArray / samplerCubemap, customSurfaceShading, transmission, and the skybox material pattern used by the black-hole pass
- 130. https://github.com/ebruneton/precomputed_atmospheric_scattering - BSD-3-Clause scattering reference implementation (atmosphere LUT generation)
- 131. <https://github.com/zeux/meshoptimizer> - MIT; LOD generation and vertex/index cache optimisation for the ISS and hero patches

RETAINED FOR REFERENCE ONLY AFTER REVISION 3 (no longer part of the runtime plan):
the NAIF/SPICE kernel pages (113, and the naif rules page), JPL Horizons (112),
and the ISS trajectory data (99). They are kept in this list because they are
the correct authority if ZIG ever adds a dynamics mode - not because the
inspector needs them.

=====